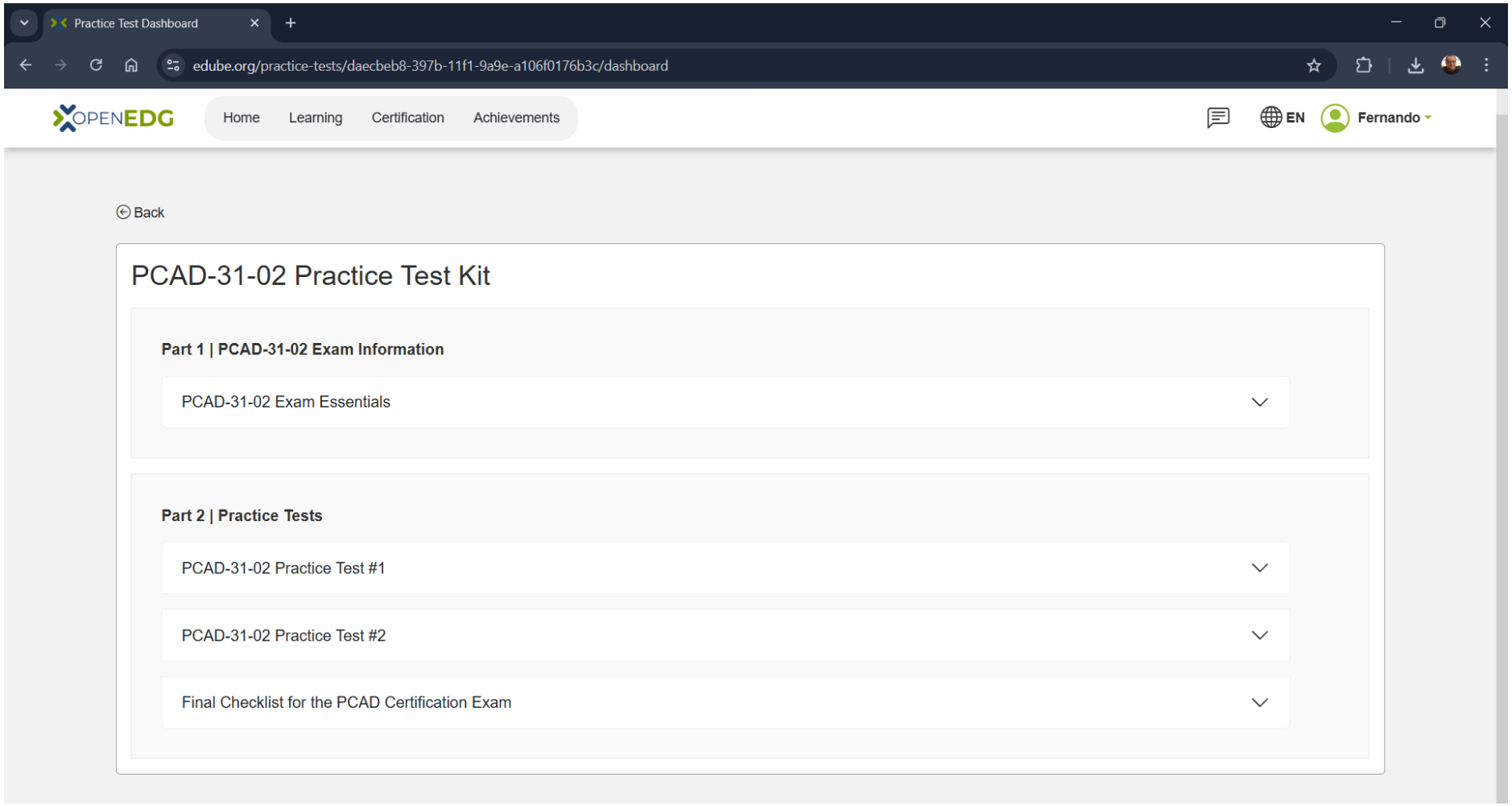
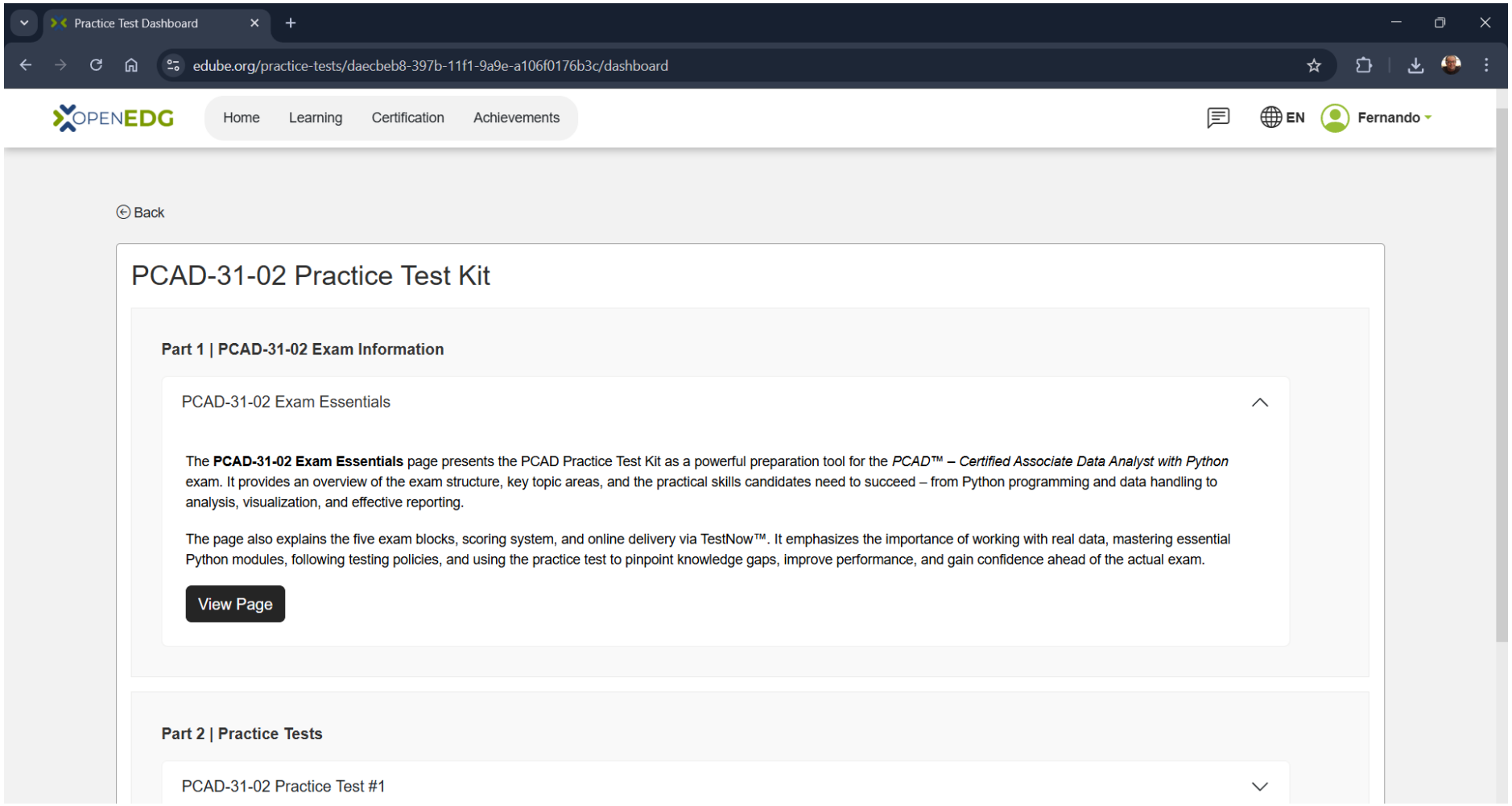
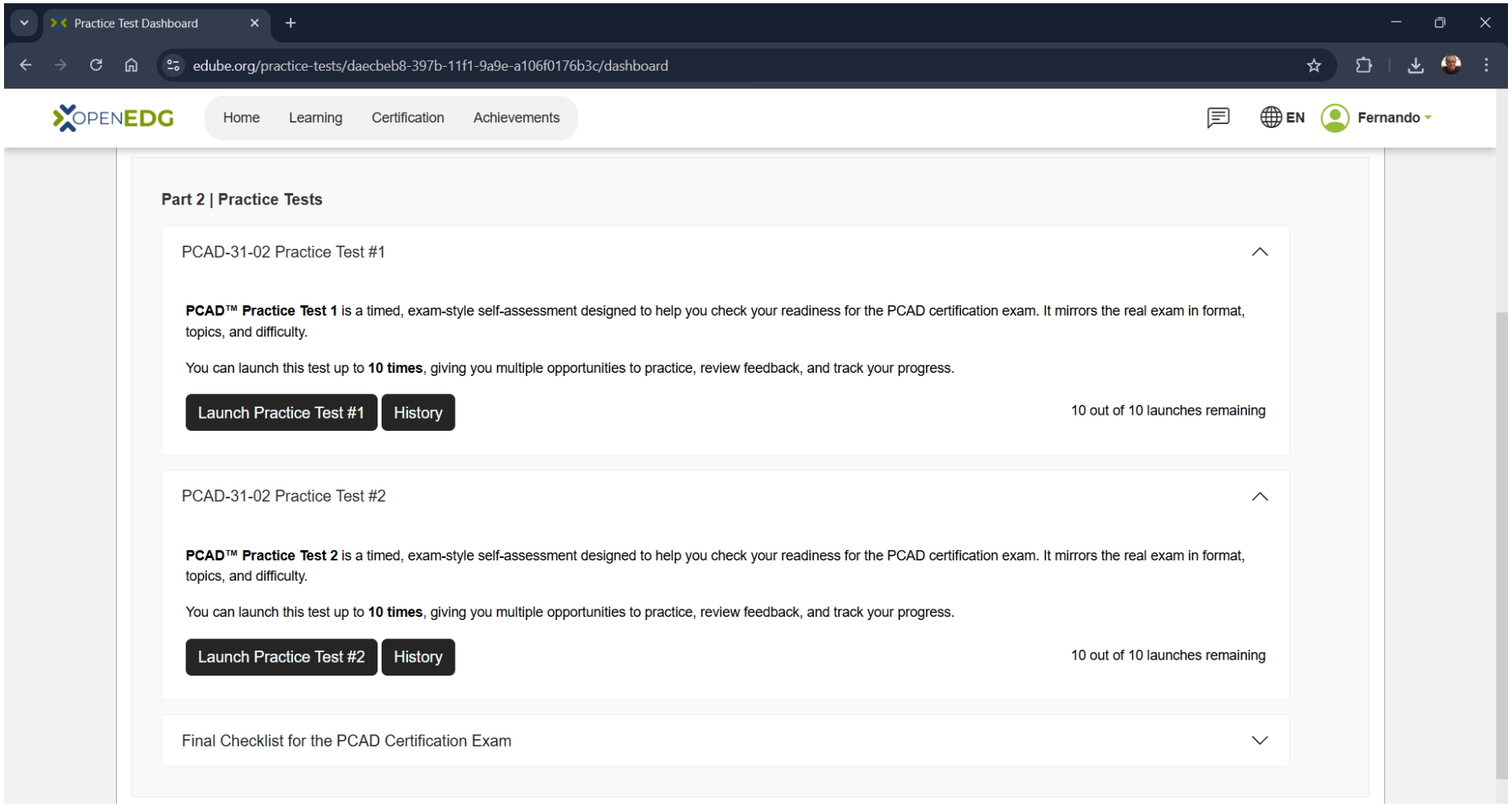
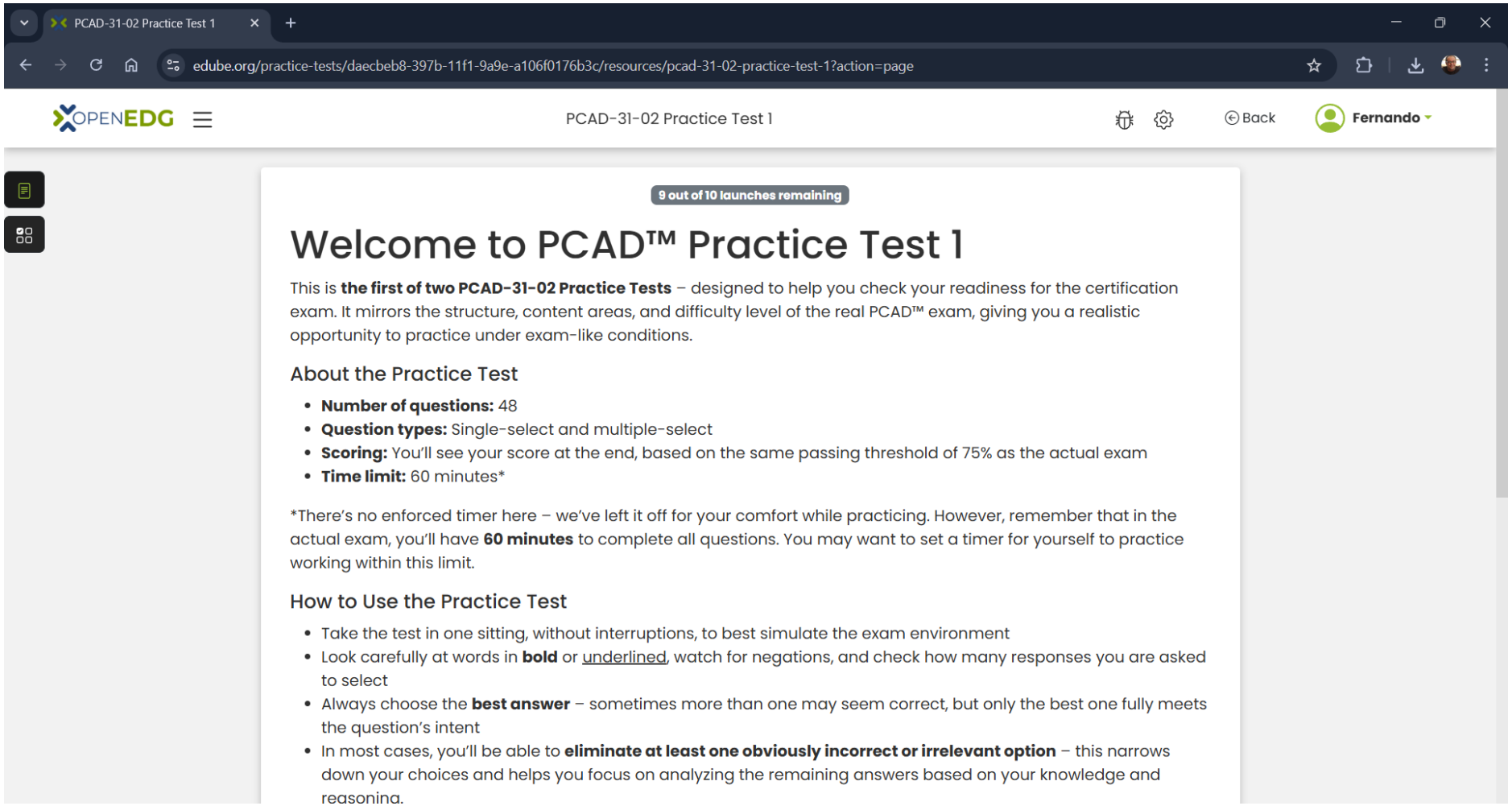
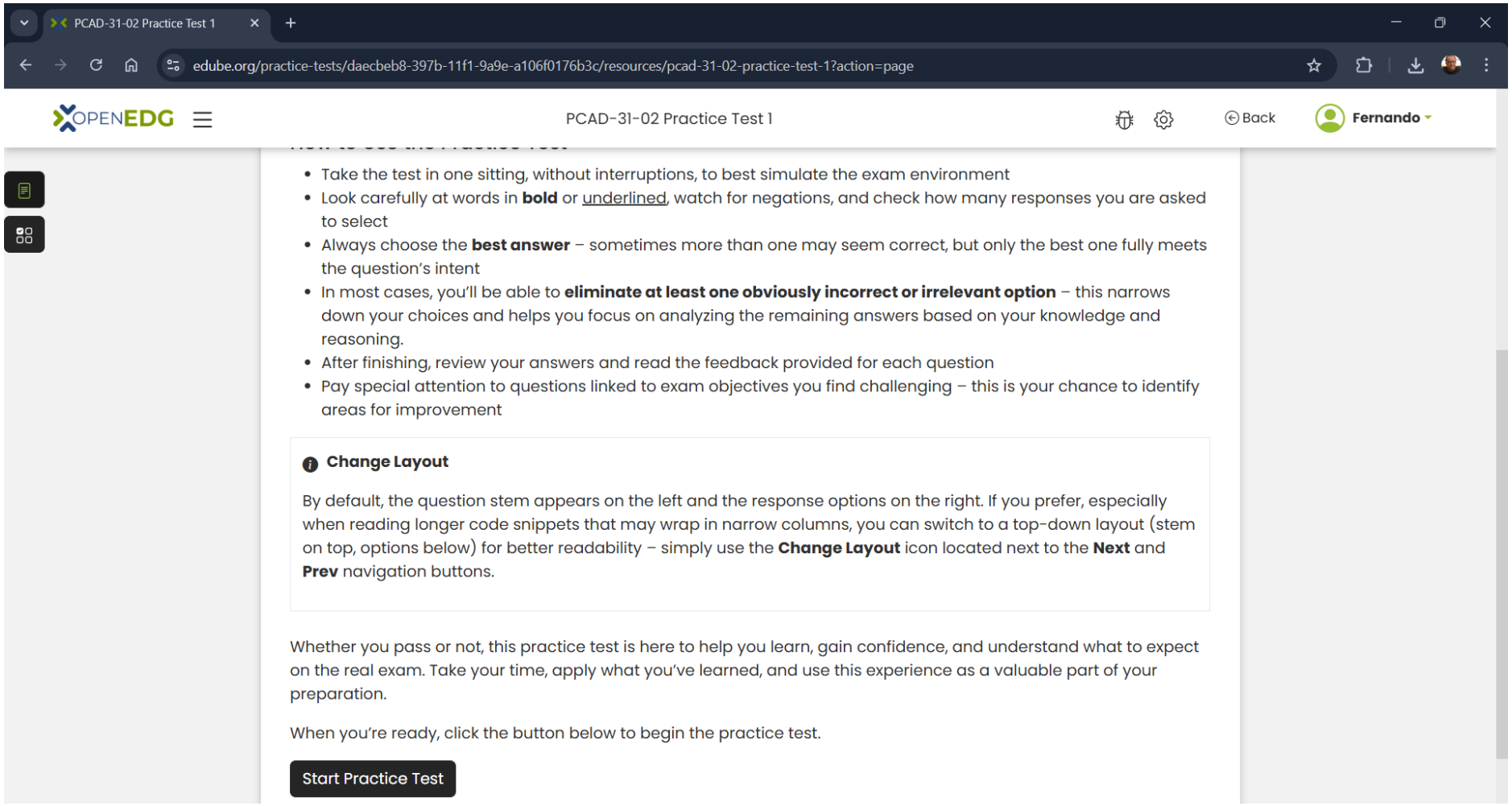
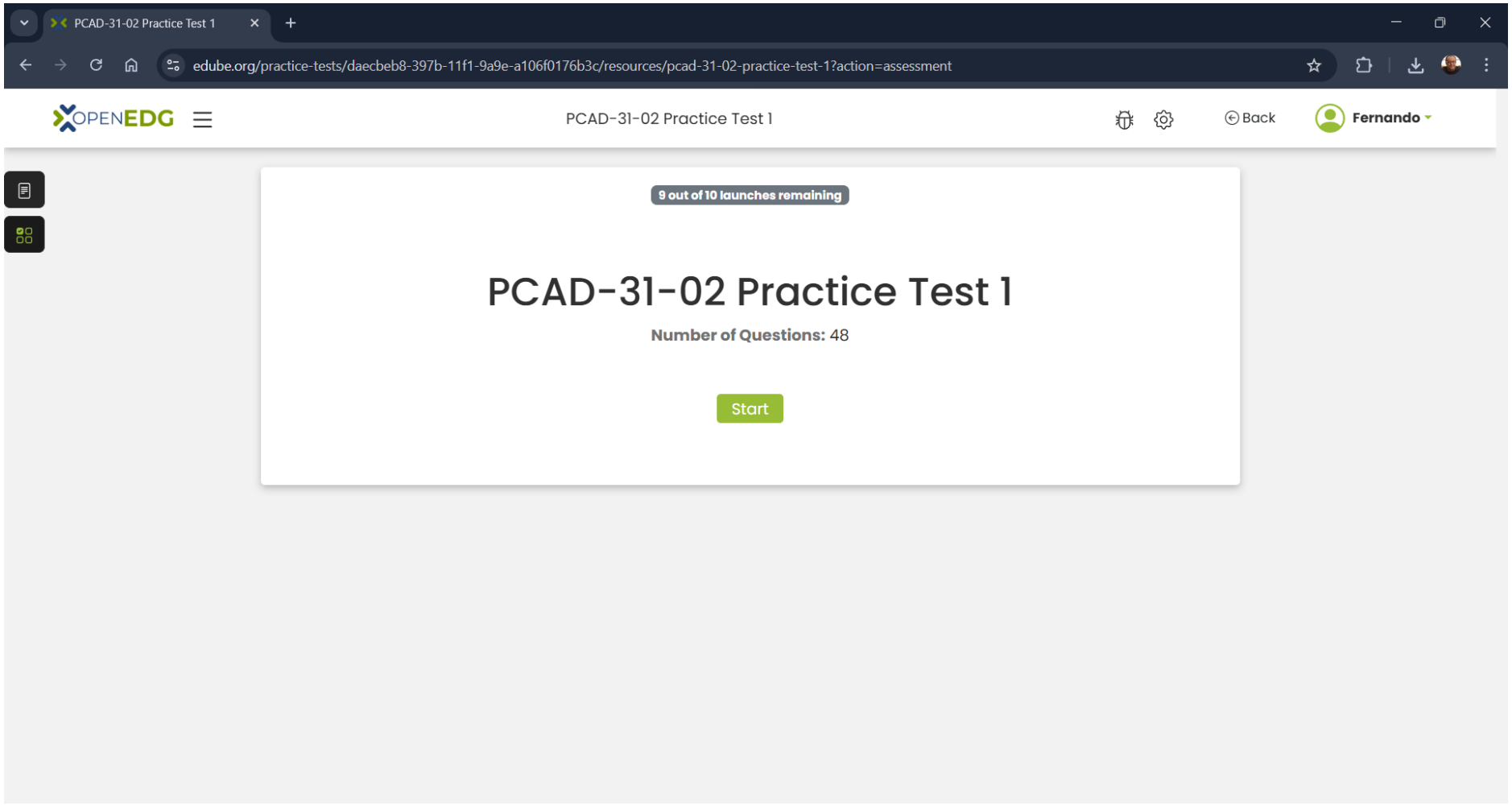

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PCAD-31-02 Exam Essentials
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Final Checklist for the PCAD Certification Exam

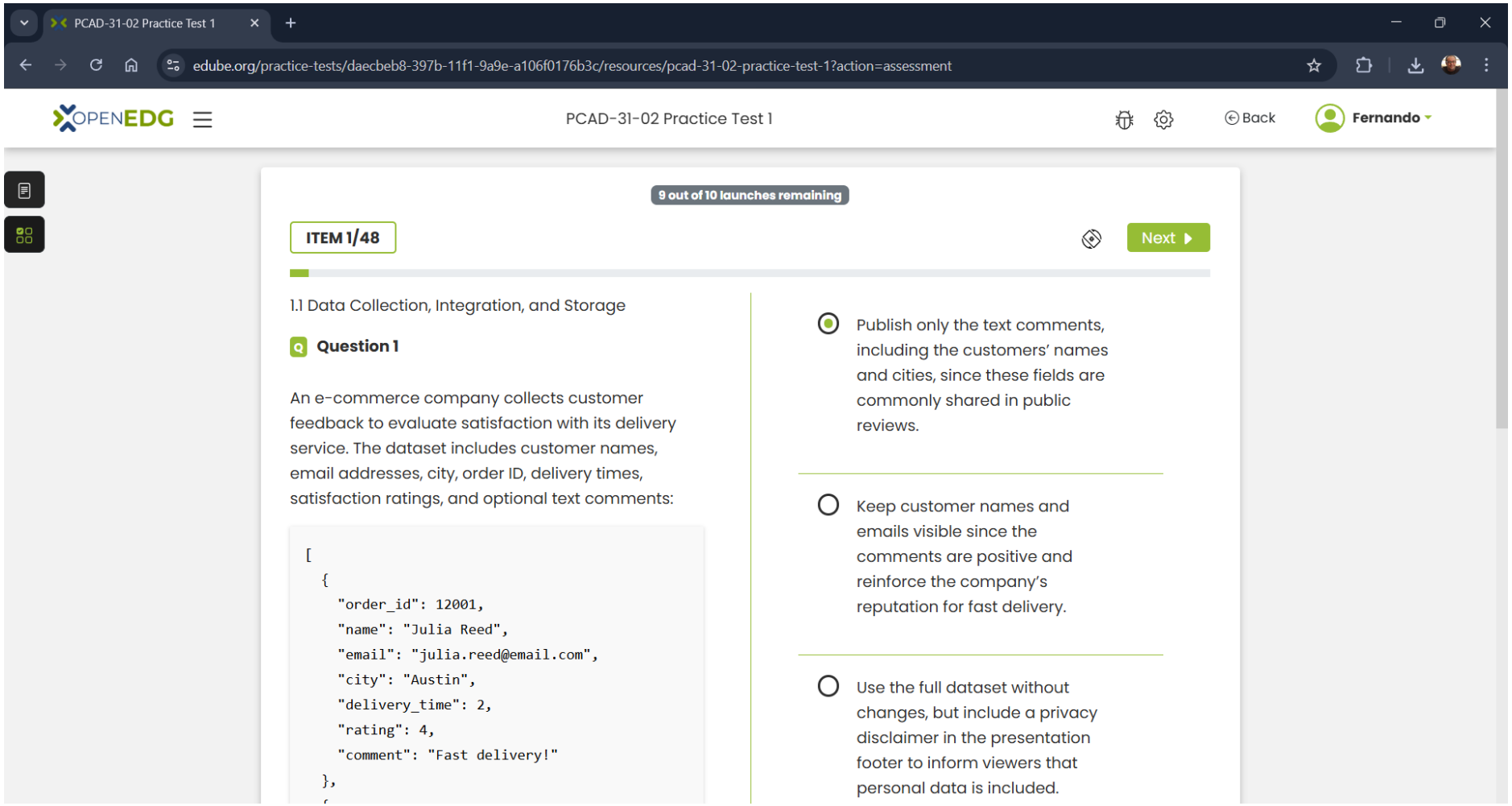

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PCAD-31-02 Practice Test Kit
Part 1 | PCAD-31-02 Exam Information
PCAD-31-02 Exam Essentials
The PCAD-31-02 Exam Essentials page presents the PCAD Practice Test Kit as a powerful preparation tool for the PCAD™ – Certified Associate Data Analyst with Python exam. It provides an overview of the exam structure, key topic areas, and the practical skills candidates need to succeed – from Python programming and data handling to analysis, visualization, and effective reporting.
The page also explains the five exam blocks, scoring system, and online delivery via TestNow™. It emphasizes the importance of working with real data, mastering essential Python modules, following testing policies, and using the practice test to pinpoint knowledge gaps, improve performance, and gain confidence ahead of the actual exam.
View Page
Part 2 | Practice Tests
PCAD-31-02 Practice Test #1

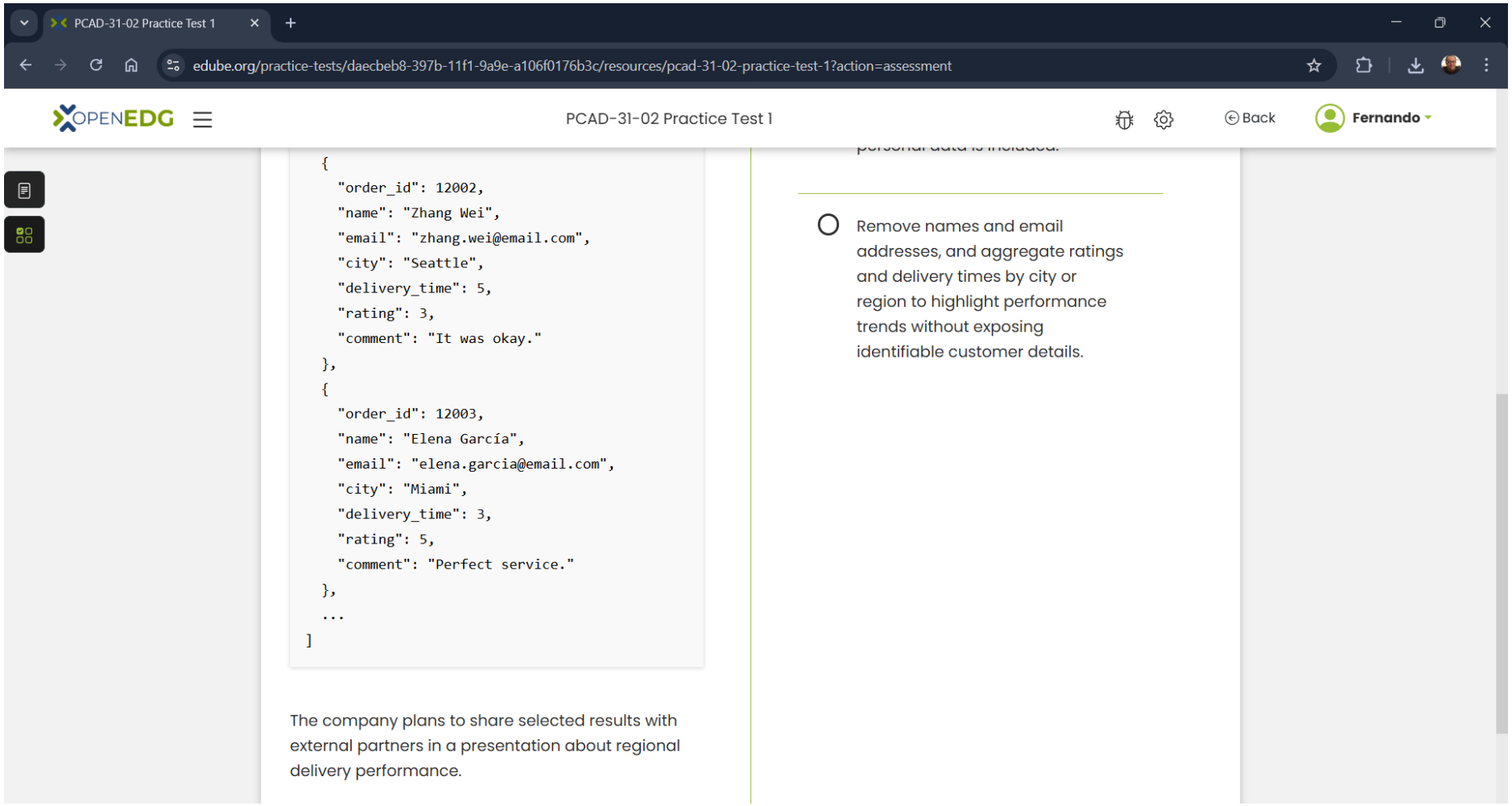

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Part 2 | Practice Tests
PCAD-31-02 Practice Test #1
PCAD™ Practice Test 1 is a timed, exam-style self-assessment designed to help you check your readiness for the PCAD certification exam. It mirrors the real exam in format, topics, and difficulty.
You can launch this test up to 10 times, giving you multiple opportunities to practice, review feedback, and track your progress.
Launch Practice Test #1
History
10 out of 10 launches remaining
PCAD-31-02 Practice Test #2
PCAD™ Practice Test 2 is a timed, exam-style self-assessment designed to help you check your readiness for the PCAD certification exam. It mirrors the real exam in format, topics, and difficulty.
You can launch this test up to 10 times, giving you multiple opportunities to practice, review feedback, and track your progress.
Launch Practice Test #2
History
10 out of 10 launches remaining
Final Checklist for the PCAD Certification Exam

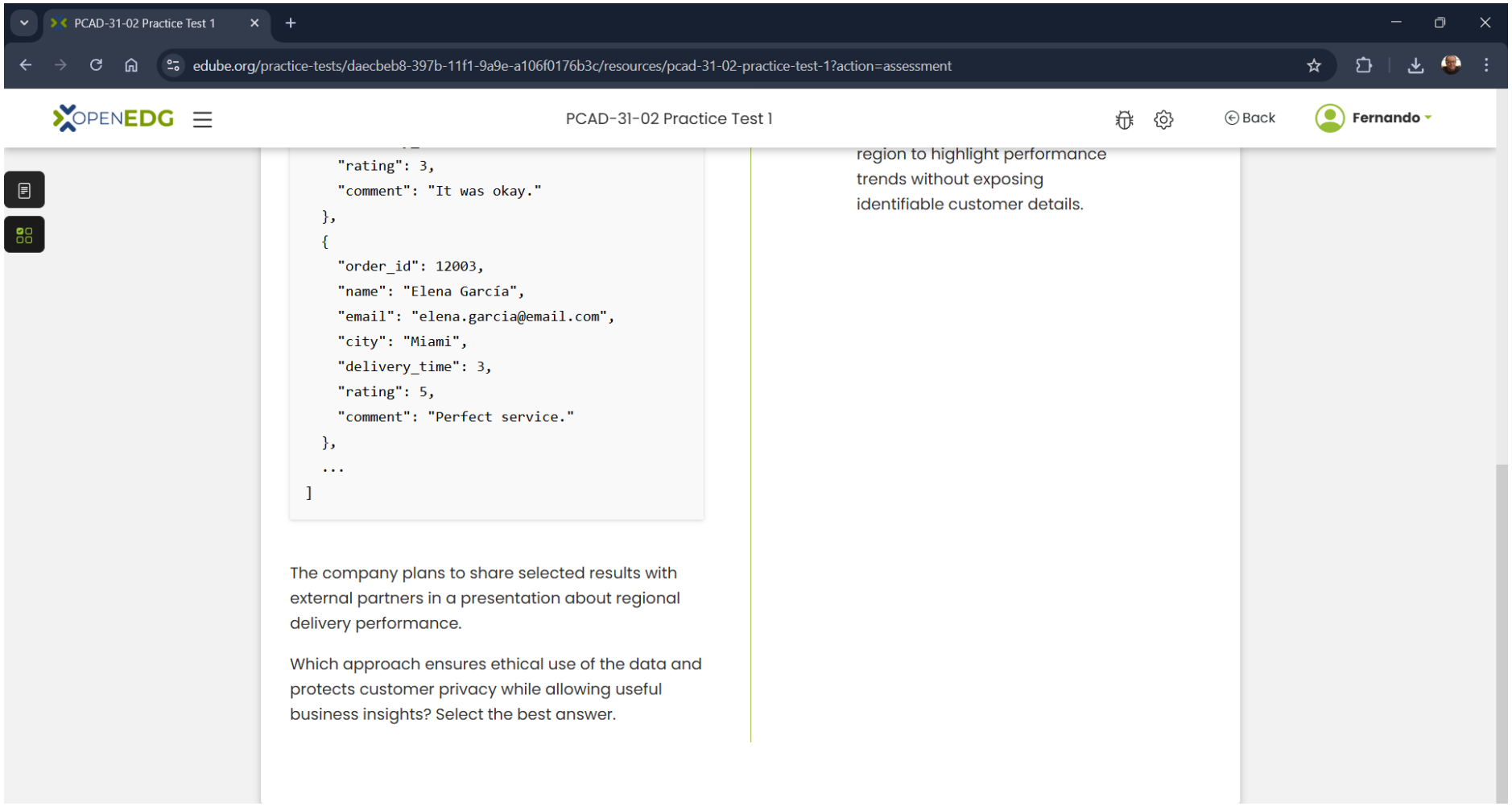

PCAD-31-02 Practice Test 1
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9 out of 10 launches remaining
Welcome to PCAD™ Practice Test 1
This is the first of two PCAD-31-02 Practice Tests – designed to help you check your readiness for the certification exam. It mirrors the structure, content areas, and difficulty level of the real PCAD™ exam, giving you a realistic opportunity to practice under exam-like conditions.
About the Practice Test
Number of questions: 48
Question types: Single-select and multiple-select
Scoring: You'll see your score at the end, based on the same passing threshold of 75% as the actual exam
Time limit: 60 minutes*
*There's no enforced timer here – we've left it off for your comfort while practicing. However, remember that in the actual exam, you'll have 60 minutes to complete all questions. You may want to set a timer for yourself to practice working within this limit.
How to Use the Practice Test
Take the test in one sitting, without interruptions, to best simulate the exam environment
Look carefully at words in bold or underlined, watch for negations, and check how many responses you are asked to select
Always choose the best answer – sometimes more than one may seem correct, but only the best one fully meets the question's intent
In most cases, you'll be able to eliminate at least one obviously incorrect or irrelevant option – this narrows down your choices and helps you focus on analyzing the remaining answers based on your knowledge and reasonina.

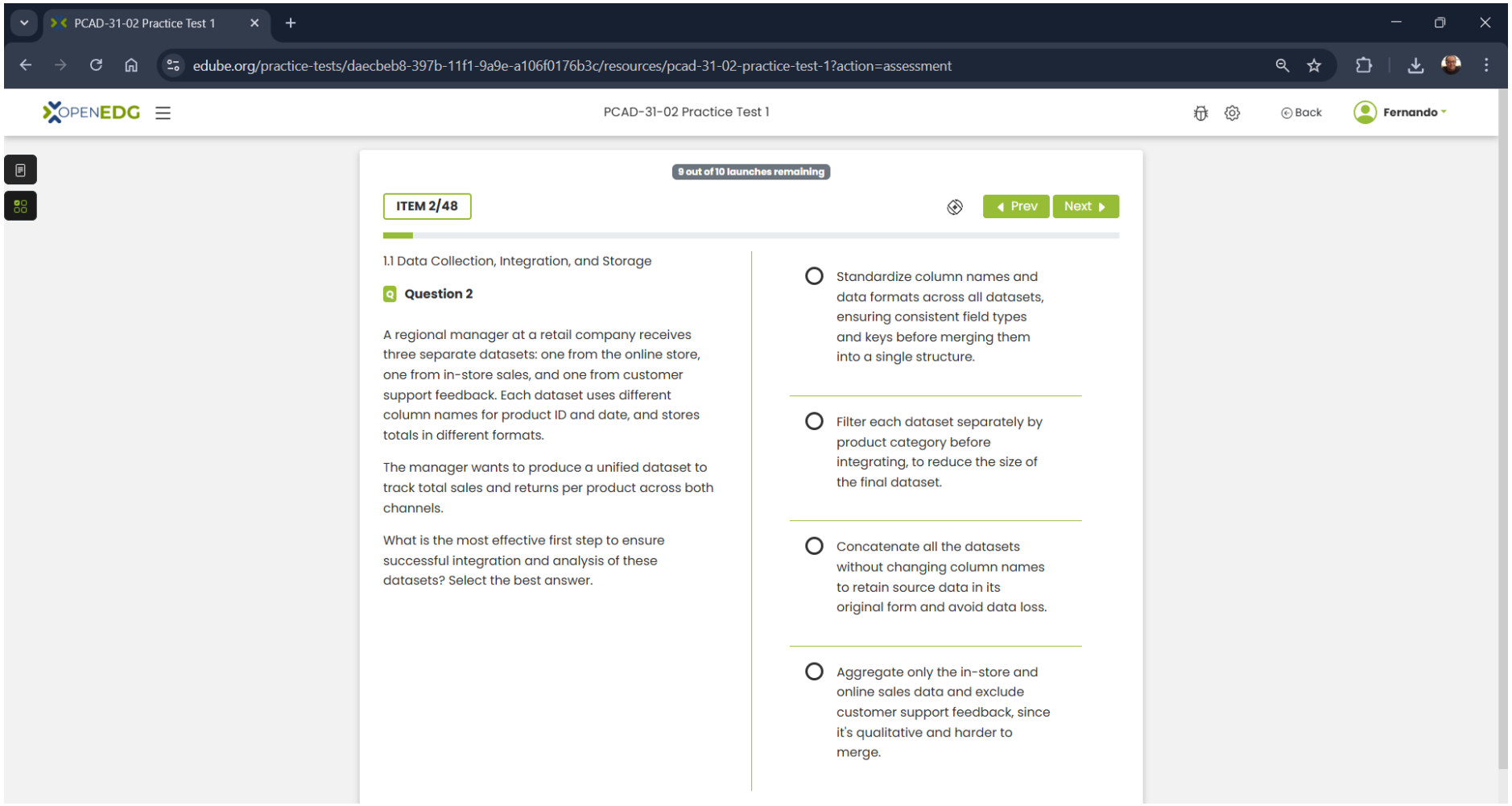

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Take the test in one sitting, without interruptions, to best simulate the exam environment
Look carefully at words in bold or underlined, watch for negations, and check how many responses you are asked to select
Always choose the best answer – sometimes more than one may seem correct, but only the best one fully meets the question's intent
In most cases, you'll be able to eliminate at least one obviously incorrect or irrelevant option – this narrows down your choices and helps you focus on analyzing the remaining answers based on your knowledge and reasoning.
After finishing, review your answers and read the feedback provided for each question
Pay special attention to questions linked to exam objectives you find challenging – this is your chance to identify areas for improvement
Change Layout
By default, the question stem appears on the left and the response options on the right. If you prefer, especially when reading longer code snippets that may wrap in narrow columns, you can switch to a top-down layout (stem on top, options below) for better readability – simply use the Change Layout icon located next to the Next and Prev navigation buttons.
Whether you pass or not, this practice test is here to help you learn, gain confidence, and understand what to expect on the real exam. Take your time, apply what you've learned, and use this experience as a valuable part of your preparation.
When you're ready, click the button below to begin the practice test.
Start Practice Test

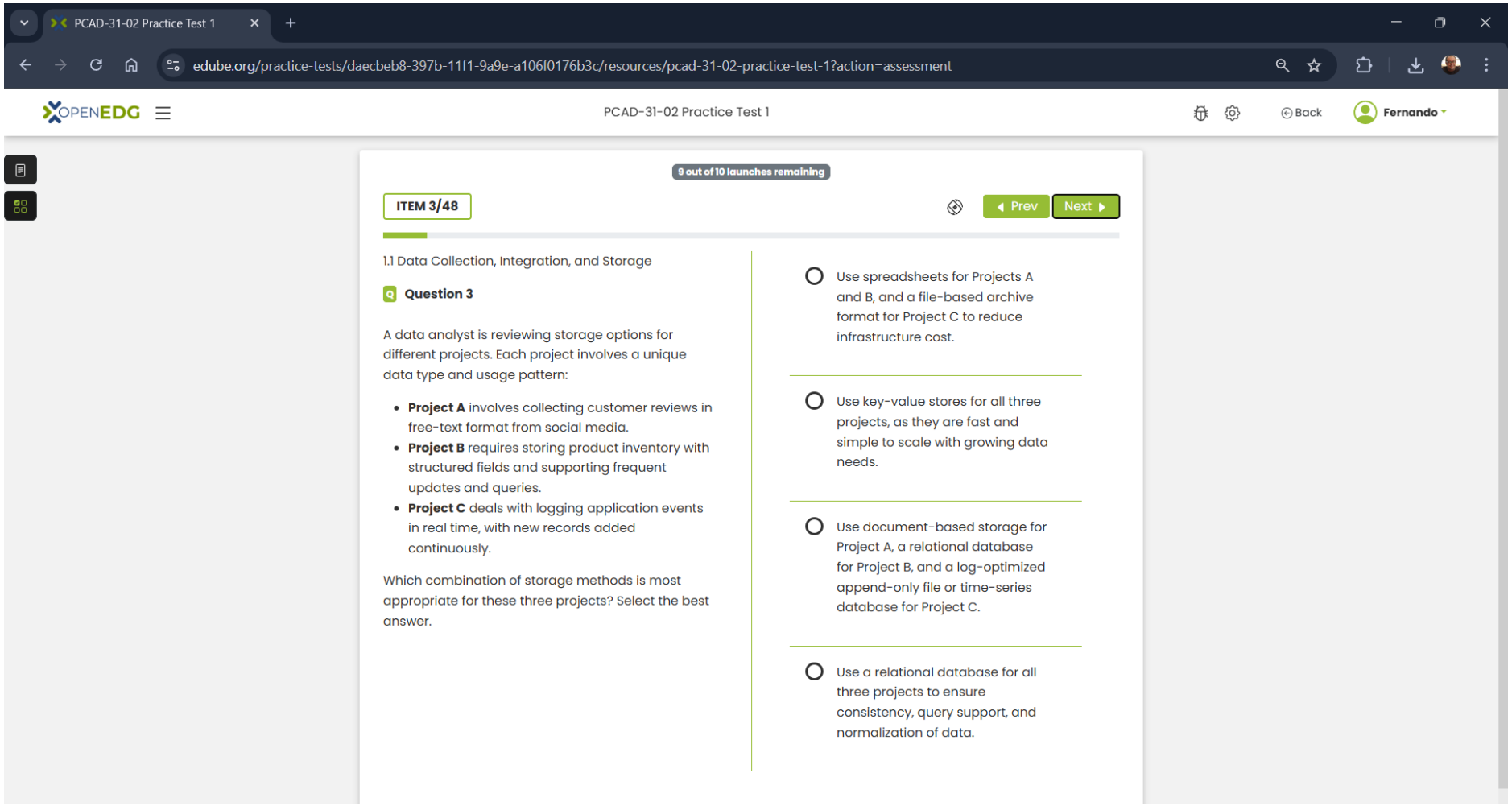

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PCAD-31-02 Practice Test 1
Number of Questions: 48
Start

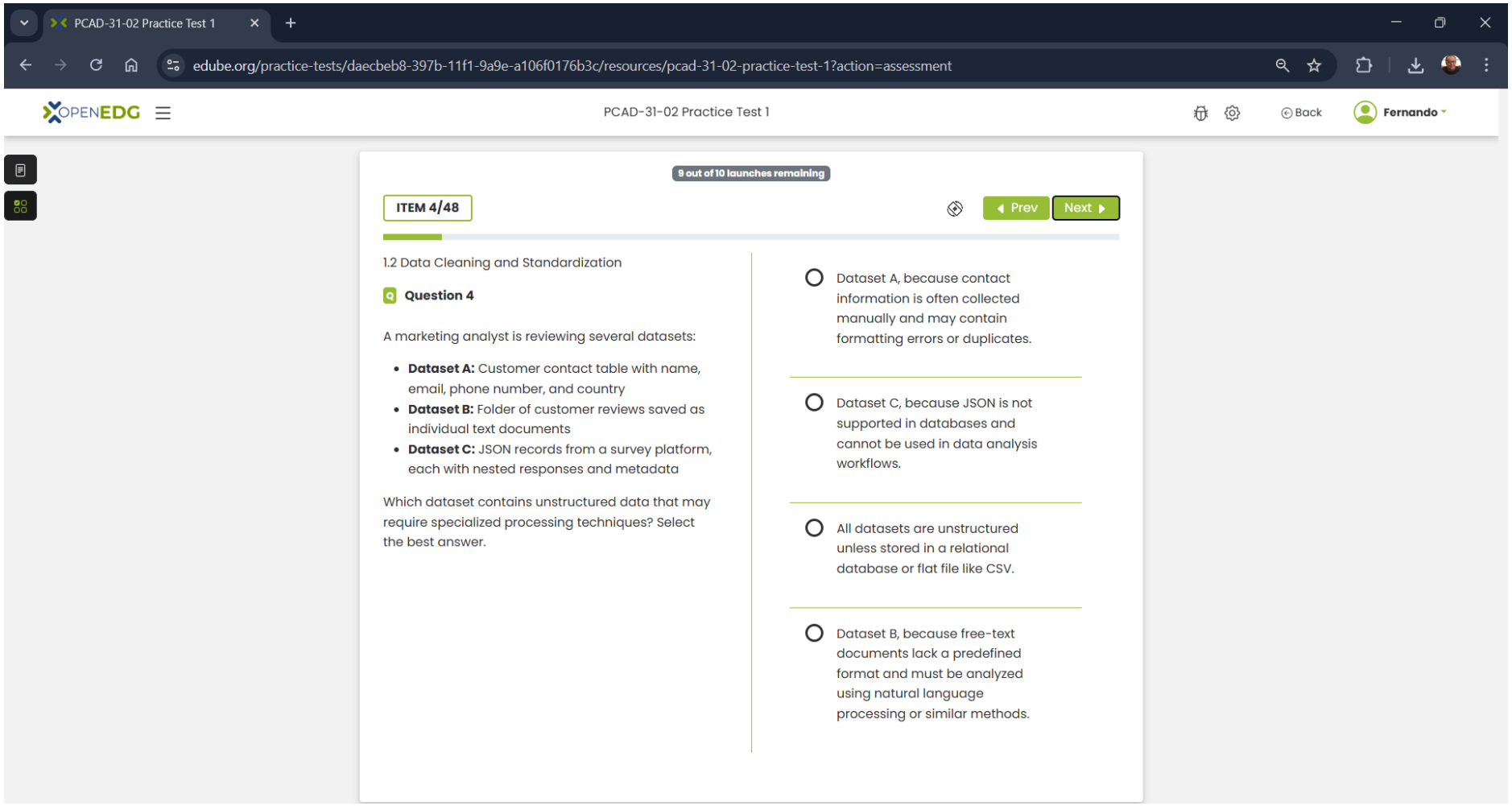

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1.1 Data Collection, Integration, and Storage
Question 1
An e-commerce company collects customer feedback to evaluate satisfaction with its delivery service. The dataset includes customer names, email addresses, city, order ID, delivery times, satisfaction ratings, and optional text comments:
[
{
"order_id": 12001,
"name": "Julia Reed",
"email": "julia.reed@email.com",
"city": "Austin",
"delivery_time": 2,
"rating": 4,
"comment": "Fast delivery!"
},
Publish only the text comments, including the customers' names and cities, since these fields are commonly shared in public reviews.
Keep customer names and emails visible since the comments are positive and reinforce the company's reputation for fast delivery.
Use the full dataset without changes, but include a privacy disclaimer in the presentation footer to inform viewers that personal data is included.

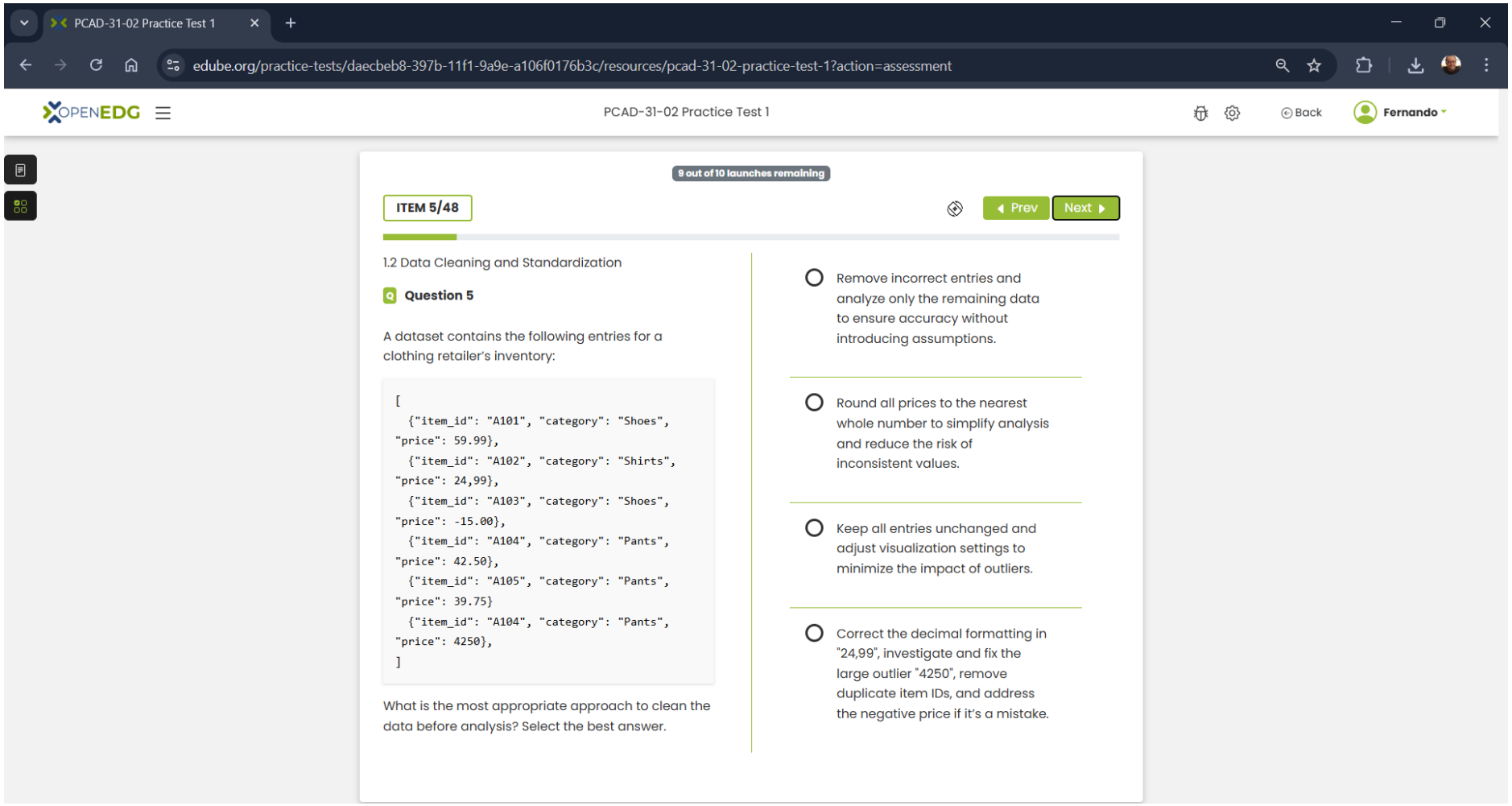

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{
"order_id": 12002,
"name": "Zhang Wei",
"email": "zhang.wei@email.com",
"city": "Seattle",
"delivery_time": 5,
"rating": 3,
"comment": "It was okay."
},
{
"order_id": 12003,
"name": "Elena García",
"email": "elena.garcia@email.com",
"city": "Miami",
"delivery_time": 3,
"rating": 5,
"comment": "Perfect service."
},
...
]
The company plans to share selected results with external partners in a presentation about regional delivery performance.
Remove names and email addresses, and aggregate ratings and delivery times by city or region to highlight performance trends without exposing identifiable customer details.

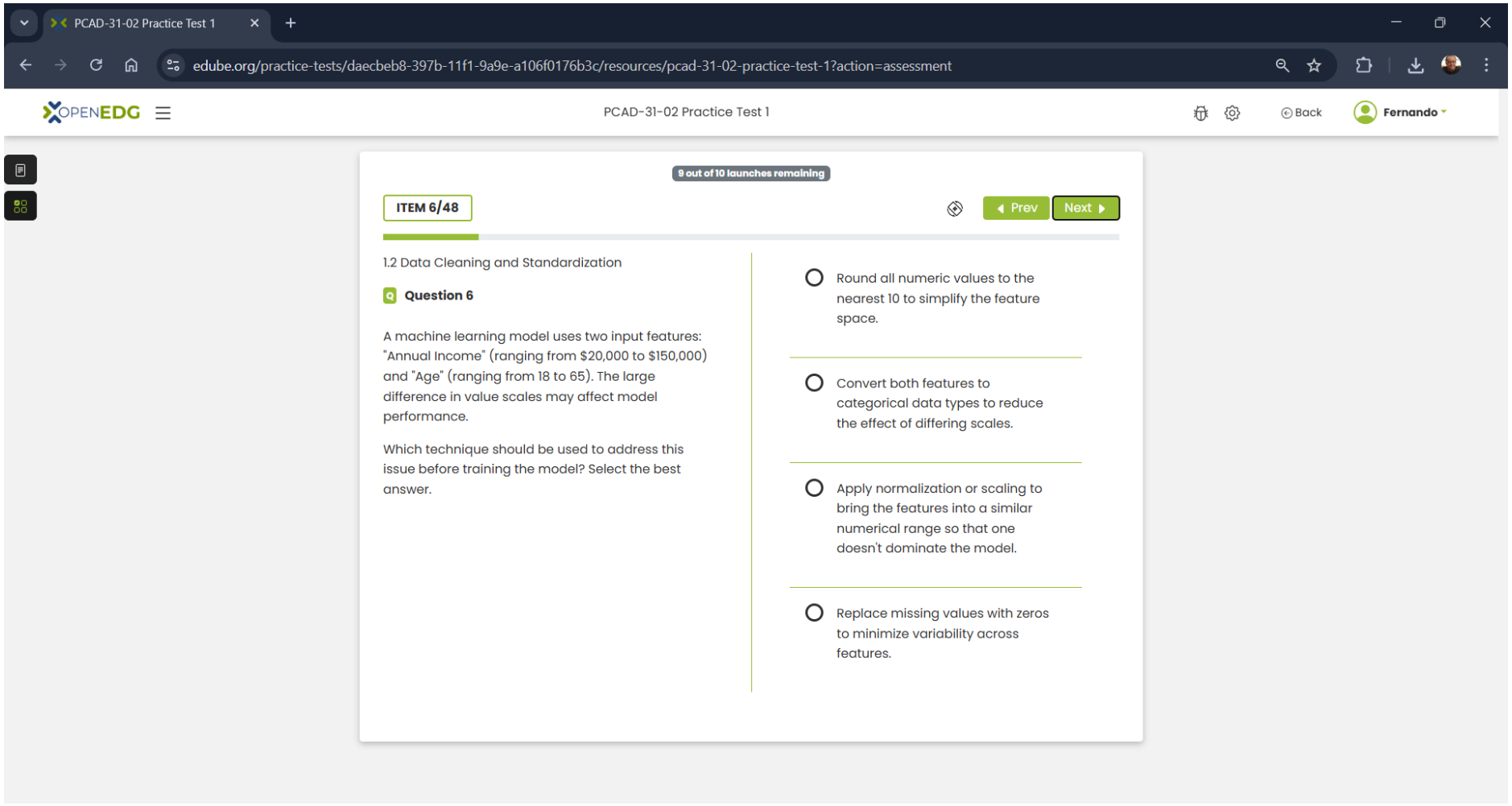

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"rating": 3,
"comment": "It was okay."
},
{
"order_id": 12003,
"name": "Elena García",
"email": "elena.garcia@email.com",
"city": "Miami",
"delivery_time": 3,
"rating": 5,
"comment": "Perfect service."
},
...
]
The company plans to share selected results with external partners in a presentation about regional delivery performance.
Which approach ensures ethical use of the data and protects customer privacy while allowing useful business insights? Select the best answer.
region to highlight performance trends without exposing identifiable customer details.

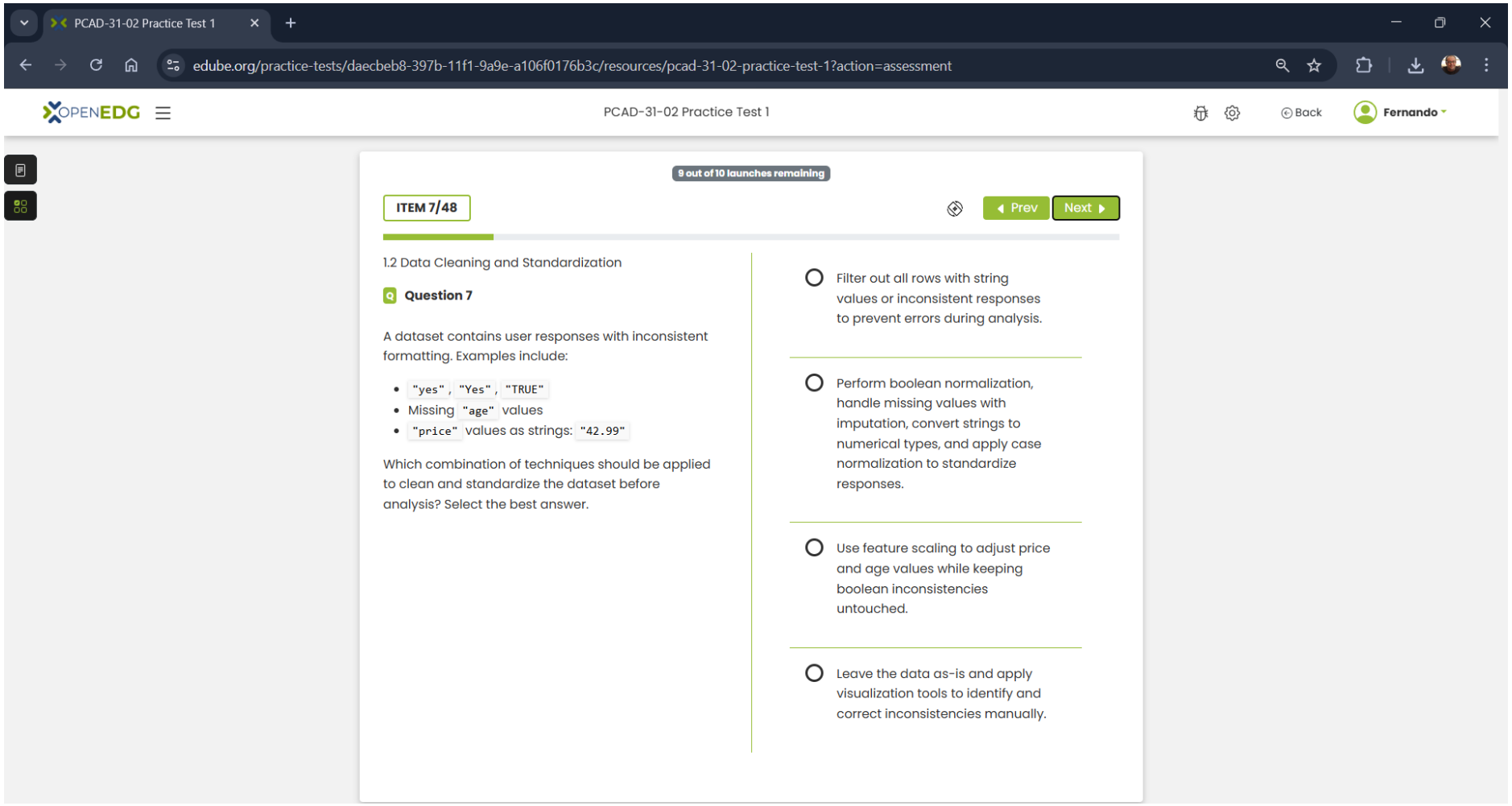

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1.1 Data Collection, Integration, and Storage
Question 2
A regional manager at a retail company receives three separate datasets: one from the online store, one from in-store sales, and one from customer support feedback. Each dataset uses different column names for product ID and date, and stores totals in different formats.
The manager wants to produce a unified dataset to track total sales and returns per product across both channels.
What is the most effective first step to ensure successful integration and analysis of these datasets? Select the best answer.
Standardize column names and data formats across all datasets, ensuring consistent field types and keys before merging them into a single structure.
Filter each dataset separately by product category before integrating, to reduce the size of the final dataset.
Concatenate all the datasets without changing column names to retain source data in its original form and avoid data loss.
Aggregate only the in-store and online sales data and exclude customer support feedback, since it's qualitative and harder to merge.

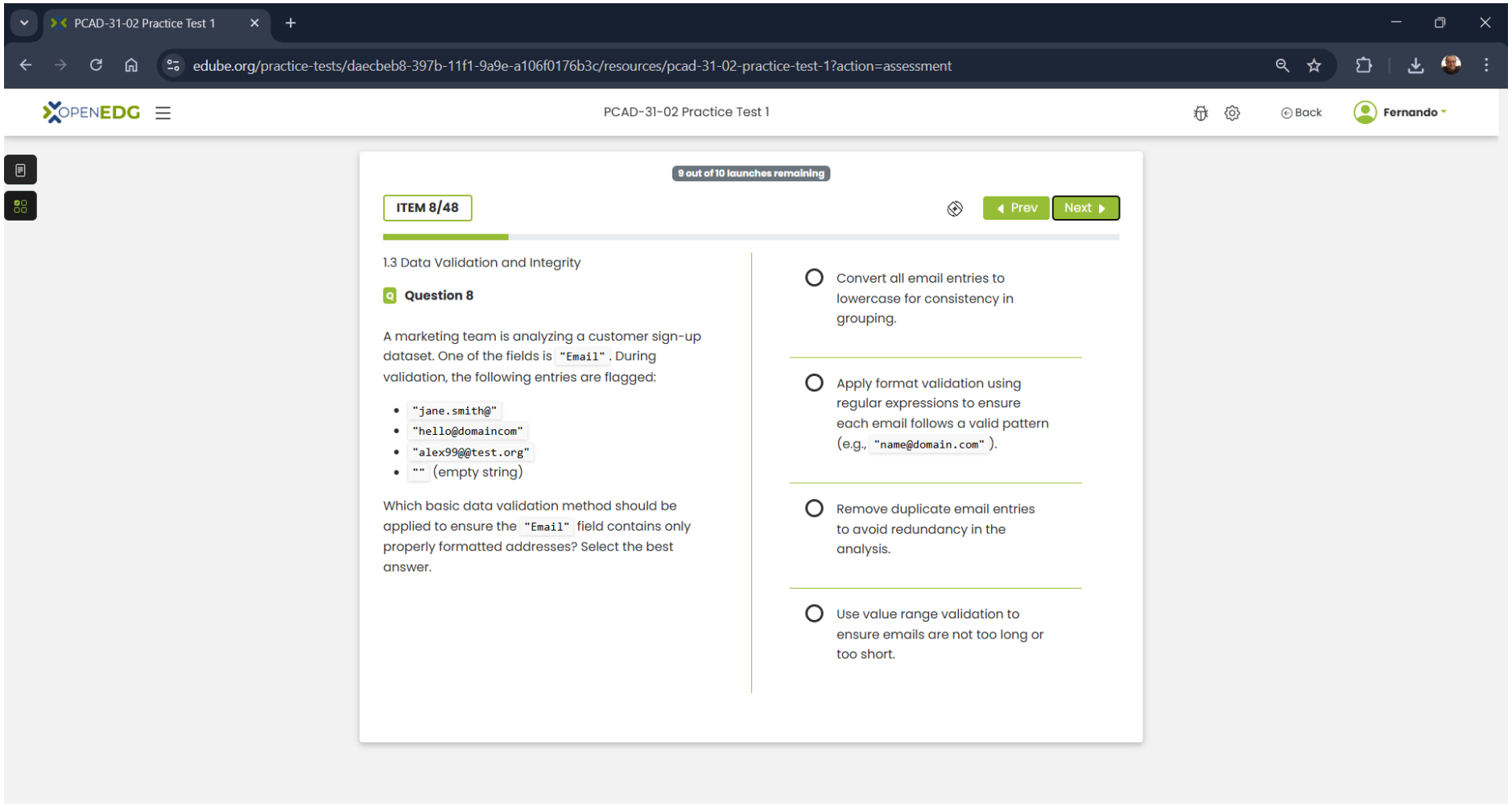

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1.1 Data Collection, Integration, and Storage
Question 3
A data analyst is reviewing storage options for different projects. Each project involves a unique data type and usage pattern:
Project A involves collecting customer reviews in free-text format from social media.
Project B requires storing product inventory with structured fields and supporting frequent updates and queries.
Project C deals with logging application events in real time, with new records added continuously.
Which combination of storage methods is most appropriate for these three projects? Select the best answer.
Use spreadsheets for Projects A and B, and a file-based archive format for Project C to reduce infrastructure cost.
Use key-value stores for all three projects, as they are fast and simple to scale with growing data needs.
Use document-based storage for Project A, a relational database for Project B, and a log-optimized append-only file or time-series database for Project C.
Use a relational database for all three projects to ensure consistency, query support, and normalization of data.

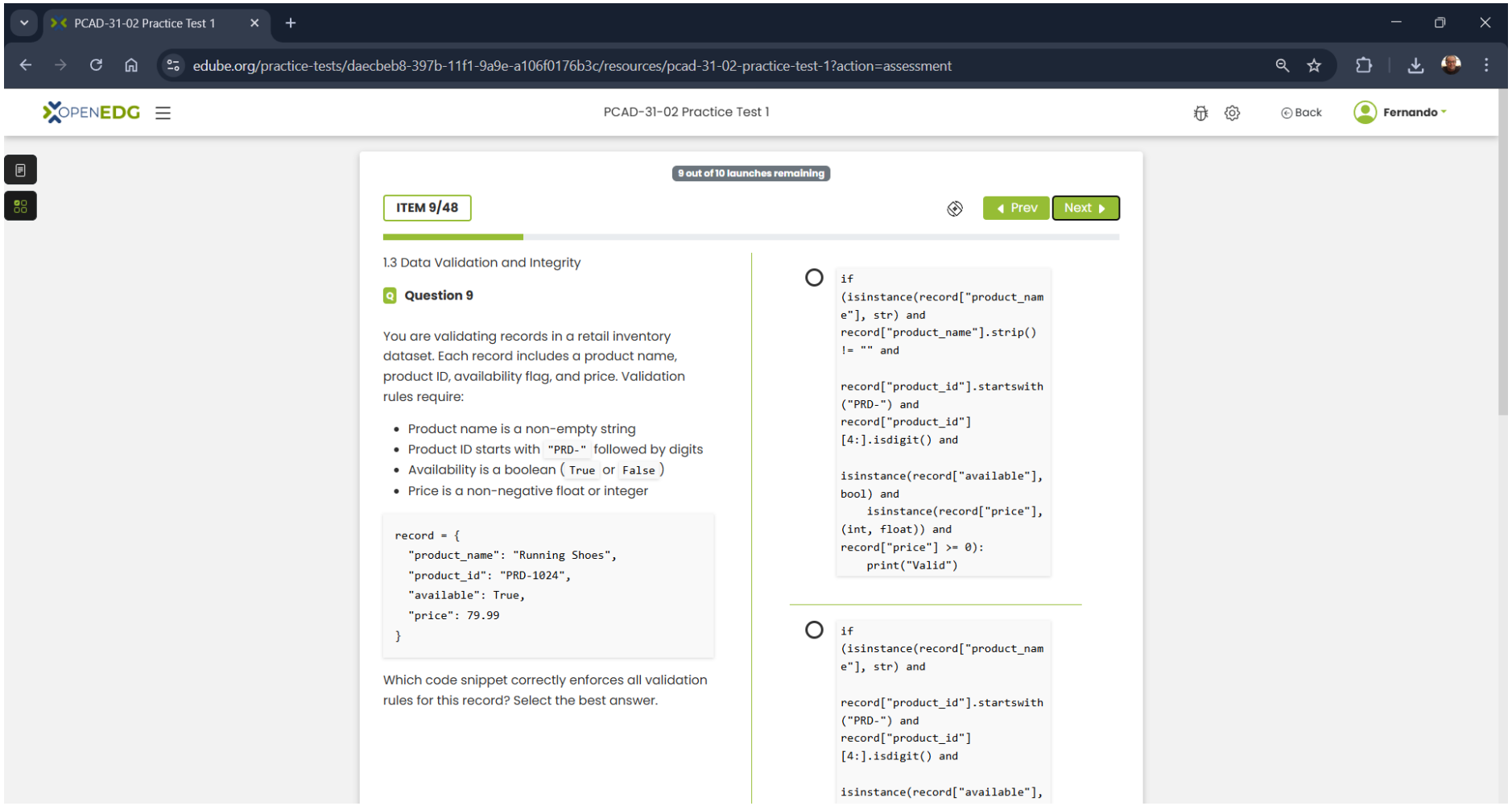

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1.2 Data Cleaning and Standardization
Question 4
A marketing analyst is reviewing several datasets:
Dataset A: Customer contact table with name, email, phone number, and country
Dataset B: Folder of customer reviews saved as individual text documents
Dataset C: JSON records from a survey platform, each with nested responses and metadata
Which dataset contains unstructured data that may require specialized processing techniques? Select the best answer.
Dataset A, because contact information is often collected manually and may contain formatting errors or duplicates.
Dataset C, because JSON is not supported in databases and cannot be used in data analysis workflows.
All datasets are unstructured unless stored in a relational database or flat file like CSV.
Dataset B, because free-text documents lack a predefined format and must be analyzed using natural language processing or similar methods.

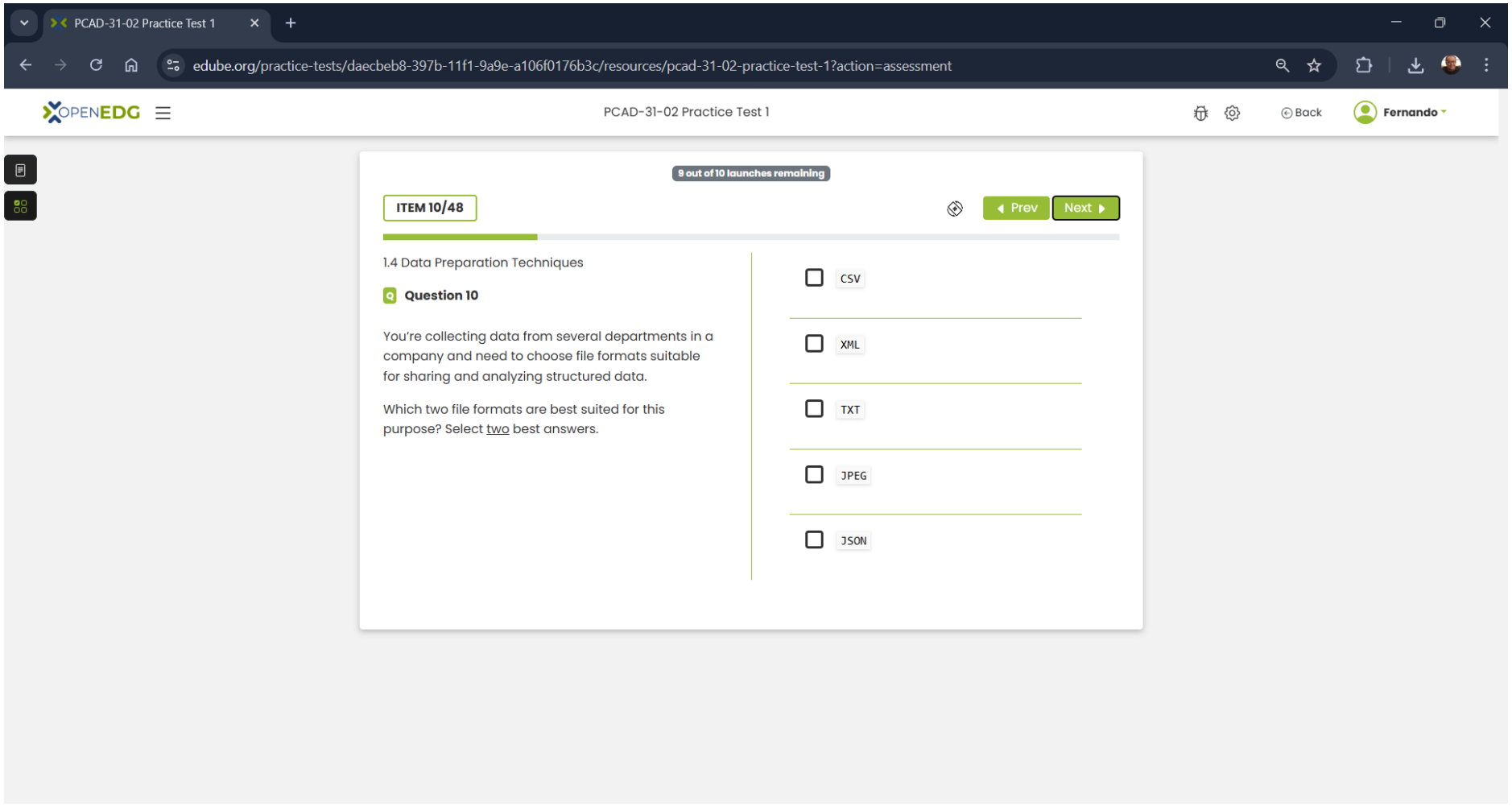

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1.2 Data Cleaning and Standardization
Question 5
A dataset contains the following entries for a clothing retailer's inventory:
[
{"item_id": "A101", "category": "Shoes", "price": 59.99},
{"item_id": "A102", "category": "Shirts", "price": 24,99},
{"item_id": "A103", "category": "Shoes", "price": -15.00},
{"item_id": "A104", "category": "Pants", "price": 42.50},
{"item_id": "A105", "category": "Pants", "price": 39.75}
{"item_id": "A104", "category": "Pants", "price": 4250},
]
What is the most appropriate approach to clean the data before analysis? Select the best answer.
Remove incorrect entries and analyze only the remaining data to ensure accuracy without introducing assumptions.
Round all prices to the nearest whole number to simplify analysis and reduce the risk of inconsistent values.
Keep all entries unchanged and adjust visualization settings to minimize the impact of outliers.
Correct the decimal formatting in "24,99", investigate and fix the large outlier "4250", remove duplicate item IDs, and address the negative price if it's a mistake.

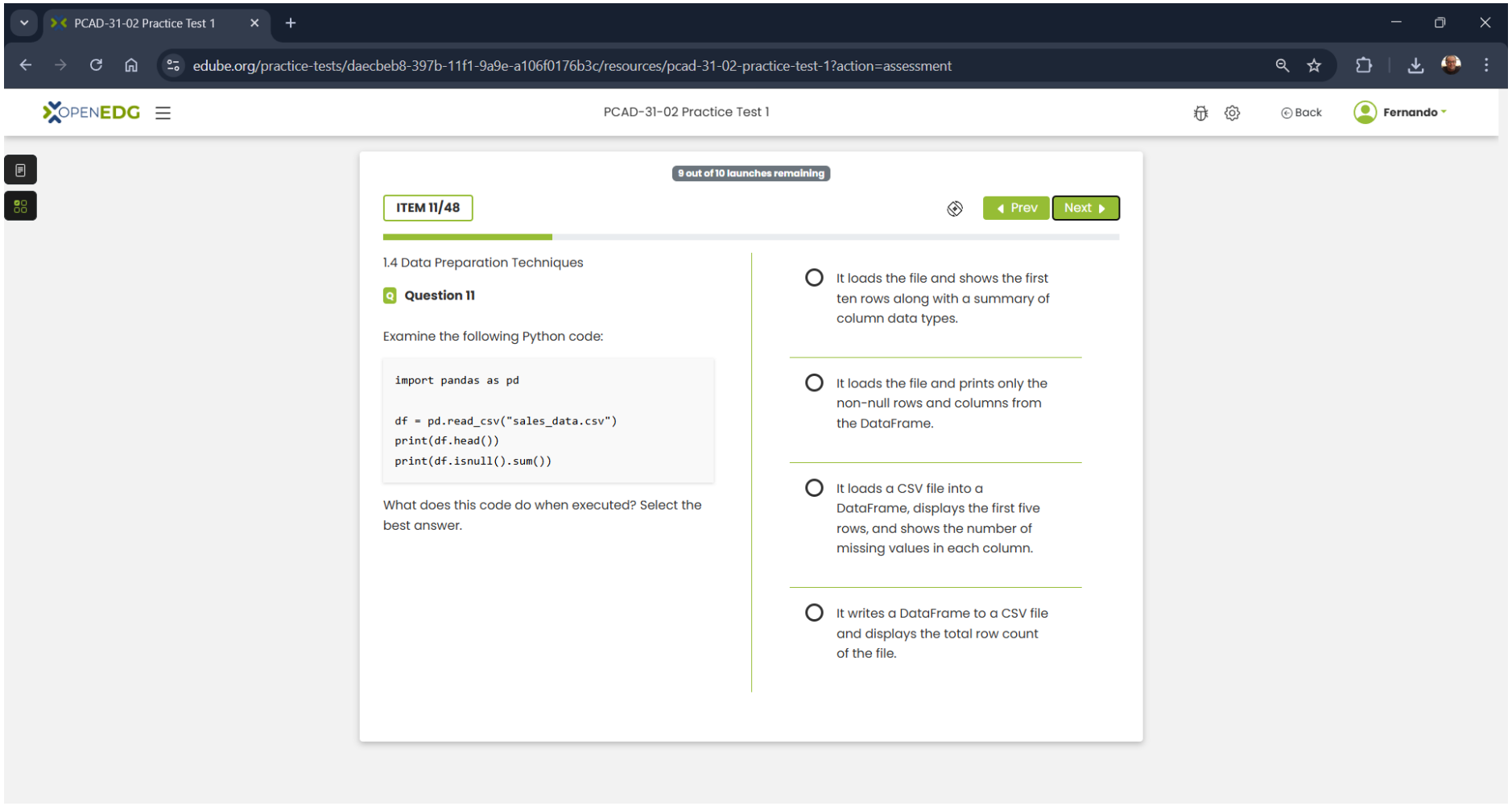

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1.2 Data Cleaning and Standardization
Question 6
A machine learning model uses two input features: "Annual Income" (ranging from $20,000 to $150,000) and "Age" (ranging from 18 to 65). The large difference in value scales may affect model performance.
Which technique should be used to address this issue before training the model? Select the best answer.
Round all numeric values to the nearest 10 to simplify the feature space.
Convert both features to categorical data types to reduce the effect of differing scales.
Apply normalization or scaling to bring the features into a similar numerical range so that one doesn't dominate the model.
Replace missing values with zeros to minimize variability across features.

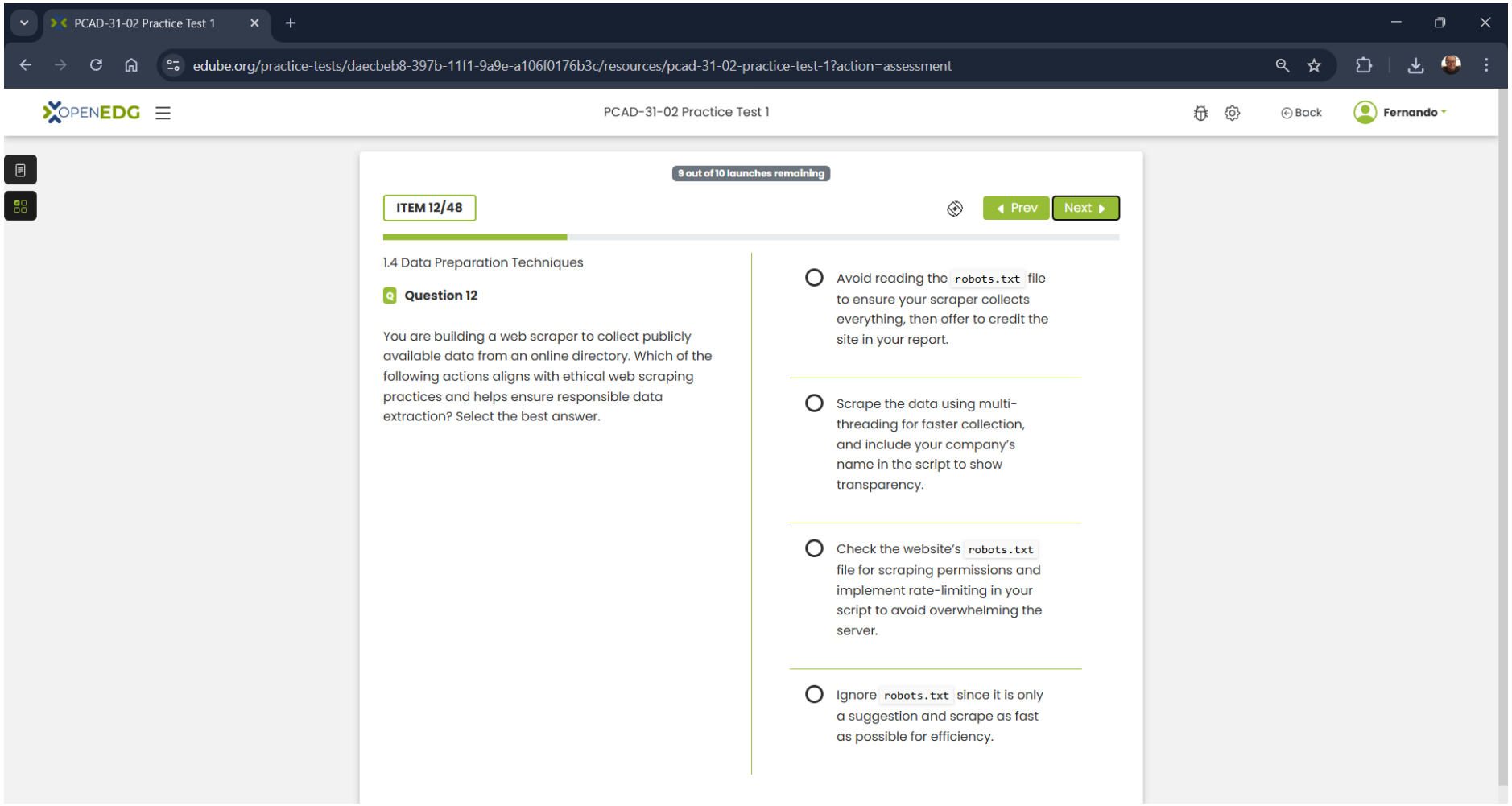

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1.2 Data Cleaning and Standardization
Question 7
A dataset contains user responses with inconsistent formatting. Examples include:
"yes", "Yes", "TRUE"
Missing "age" values
"price" values as strings: "42.99"
Which combination of techniques should be applied to clean and standardize the dataset before analysis? Select the best answer.
Filter out all rows with string values or inconsistent responses to prevent errors during analysis.
Perform boolean normalization, handle missing values with imputation, convert strings to numerical types, and apply case normalization to standardize responses.
Use feature scaling to adjust price and age values while keeping boolean inconsistencies untouched.
Leave the data as-is and apply visualization tools to identify and correct inconsistencies manually.

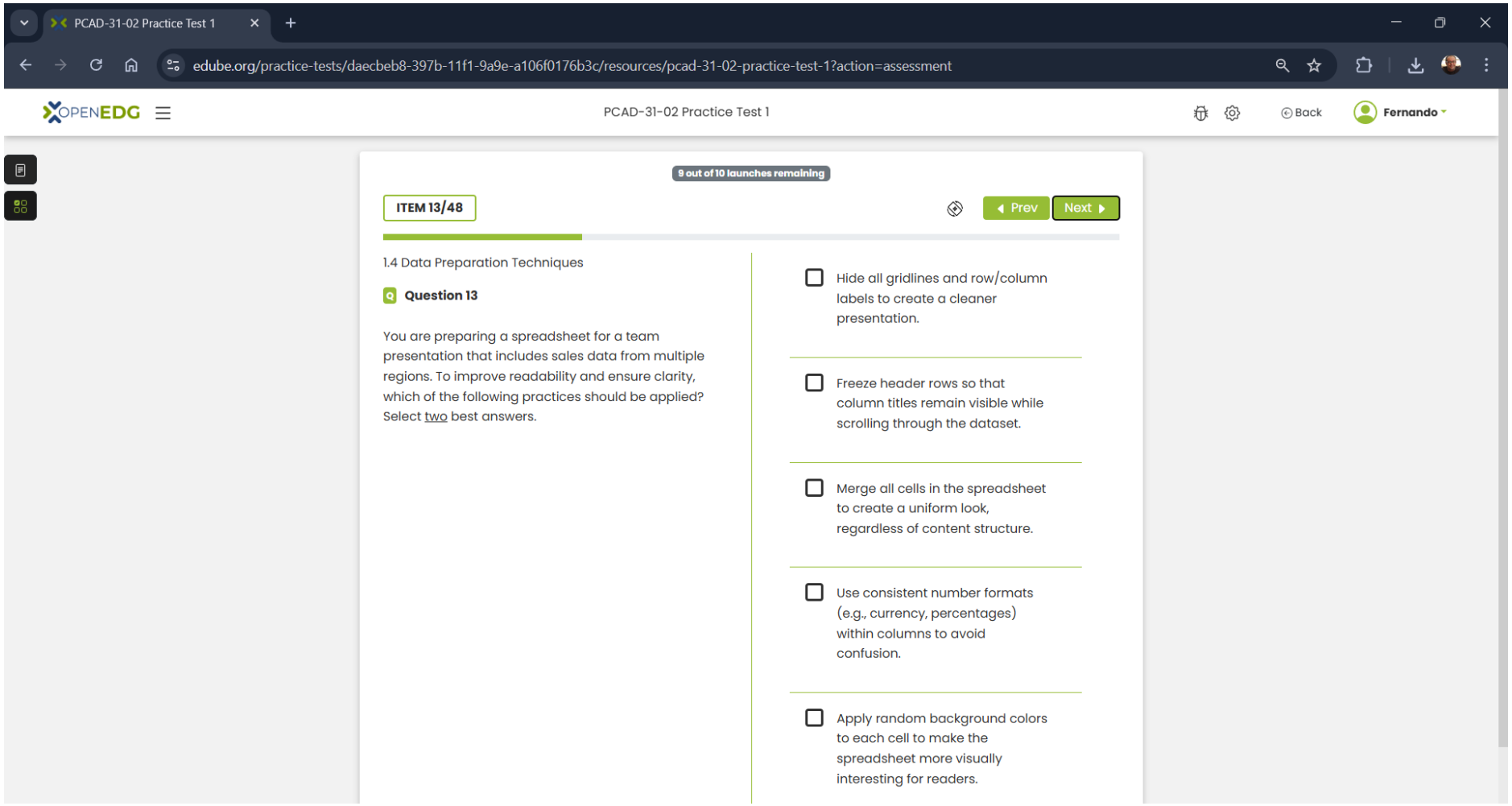

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1.3 Data Validation and Integrity
Question 8
A marketing team is analyzing a customer sign-up dataset. One of the fields is "Email". During validation, the following entries are flagged:
"jane.smith@"
"hello@domaincom"
"alex99@@test.org"
"" (empty string)
Which basic data validation method should be applied to ensure the "Email" field contains only properly formatted addresses? Select the best answer.
Convert all email entries to lowercase for consistency in grouping.
Apply format validation using regular expressions to ensure each email follows a valid pattern (e.g., "name@domain.com").
Remove duplicate email entries to avoid redundancy in the analysis.
Use value range validation to ensure emails are not too long or too short.

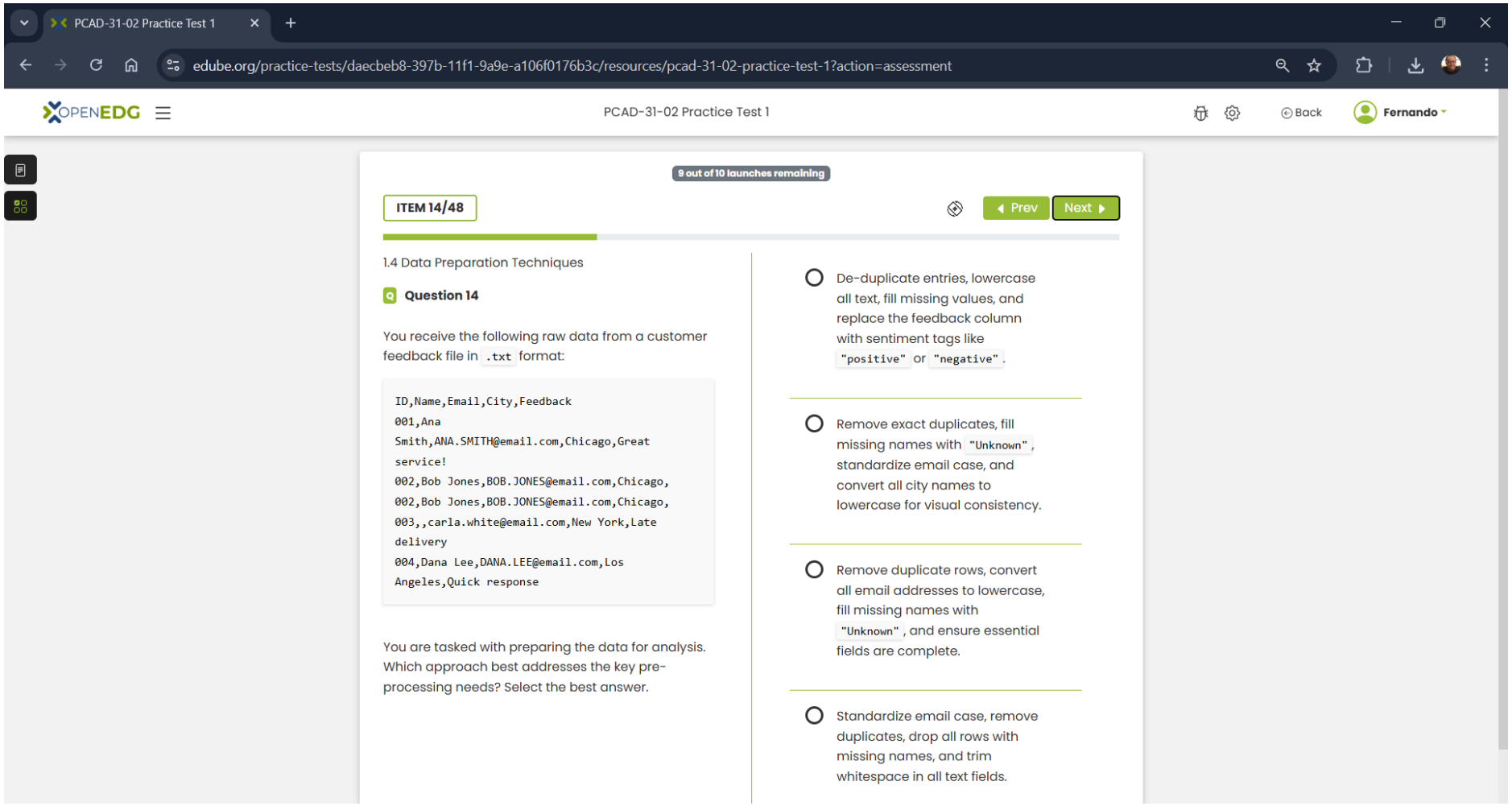

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1.3 Data Validation and Integrity
Question 9
You are validating records in a retail inventory dataset. Each record includes a product name, product ID, availability flag, and price. Validation rules require:
Product name is a non-empty string
Product ID starts with "PRD-" followed by digits
Availability is a boolean (True or False)
Price is a non-negative float or integer
record = {
"product_name": "Running Shoes",
"product_id": "PRD-1024",
"available": True,
"price": 79.99
}
Which code snippet correctly enforces all validation rules for this record? Select the best answer.
if (isinstance(record["product_name"], str) and record["product_name"].strip() != "" and
record["product_id"].startswith("PRD-") and record["product_id"][4:].isdigit() and
isinstance(record["available"], bool) and
isinstance(record["price"], (int, float)) and record["price"] >= 0):
print("Valid")
if (isinstance(record["product_name"], str) and
record["product_id"].startswith("PRD-") and record["product_id"][4:].isdigit() and
isinstance(record["available"],

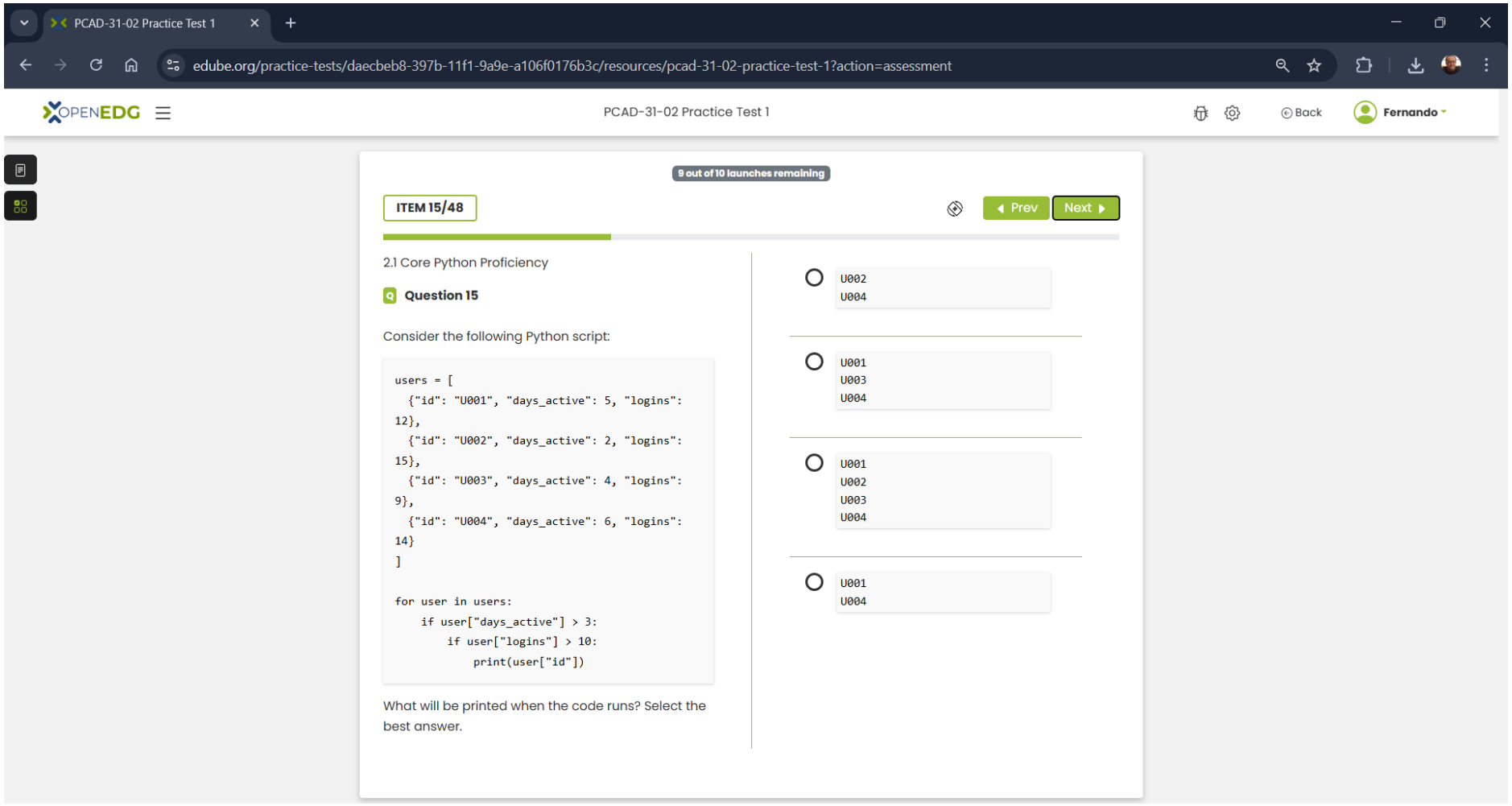

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1.4 Data Preparation Techniques
Question 10
You're collecting data from several departments in a company and need to choose file formats suitable for sharing and analyzing structured data.
Which two file formats are best suited for this purpose? Select two best answers.
CSV
XML
TXT
JPEG
JSON

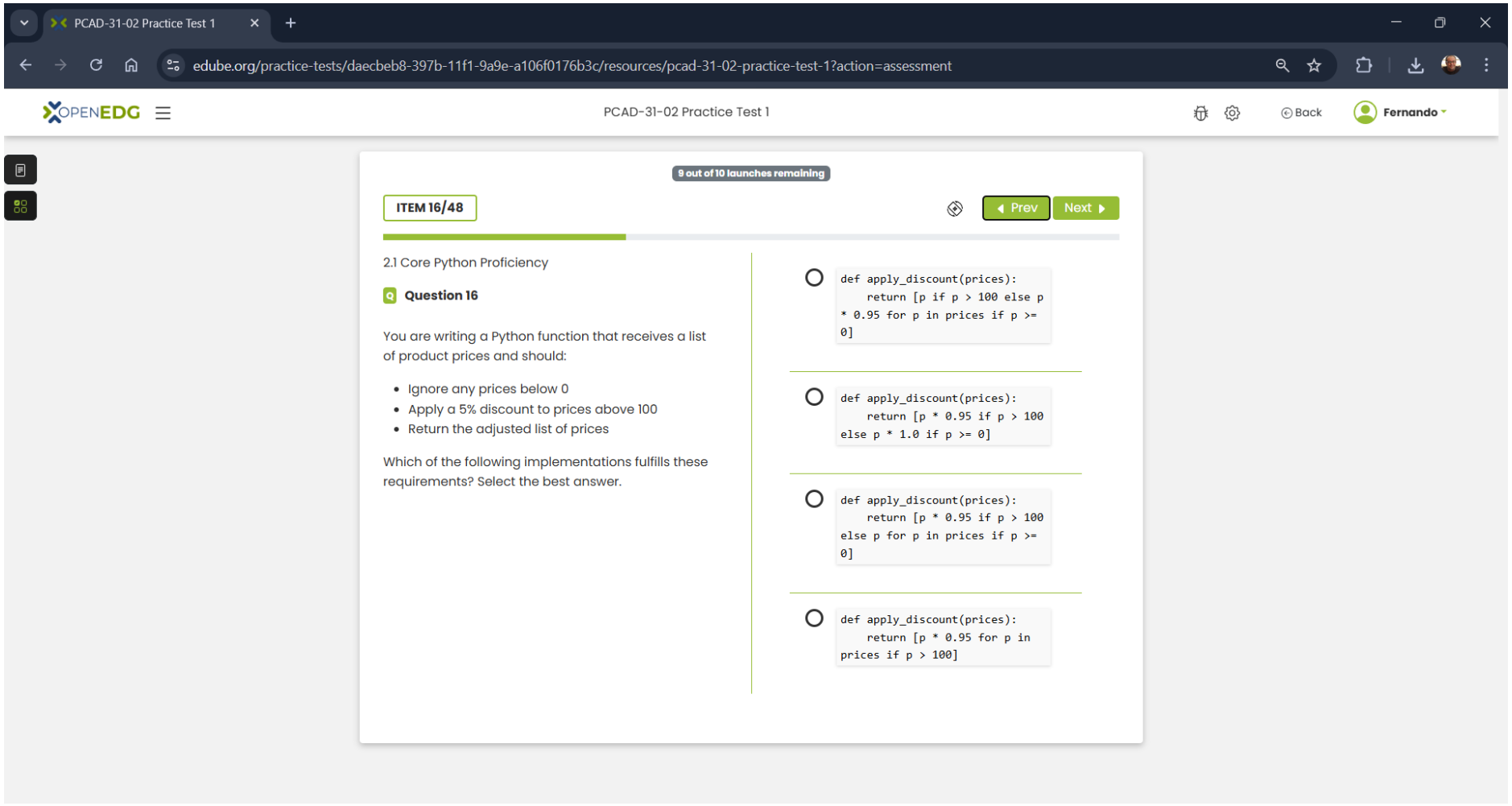

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1.4 Data Preparation Techniques
Question 11
Examine the following Python code:
import pandas as pd
df = pd.read_csv("sales_data.csv")
print(df.head())
print(df.isnull().sum())
What does this code do when executed? Select the best answer.
It loads the file and shows the first ten rows along with a summary of column data types.
It loads the file and prints only the non-null rows and columns from the DataFrame.
It loads a CSV file into a DataFrame, displays the first five rows, and shows the number of missing values in each column.
It writes a DataFrame to a CSV file and displays the total row count of the file.

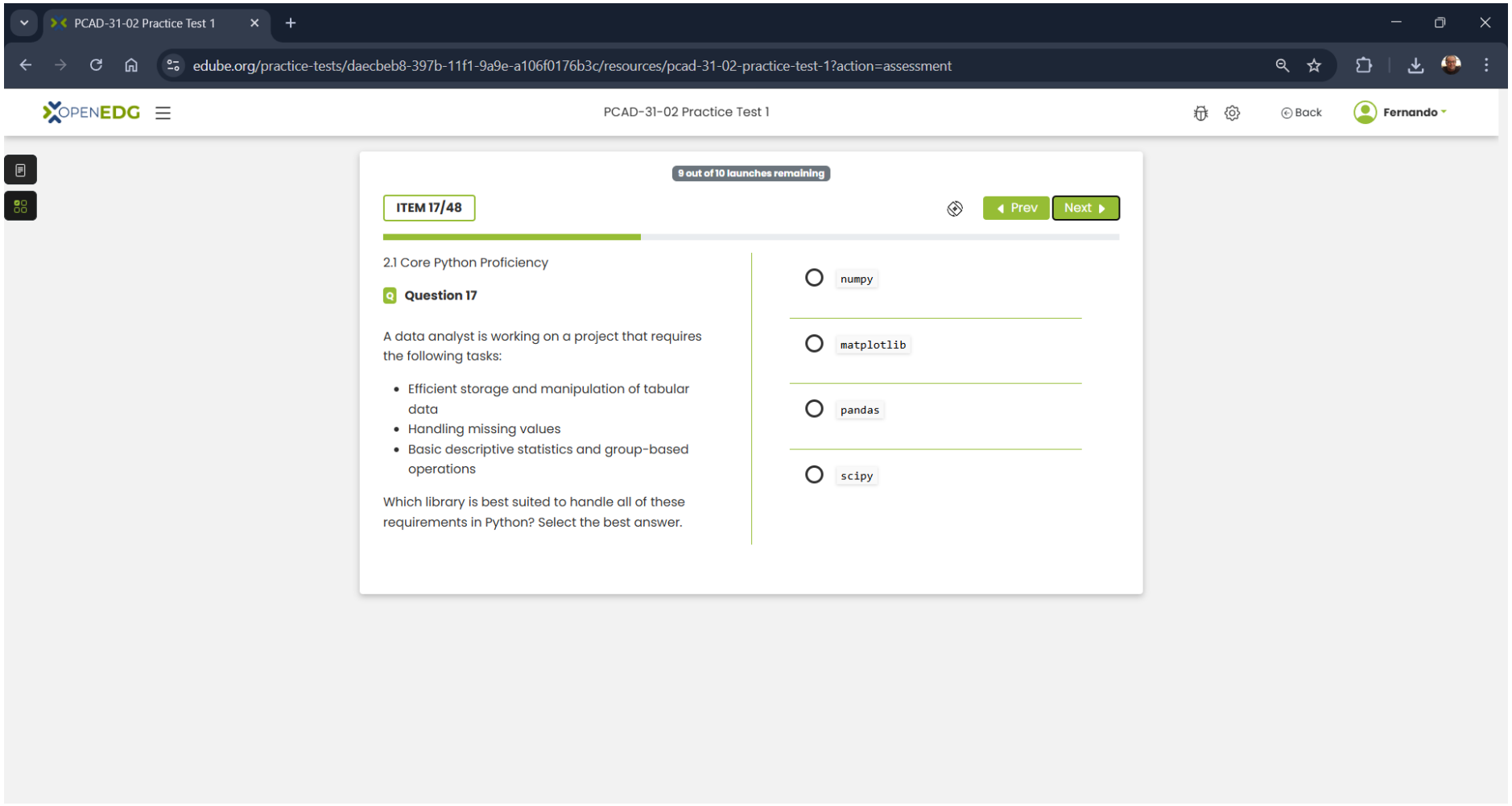

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1.4 Data Preparation Techniques
Question 12
You are building a web scraper to collect publicly available data from an online directory. Which of the following actions aligns with ethical web scraping practices and helps ensure responsible data extraction? Select the best answer.
Avoid reading the robots.txt file to ensure your scraper collects everything, then offer to credit the site in your report.
Scrape the data using multi-threading for faster collection, and include your company's name in the script to show transparency.
Check the website's robots.txt file for scraping permissions and implement rate-limiting in your script to avoid overwhelming the server.
Ignore robots.txt since it is only a suggestion and scrape as fast as possible for efficiency.

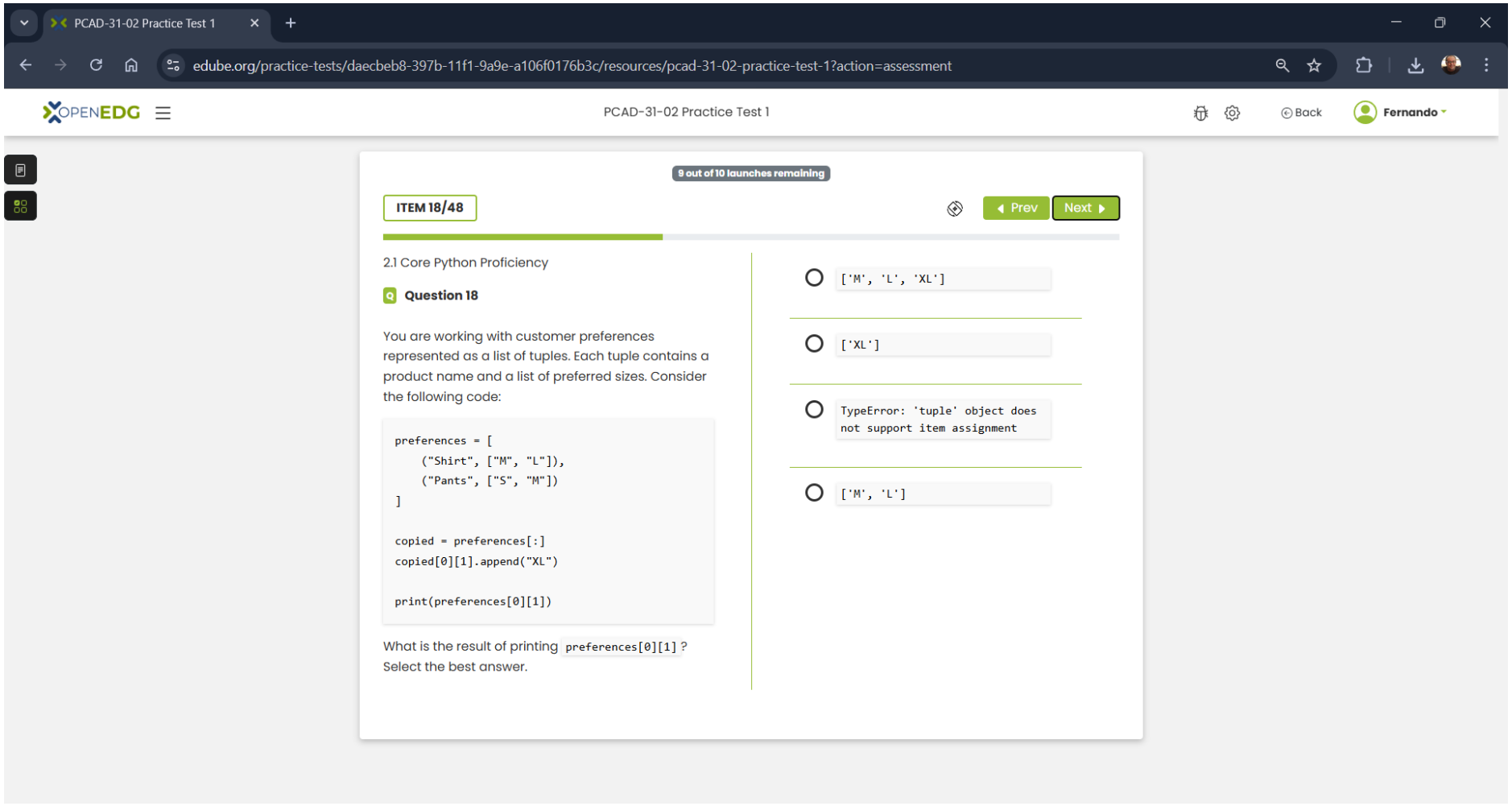

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1.4 Data Preparation Techniques
Question 13
You are preparing a spreadsheet for a team presentation that includes sales data from multiple regions. To improve readability and ensure clarity, which of the following practices should be applied? Select two best answers.
Hide all gridlines and row/column labels to create a cleaner presentation.
Freeze header rows so that column titles remain visible while scrolling through the dataset.
Merge all cells in the spreadsheet to create a uniform look, regardless of content structure.
Use consistent number formats (e.g., currency, percentages) within columns to avoid confusion.
Apply random background colors to each cell to make the spreadsheet more visually interesting for readers.

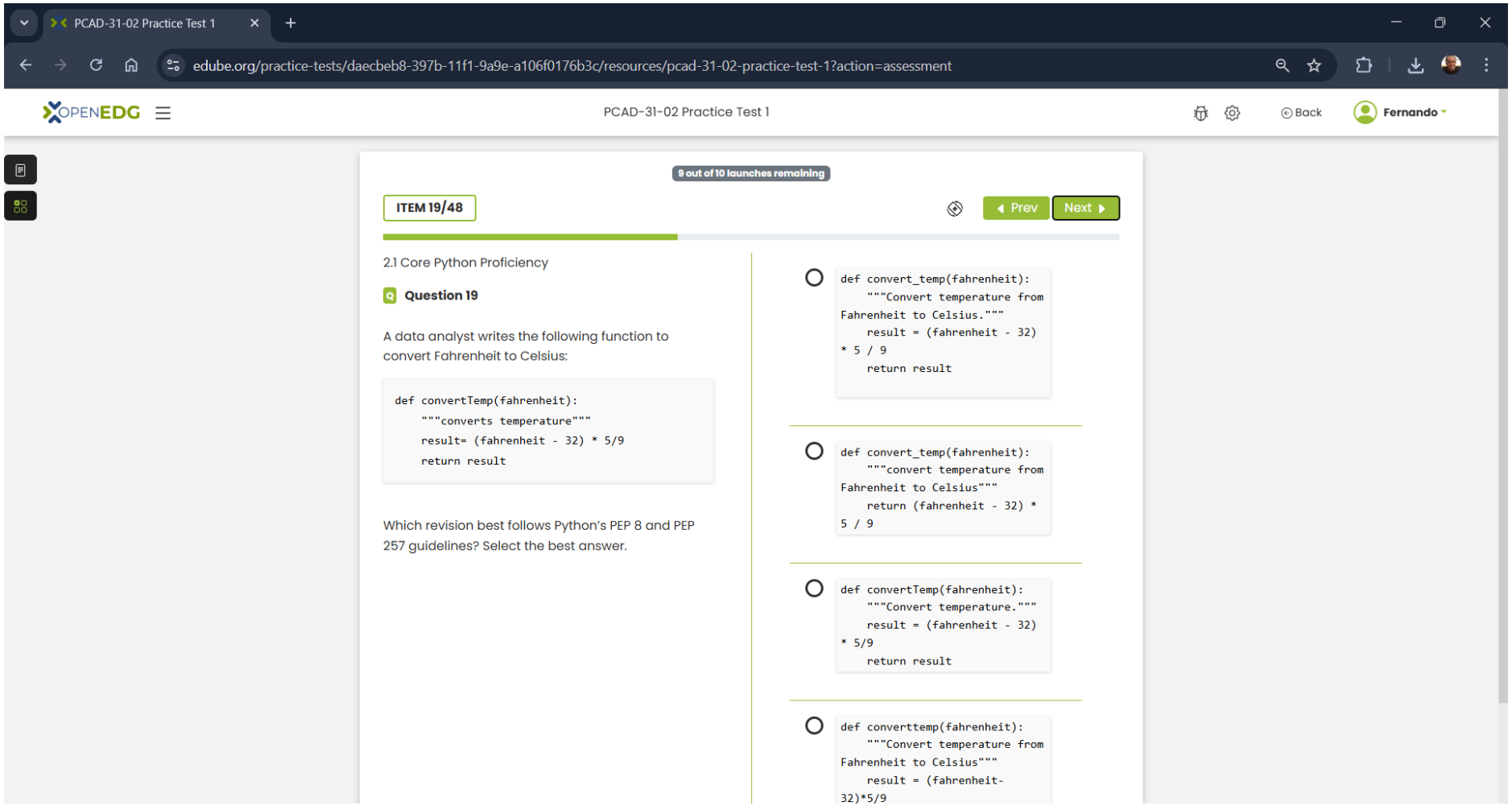

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1.4 Data Preparation Techniques
Question 14
You receive the following raw data from a customer feedback file in .txt format:
ID,Name,Email,City,Feedback
001,Ana Smith,ANA.SMITH@email.com,Chicago,Great service!
002,Bob Jones,BOB.JONES@email.com,Chicago,
002,Bob Jones,BOB.JONES@email.com,Chicago,
003,,carla.white@email.com,New York,Late delivery
004,Dana Lee,DANA.LEE@email.com,Los Angeles,Quick response
You are tasked with preparing the data for analysis. Which approach best addresses the key pre-processing needs? Select the best answer.
De-duplicate entries, lowercase all text, fill missing values, and replace the feedback column with sentiment tags like "positive" or "negative".
Remove exact duplicates, fill missing names with "Unknown", standardize email case, and convert all city names to lowercase for visual consistency.
Remove duplicate rows, convert all email addresses to lowercase, fill missing names with "Unknown", and ensure essential fields are complete.
Standardize email case, remove duplicates, drop all rows with missing names, and trim whitespace in all text fields.

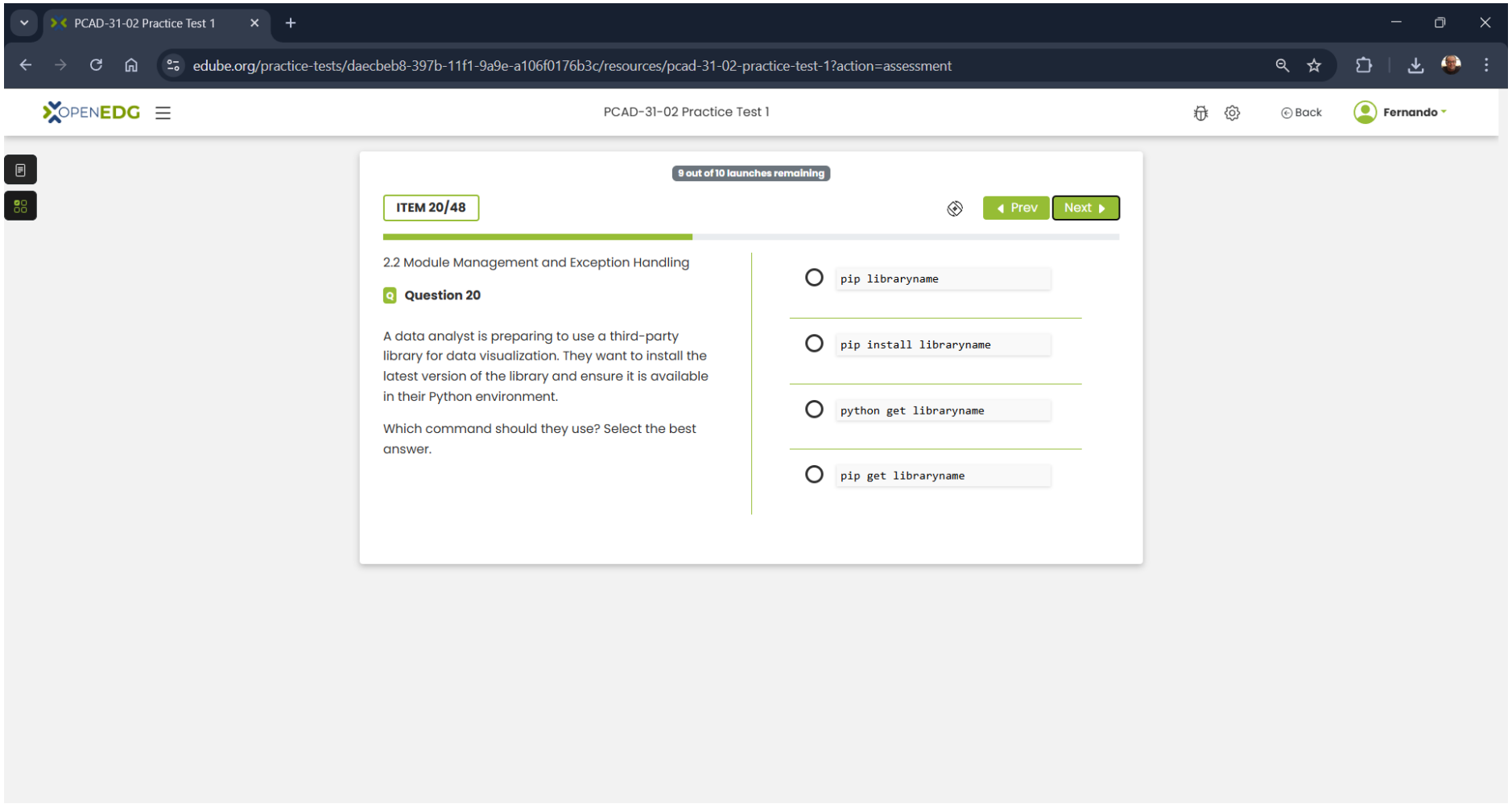

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2.1 Core Python Proficiency
Question 15
Consider the following Python script:
users = [
{"id": "U001", "days_active": 5, "logins": 12},
{"id": "U002", "days_active": 2, "logins": 15},
{"id": "U003", "days_active": 4, "logins": 9},
{"id": "U004", "days_active": 6, "logins": 14}
]
for user in users:
if user["days_active"] > 3:
if user["logins"] > 10:
print(user["id"])
What will be printed when the code runs? Select the best answer.
U002
U004
U001
U003
U004
U001
U002
U003
U004
U001
U004

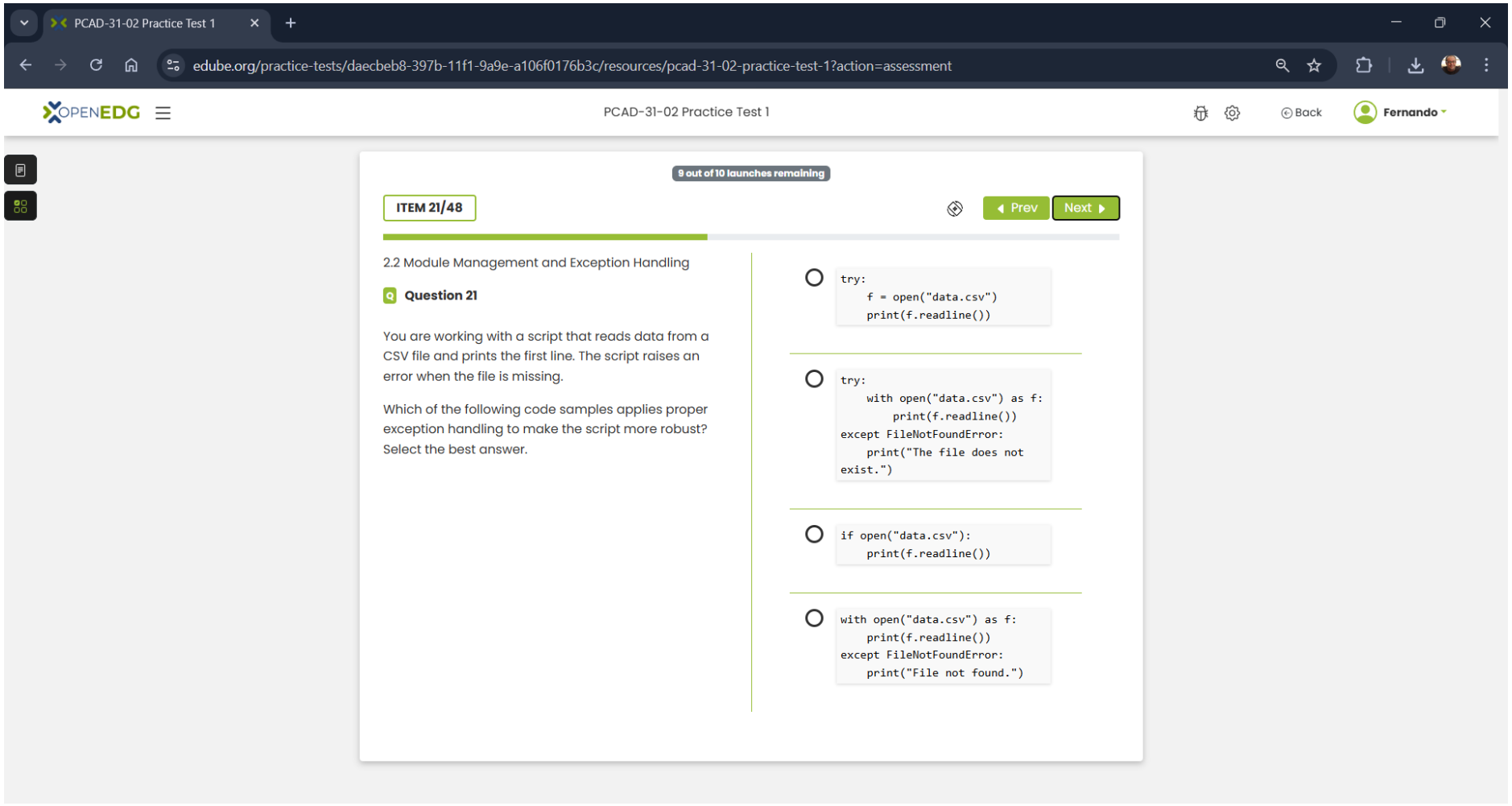

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2.1 Core Python Proficiency
Question 16
You are writing a Python function that receives a list of product prices and should:
Ignore any prices below 0
Apply a 5% discount to prices above 100
Return the adjusted list of prices
Which of the following implementations fulfills these requirements? Select the best answer.
def apply_discount(prices):
return [p if p > 100 else p * 0.95 for p in prices if p >= 0]
def apply_discount(prices):
return [p * 0.95 if p > 100 else p * 1.0 if p >= 0]
def apply_discount(prices):
return [p * 0.95 if p > 100 else p for p in prices if p >= 0]
def apply_discount(prices):
return [p * 0.95 for p in prices if p > 100]

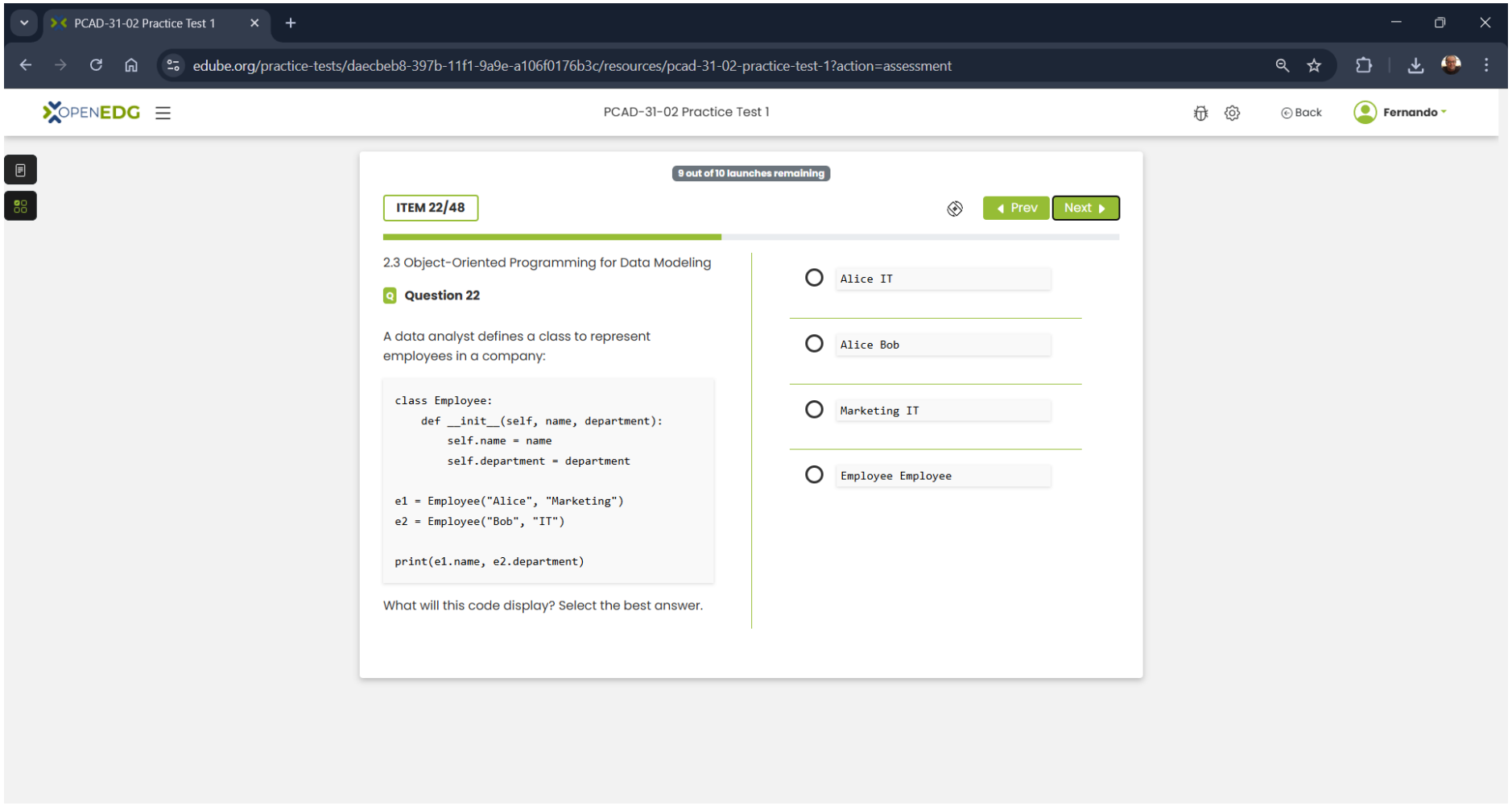

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2.1 Core Python Proficiency
Question 17
A data analyst is working on a project that requires the following tasks:
Efficient storage and manipulation of tabular data
Handling missing values
Basic descriptive statistics and group-based operations
Which library is best suited to handle all of these requirements in Python? Select the best answer.
numpy
matplotlib
pandas
scipy

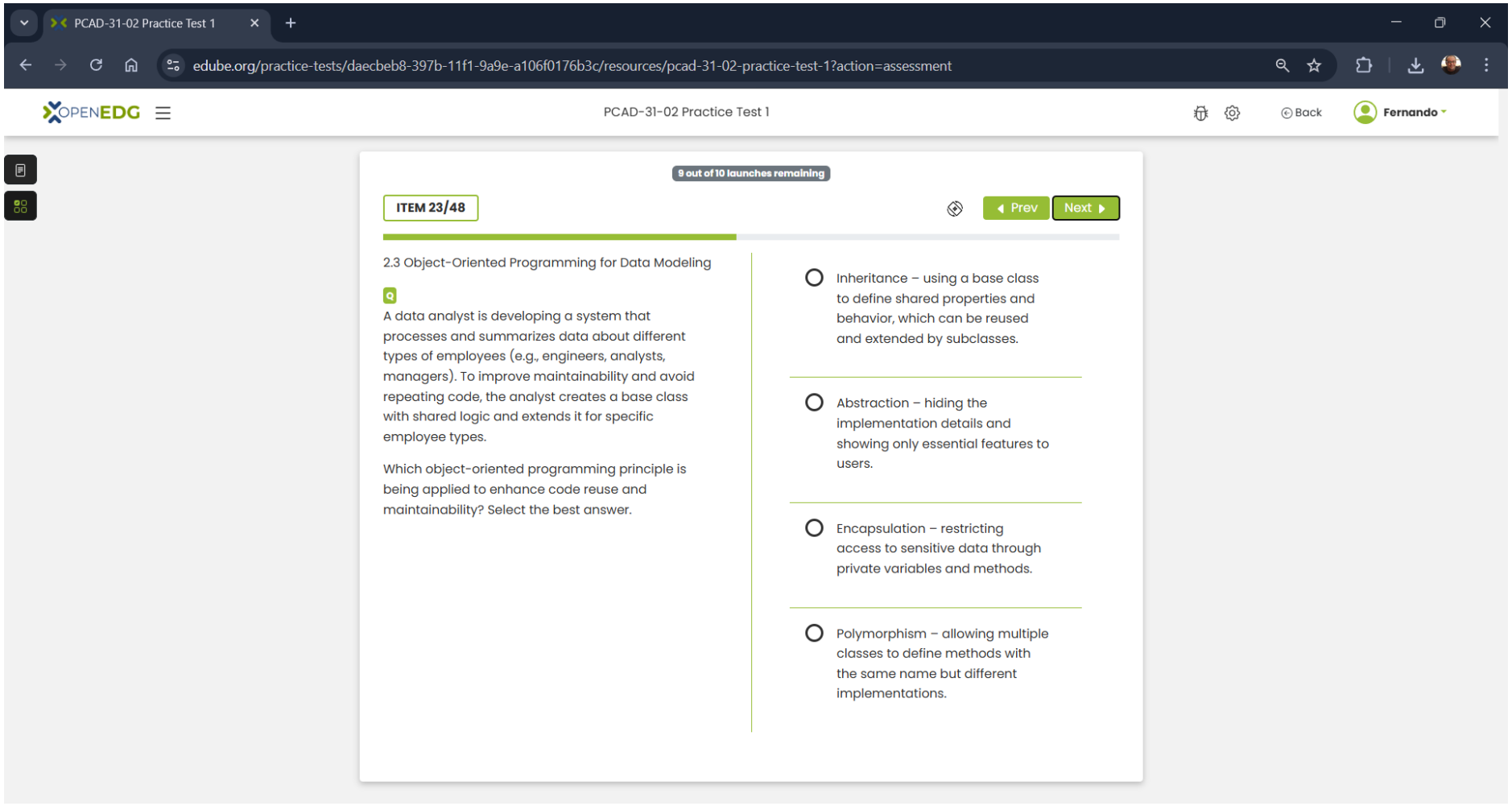

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2.1 Core Python Proficiency
Question 18
You are working with customer preferences represented as a list of tuples. Each tuple contains a product name and a list of preferred sizes. Consider the following code:
preferences = [
("Shirt", ["M", "L"]),
("Pants", ["S", "M"])
]
copied = preferences[:]
copied[0][1].append("XL")
print(preferences[0][1])
What is the result of printing preferences[0][1] ?
Select the best answer.
['M', 'L', 'XL']
['XL']
TypeError: 'tuple' object does not support item assignment
['M', 'L']

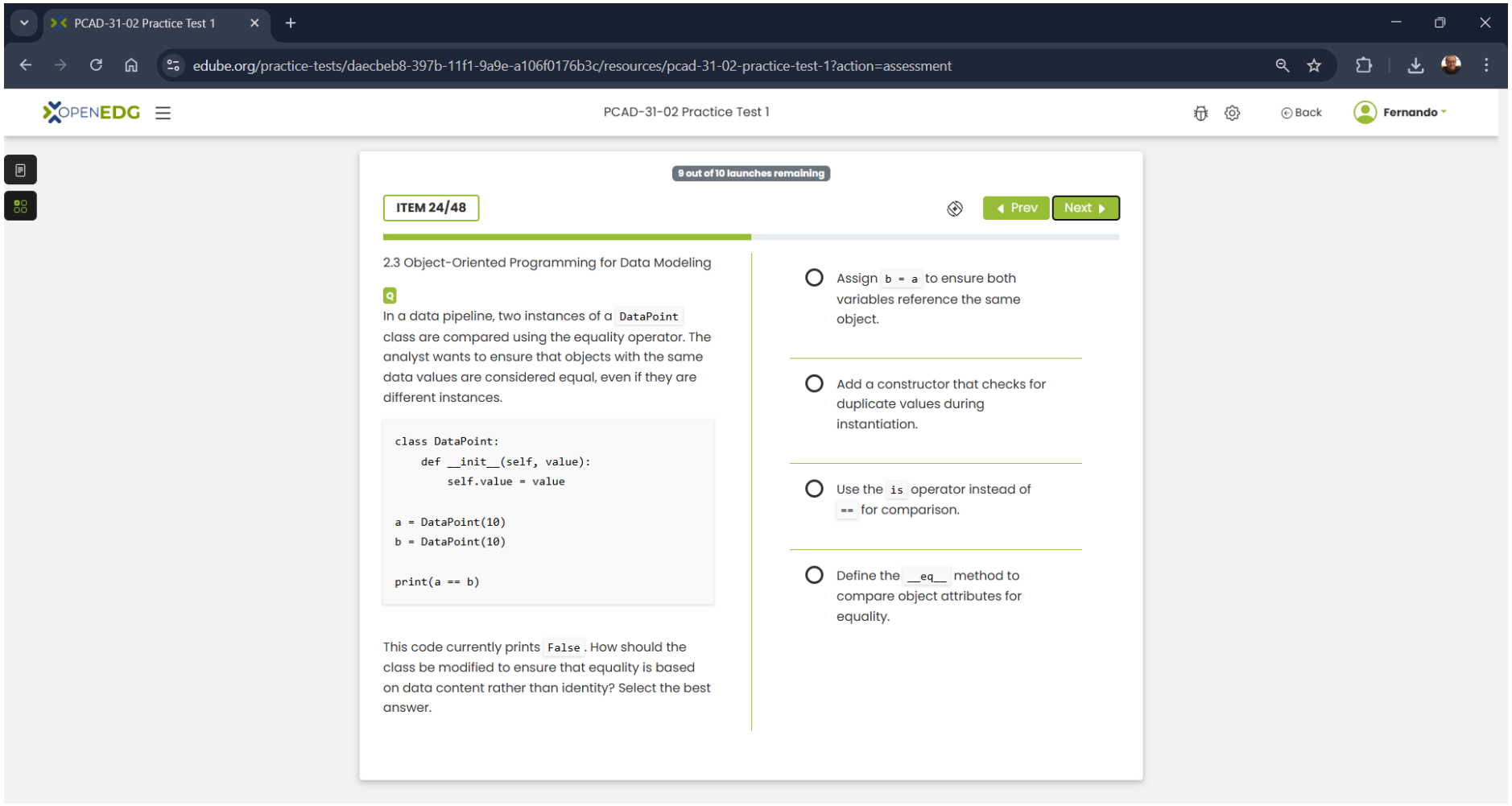

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2.1 Core Python Proficiency
Question 19
A data analyst writes the following function to convert Fahrenheit to Celsius:
def convertTemp(fahrenheit):
"""converts temperature"""
result= (fahrenheit - 32) * 5/9
return result
Which revision best follows Python's PEP 8 and PEP 257 guidelines? Select the best answer.
def convert_temp(fahrenheit):
"""Convert temperature from Fahrenheit to Celsius."""
result = (fahrenheit - 32) * 5 / 9
return result
def convert_temp(fahrenheit):
"""convert temperature from Fahrenheit to Celsius"""
return (fahrenheit - 32) * 5 / 9
def convertTemp(fahrenheit):
"""Convert temperature."""
result = (fahrenheit - 32) * 5/9
return result
def converttemp(fahrenheit):
"""Convert temperature from Fahrenheit to Celsius"""
result = (fahrenheit-32)*5/9

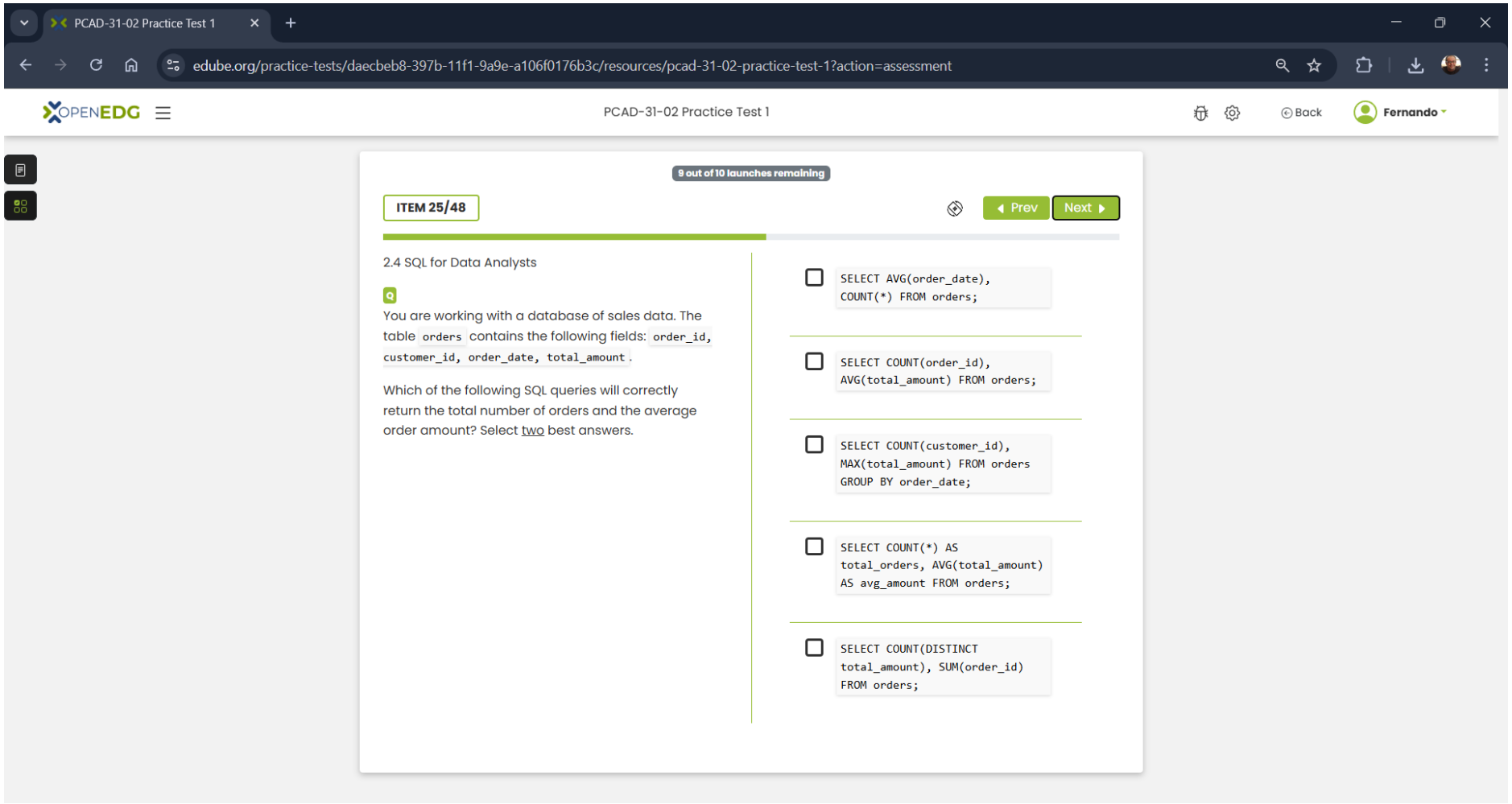

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2.2 Module Management and Exception Handling
Question 20
A data analyst is preparing to use a third-party library for data visualization. They want to install the latest version of the library and ensure it is available in their Python environment.
Which command should they use? Select the best answer.
pip libraryname
pip install libraryname
python get libraryname
pip get libraryname

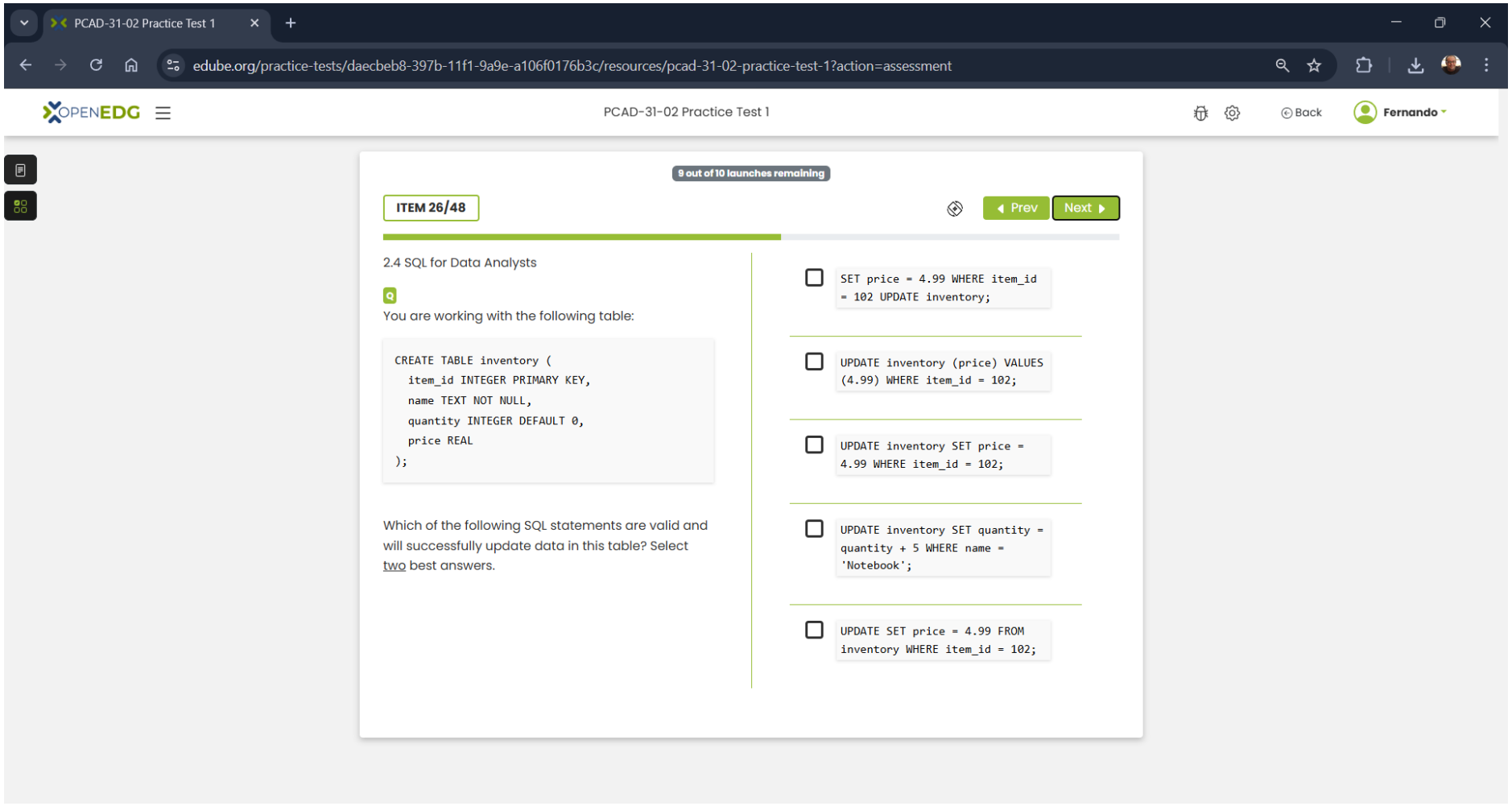

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2.2 Module Management and Exception Handling
Question 21
You are working with a script that reads data from a CSV file and prints the first line. The script raises an error when the file is missing.
Which of the following code samples applies proper exception handling to make the script more robust? Select the best answer.
try:
f = open("data.csv")
print(f.readline())
try:
with open("data.csv") as f:
print(f.readline())
except FileNotFoundError:
print("The file does not exist.")
if open("data.csv"):
print(f.readline())
with open("data.csv") as f:
print(f.readline())
except FileNotFoundError:
print("File not found.")

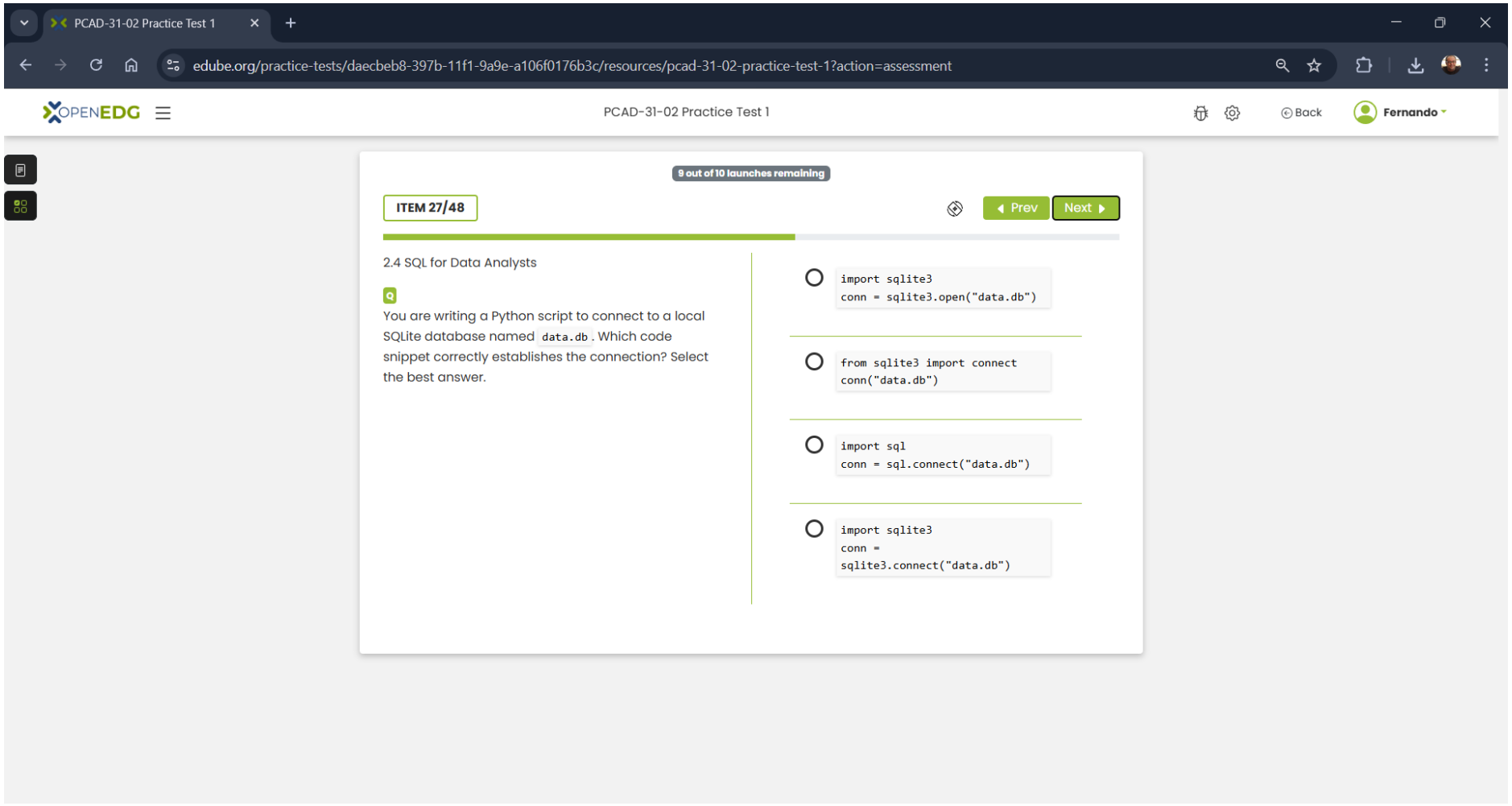

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2.3 Object-Oriented Programming for Data Modeling
Question 22
A data analyst defines a class to represent employees in a company:
class Employee:
def __init__(self, name, department):
self.name = name
self.department = department
e1 = Employee("Alice", "Marketing")
e2 = Employee("Bob", "IT")
print(e1.name, e2.department)
What will this code display? Select the best answer.
Alice IT
Alice Bob
Marketing IT
Employee Employee

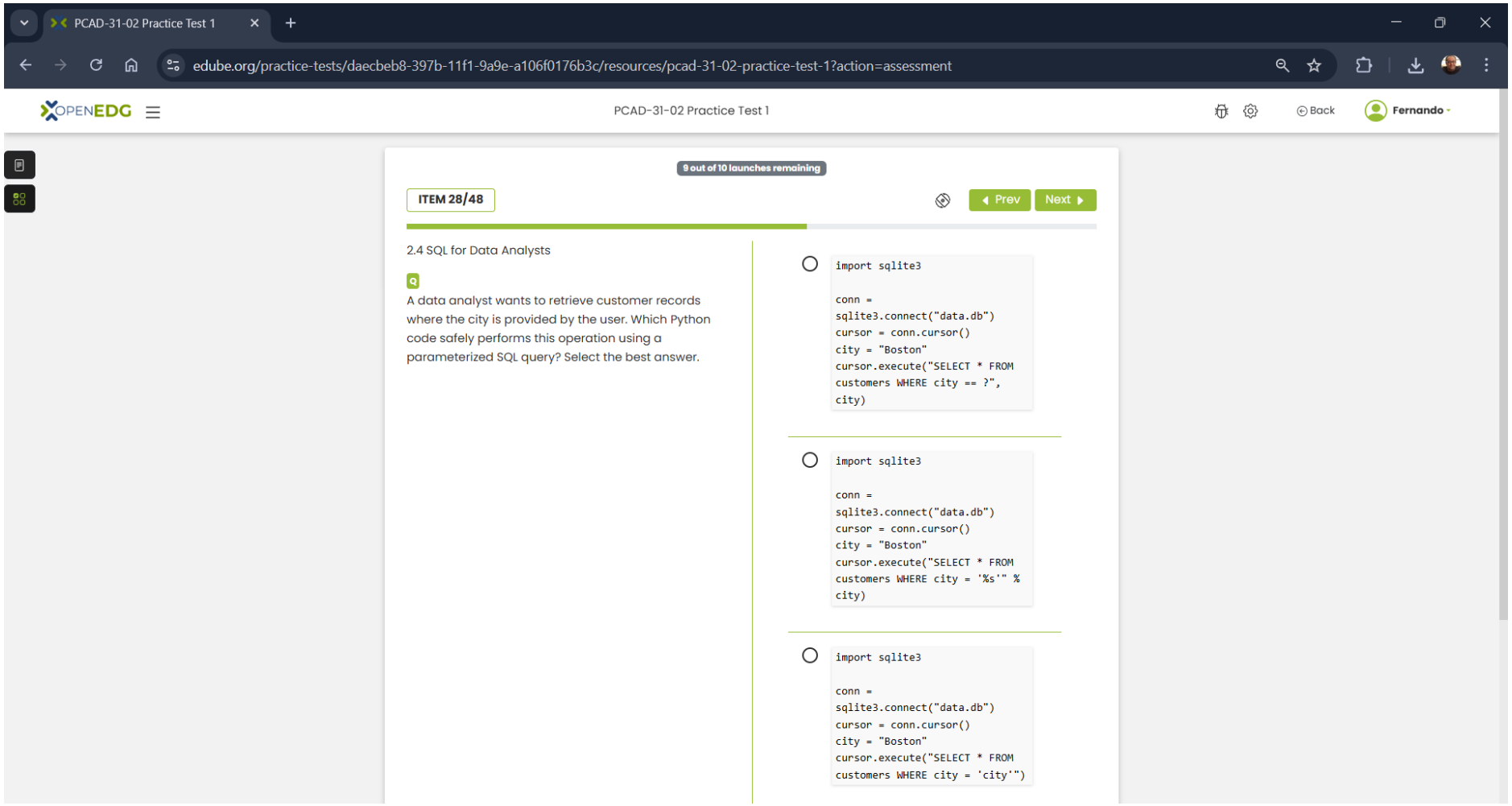

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2.3 Object-Oriented Programming for Data Modeling
Q
A data analyst is developing a system that processes and summarizes data about different types of employees (e.g., engineers, analysts, managers). To improve maintainability and avoid repeating code, the analyst creates a base class with shared logic and extends it for specific employee types.
Which object-oriented programming principle is being applied to enhance code reuse and maintainability? Select the best answer.
Inheritance – using a base class to define shared properties and behavior, which can be reused and extended by subclasses.
Abstraction – hiding the implementation details and showing only essential features to users.
Encapsulation – restricting access to sensitive data through private variables and methods.
Polymorphism – allowing multiple classes to define methods with the same name but different implementations.

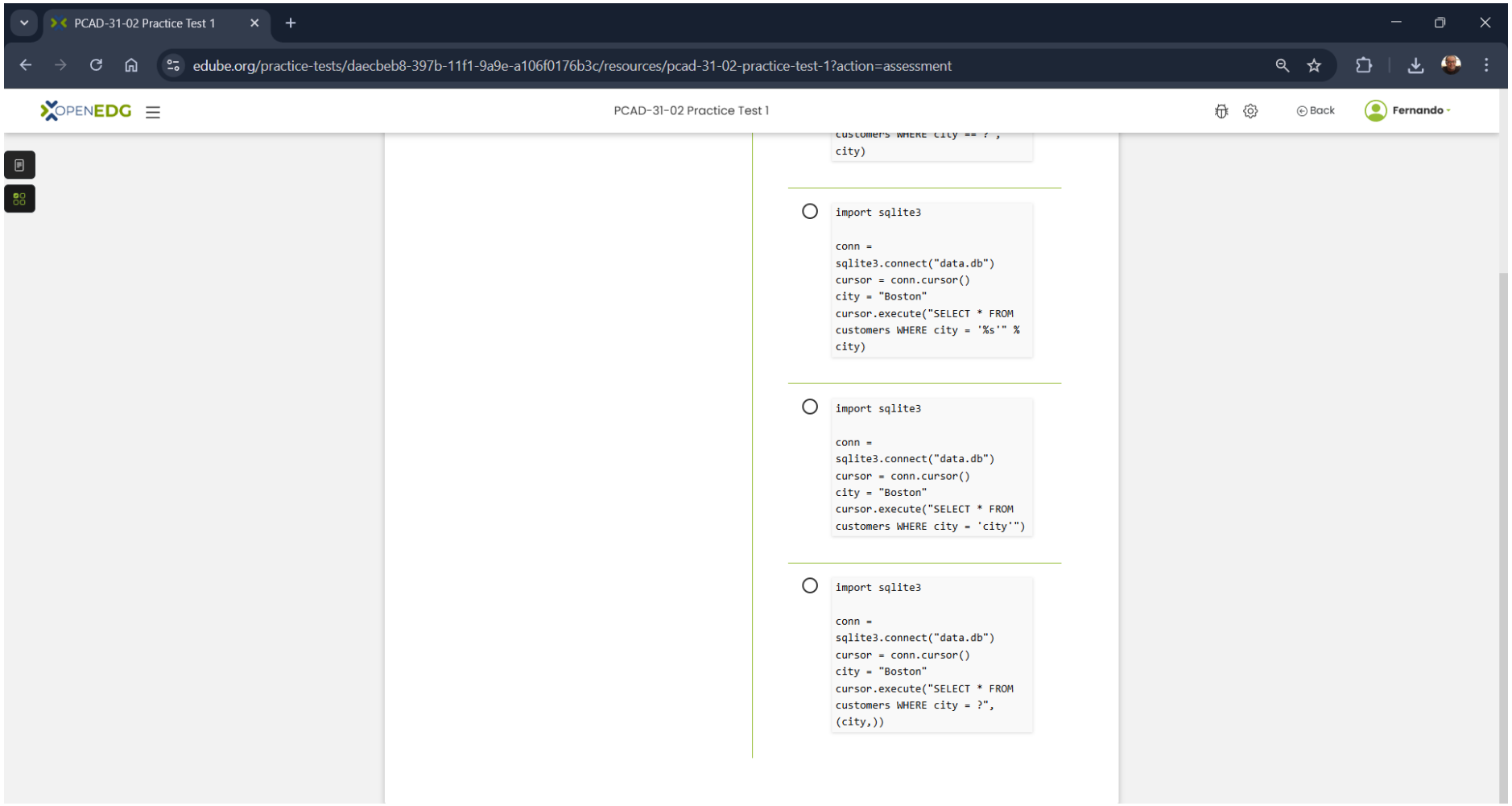

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2.3 Object-Oriented Programming for Data Modeling
Q
In a data pipeline, two instances of a DataPoint class are compared using the equality operator. The analyst wants to ensure that objects with the same data values are considered equal, even if they are different instances.
class DataPoint:
def __init__(self, value):
self.value = value
a = DataPoint(10)
b = DataPoint(10)
print(a == b)
This code currently prints False. How should the class be modified to ensure that equality is based on data content rather than identity? Select the best answer.
Assign b = a to ensure both variables reference the same object.
Add a constructor that checks for duplicate values during instantiation.
Use the is operator instead of == for comparison.
Define the __eq__ method to compare object attributes for equality.

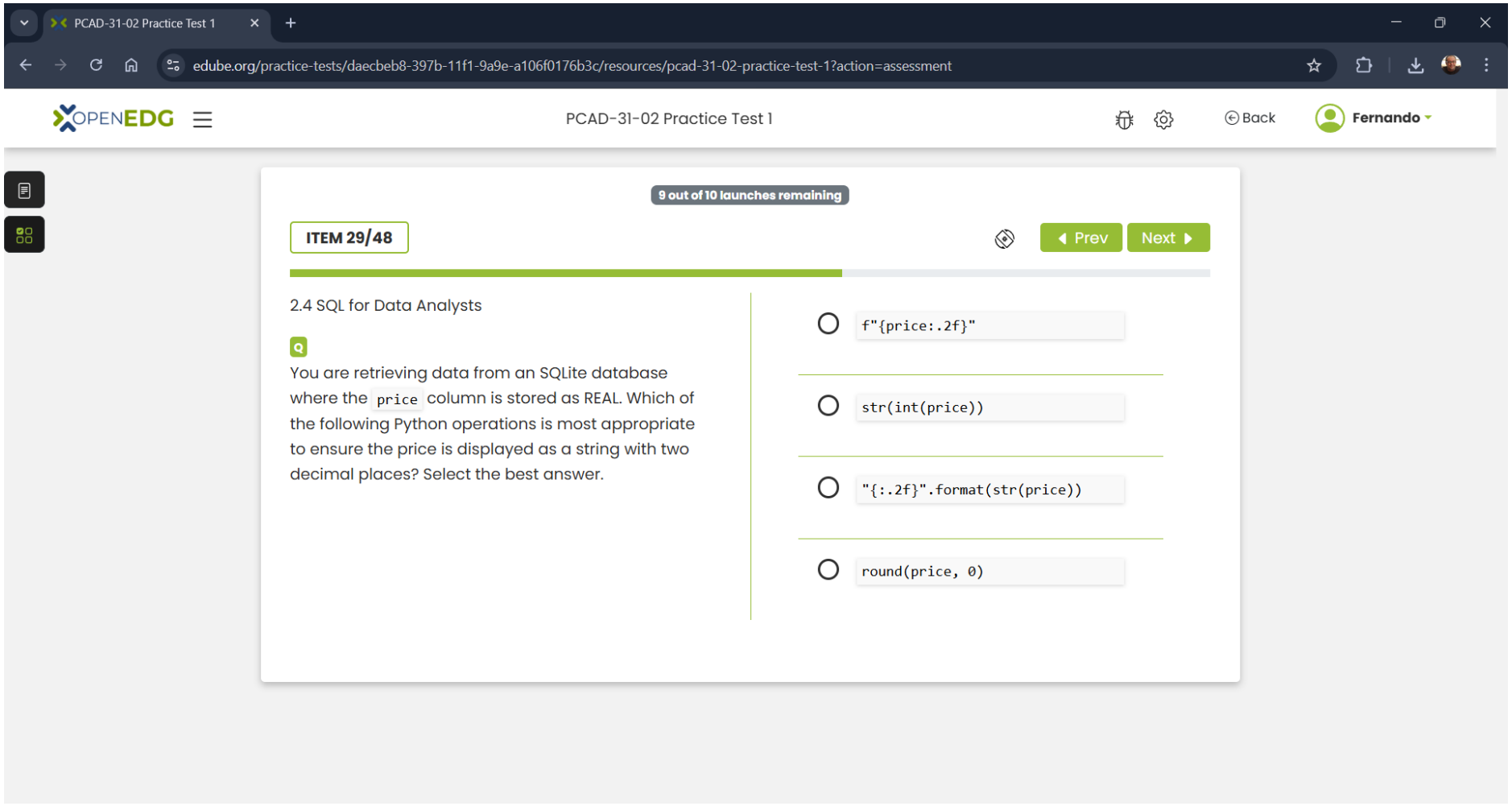

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2.4 SQL for Data Analysts
Q
You are working with a database of sales data. The table orders contains the following fields: order_id, customer_id, order_date, total_amount.
Which of the following SQL queries will correctly return the total number of orders and the average order amount? Select two best answers.
SELECT AVG(order_date), COUNT(*) FROM orders;
SELECT COUNT(order_id), AVG(total_amount) FROM orders;
SELECT COUNT(customer_id), MAX(total_amount) FROM orders GROUP BY order_date;
SELECT COUNT(*) AS total_orders, AVG(total_amount) AS avg_amount FROM orders;
SELECT COUNT(DISTINCT total_amount), SUM(order_id) FROM orders;

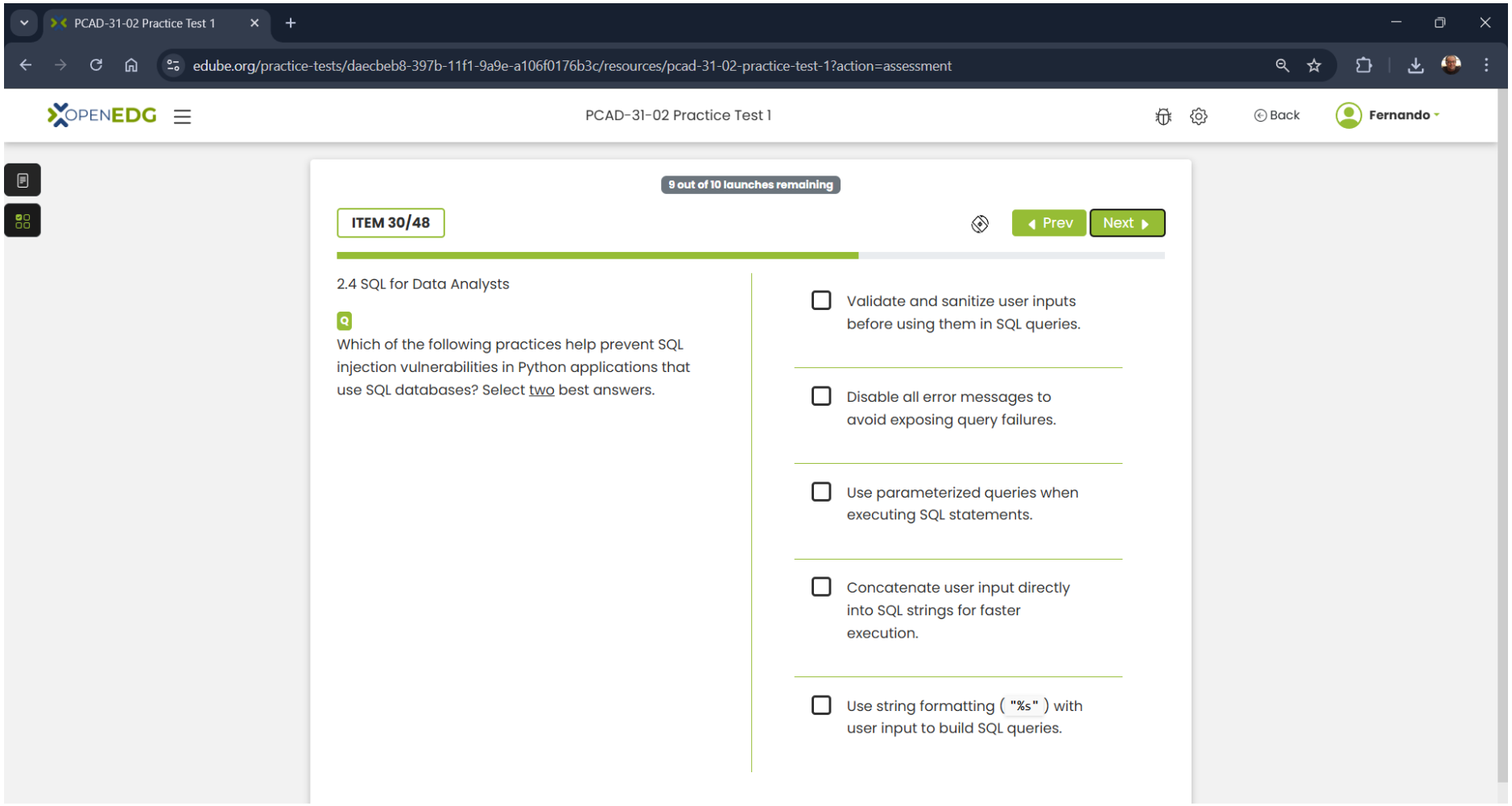

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2.4 SQL for Data Analysts
Q
You are working with the following table:
CREATE TABLE inventory (
item_id INTEGER PRIMARY KEY,
name TEXT NOT NULL,
quantity INTEGER DEFAULT 0,
price REAL
);
Which of the following SQL statements are valid and will successfully update data in this table? Select two best answers.
SET price = 4.99 WHERE item_id = 102 UPDATE inventory;
UPDATE inventory (price) VALUES (4.99) WHERE item_id = 102;
UPDATE inventory SET price = 4.99 WHERE item_id = 102;
UPDATE inventory SET quantity = quantity + 5 WHERE name = 'Notebook';
UPDATE SET price = 4.99 FROM inventory WHERE item_id = 102;

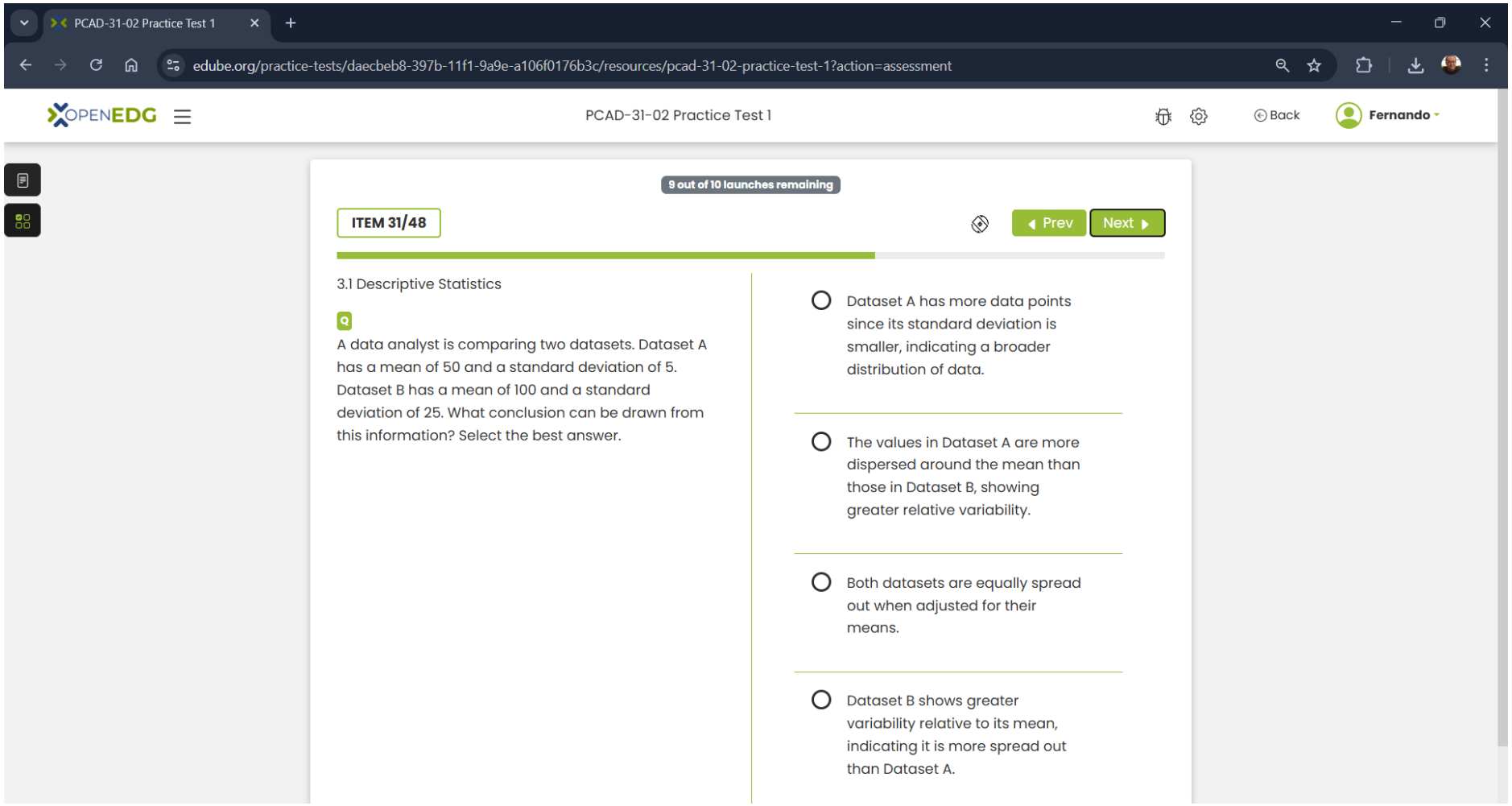

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2.4 SQL for Data Analysts
Q
You are writing a Python script to connect to a local SQLite database named data.db. Which code snippet correctly establishes the connection? Select the best answer.
import sqlite3
conn = sqlite3.open("data.db")
from sqlite3 import connect
conn("data.db")
import sql
conn = sql.connect("data.db")
import sqlite3
conn =
sqlite3.connect("data.db")


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2.4 SQL for Data Analysts
Q
A data analyst wants to retrieve customer records where the city is provided by the user. Which Python code safely performs this operation using a parameterized SQL query? Select the best answer.
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
city = "Boston"
cursor.execute("SELECT * FROM customers WHERE city == ?", city)
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
city = "Boston"
cursor.execute("SELECT * FROM customers WHERE city = '%s'" % city)
import sqlite3
conn = sqlite3.connect("data.db")
cursor = conn.cursor()
city = "Boston"
cursor.execute("SELECT * FROM customers WHERE city = 'city'")


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customers WHERE city == ? ",
city)
import sqlite3
conn =
sqlite3.connect("data.db")
cursor = conn.cursor()
city = "Boston"
cursor.execute("SELECT * FROM
customers WHERE city = '%s'" %
city)
import sqlite3
conn =
sqlite3.connect("data.db")
cursor = conn.cursor()
city = "Boston"
cursor.execute("SELECT * FROM
customers WHERE city = 'city'")
import sqlite3
conn =
sqlite3.connect("data.db")
cursor = conn.cursor()
city = "Boston"
cursor.execute("SELECT * FROM
customers WHERE city = ?",
(city,))

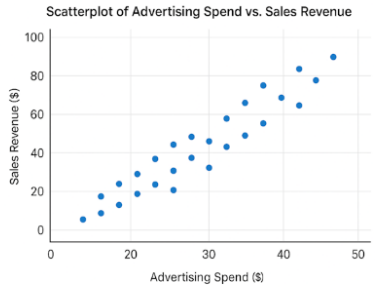

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2.4 SQL for Data Analysts
Q
You are retrieving data from an SQLite database where the price column is stored as REAL. Which of the following Python operations is most appropriate to ensure the price is displayed as a string with two decimal places? Select the best answer.
f"{price:.2f}"
str(int(price))
"{:.2f}".format(str(price))
round(price, 0)


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2.4 SQL for Data Analysts
Q
Which of the following practices help prevent SQL injection vulnerabilities in Python applications that use SQL databases? Select two best answers.
Validate and sanitize user inputs before using them in SQL queries.
Disable all error messages to avoid exposing query failures.
Use parameterized queries when executing SQL statements.
Concatenate user input directly into SQL strings for faster execution.
Use string formatting ("%s") with user input to build SQL queries.


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3.1 Descriptive Statistics
Q
A data analyst is comparing two datasets. Dataset A has a mean of 50 and a standard deviation of 5. Dataset B has a mean of 100 and a standard deviation of 25. What conclusion can be drawn from this information? Select the best answer.
Dataset A has more data points since its standard deviation is smaller, indicating a broader distribution of data.
The values in Dataset A are more dispersed around the mean than those in Dataset B, showing greater relative variability.
Both datasets are equally spread out when adjusted for their means.
Dataset B shows greater variability relative to its mean, indicating it is more spread out than Dataset A.

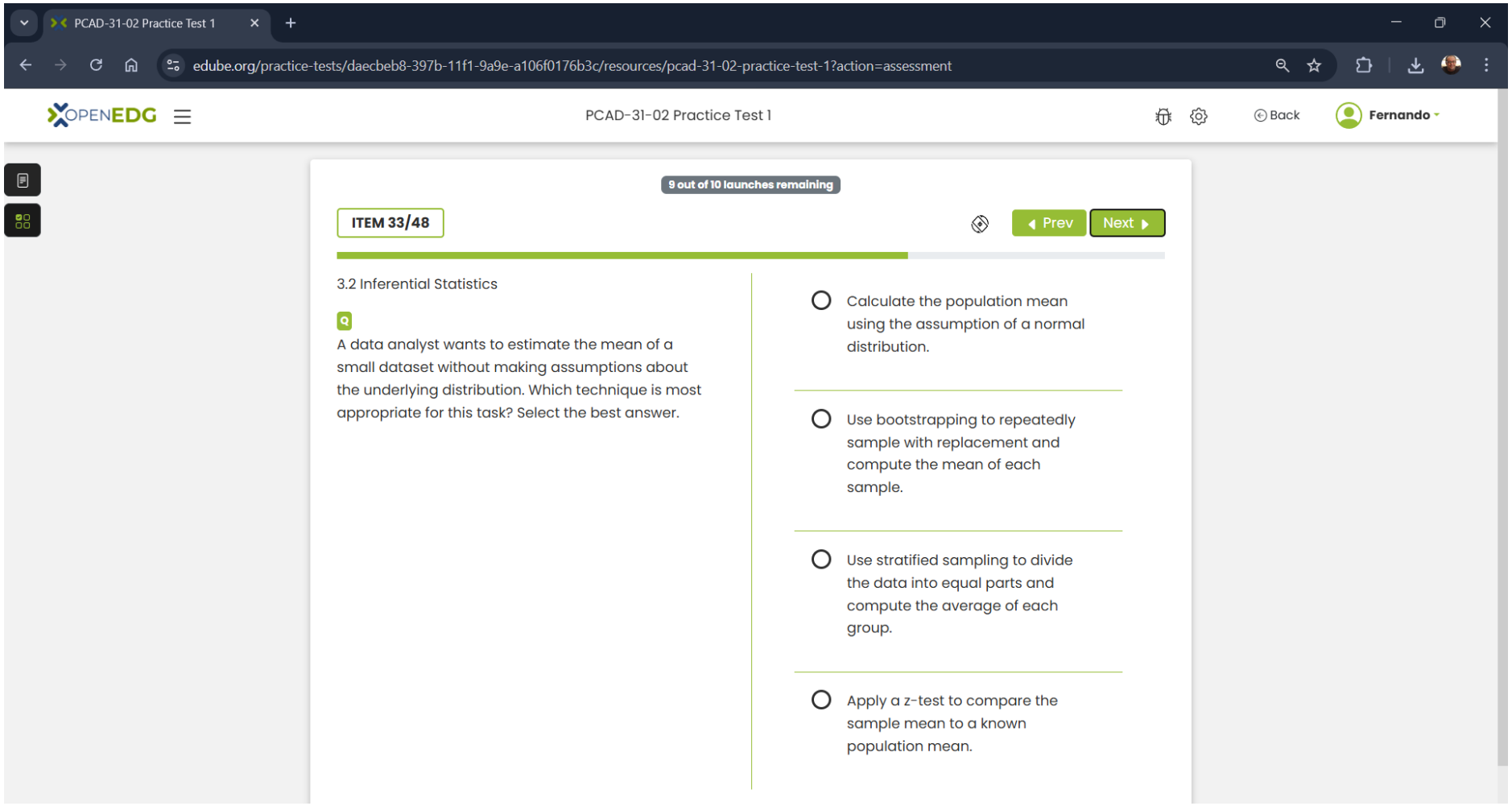
3.1 Descriptive Statistics

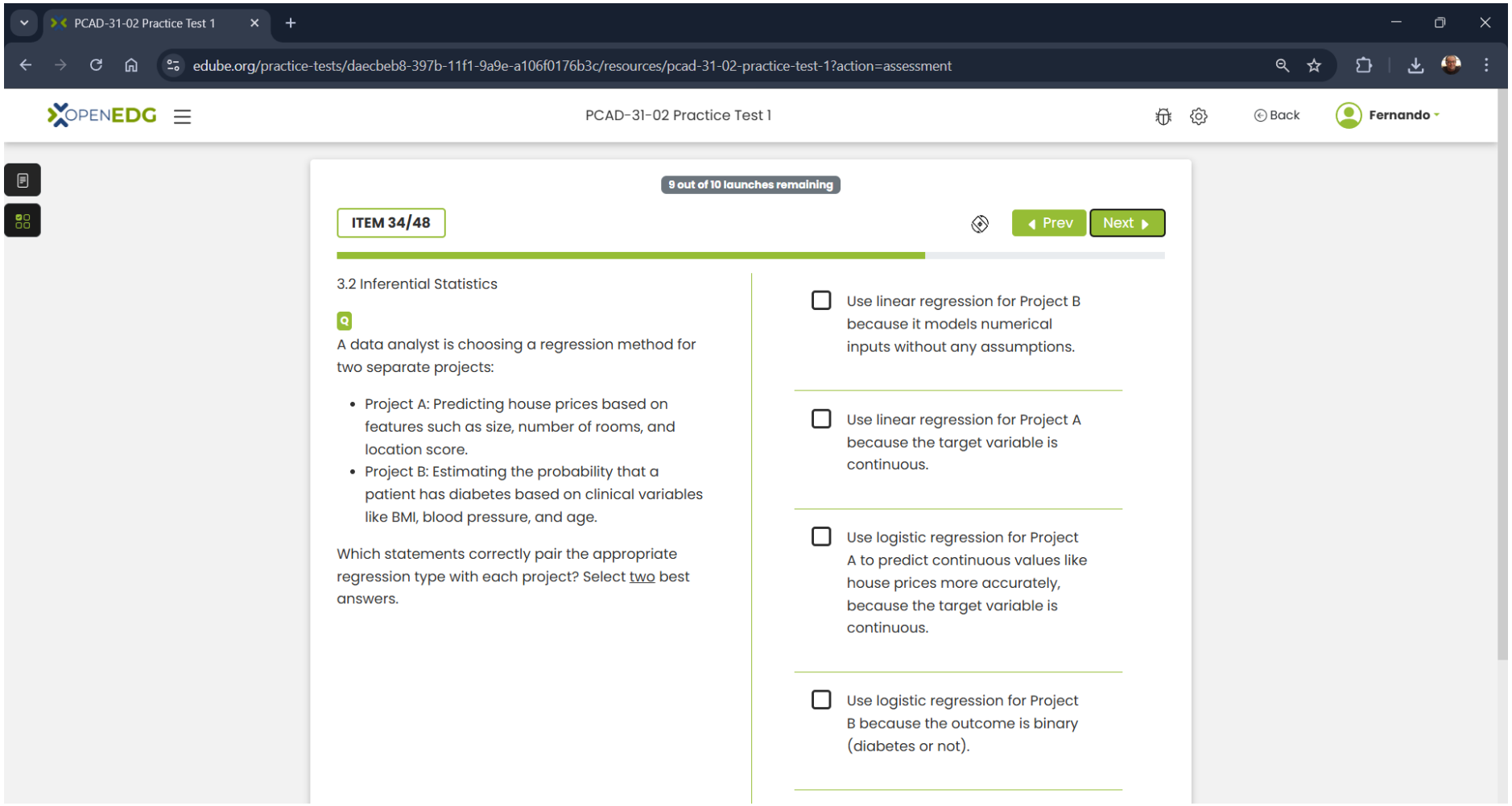
A data analyst creates the following scatterplot to study the relationship between advertising spend and sales revenue:

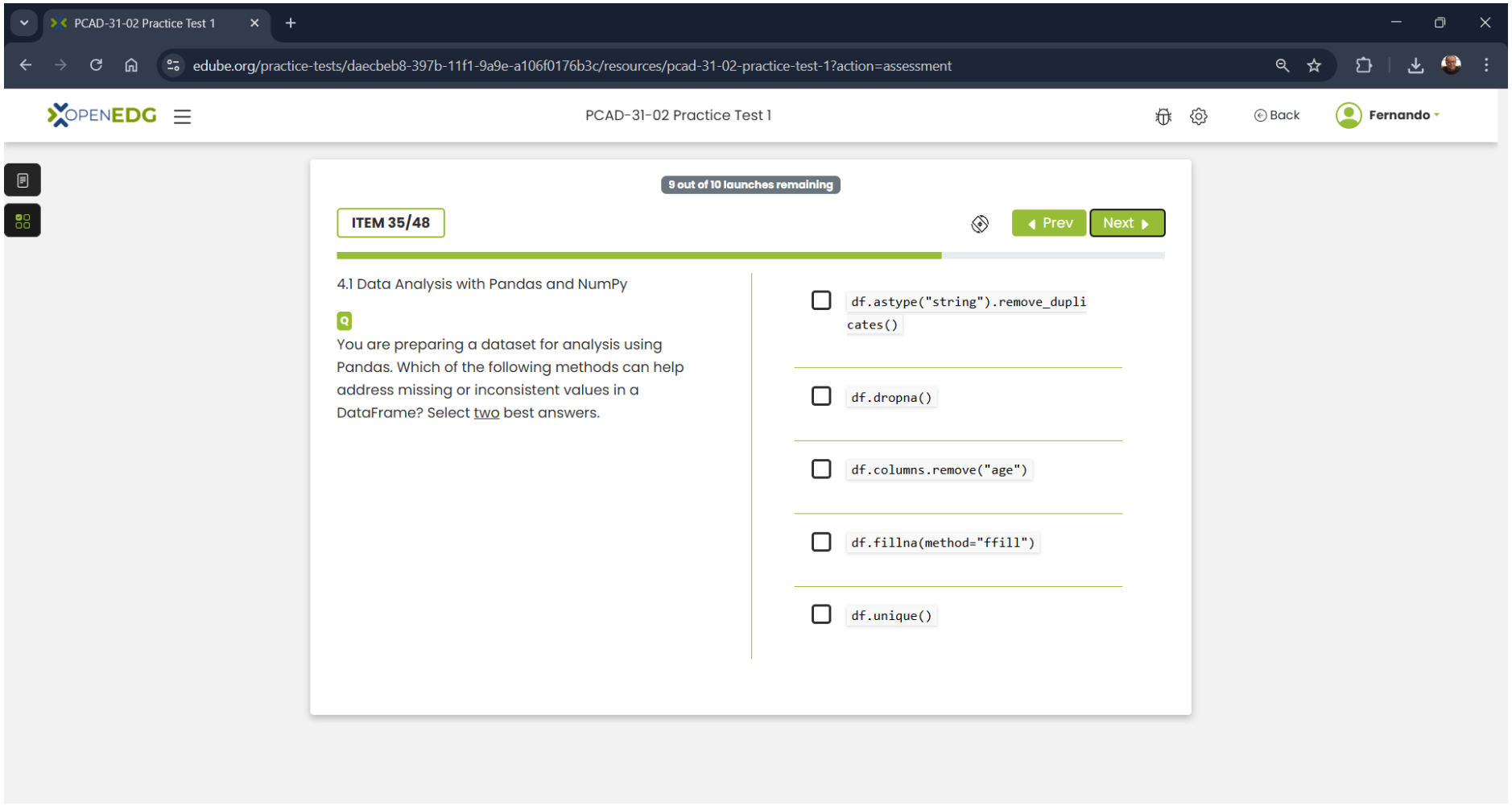


Which of the following conclusions is most appropriate? Select the best answer.

- The relationship is non-linear, so correlation cannot be determined from this plot.
- There is a strong negative correlation; as advertising spend increases, revenue decreases.
- The plot shows no clear trend; the variables are not correlated.
- The variables show a strong positive correlation, meaning higher advertising spend is associated with increased revenue.


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3.2 Inferential Statistics
A data analyst wants to estimate the mean of a small dataset without making assumptions about the underlying distribution. Which technique is most appropriate for this task? Select the best answer.
Calculate the population mean using the assumption of a normal distribution.
Use bootstrapping to repeatedly sample with replacement and compute the mean of each sample.
Use stratified sampling to divide the data into equal parts and compute the average of each group.
Apply a z-test to compare the sample mean to a known population mean.


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3.2 Inferential Statistics
Q
A data analyst is choosing a regression method for two separate projects:
Project A: Predicting house prices based on features such as size, number of rooms, and location score.
Project B: Estimating the probability that a patient has diabetes based on clinical variables like BMI, blood pressure, and age.
Which statements correctly pair the appropriate regression type with each project? Select two best answers.
Use linear regression for Project B because it models numerical inputs without any assumptions.
Use linear regression for Project A because the target variable is continuous.
Use logistic regression for Project A to predict continuous values like house prices more accurately, because the target variable is continuous.
Use logistic regression for Project B because the outcome is binary (diabetes or not).


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4.1 Data Analysis with Pandas and NumPy
You are preparing a dataset for analysis using Pandas. Which of the following methods can help address missing or inconsistent values in a DataFrame? Select two best answers.
df.astype("string").remove_duplicates()
df.dropna()
df.columns.remove("age")
df.fillna(method="ffill")
df.unique()

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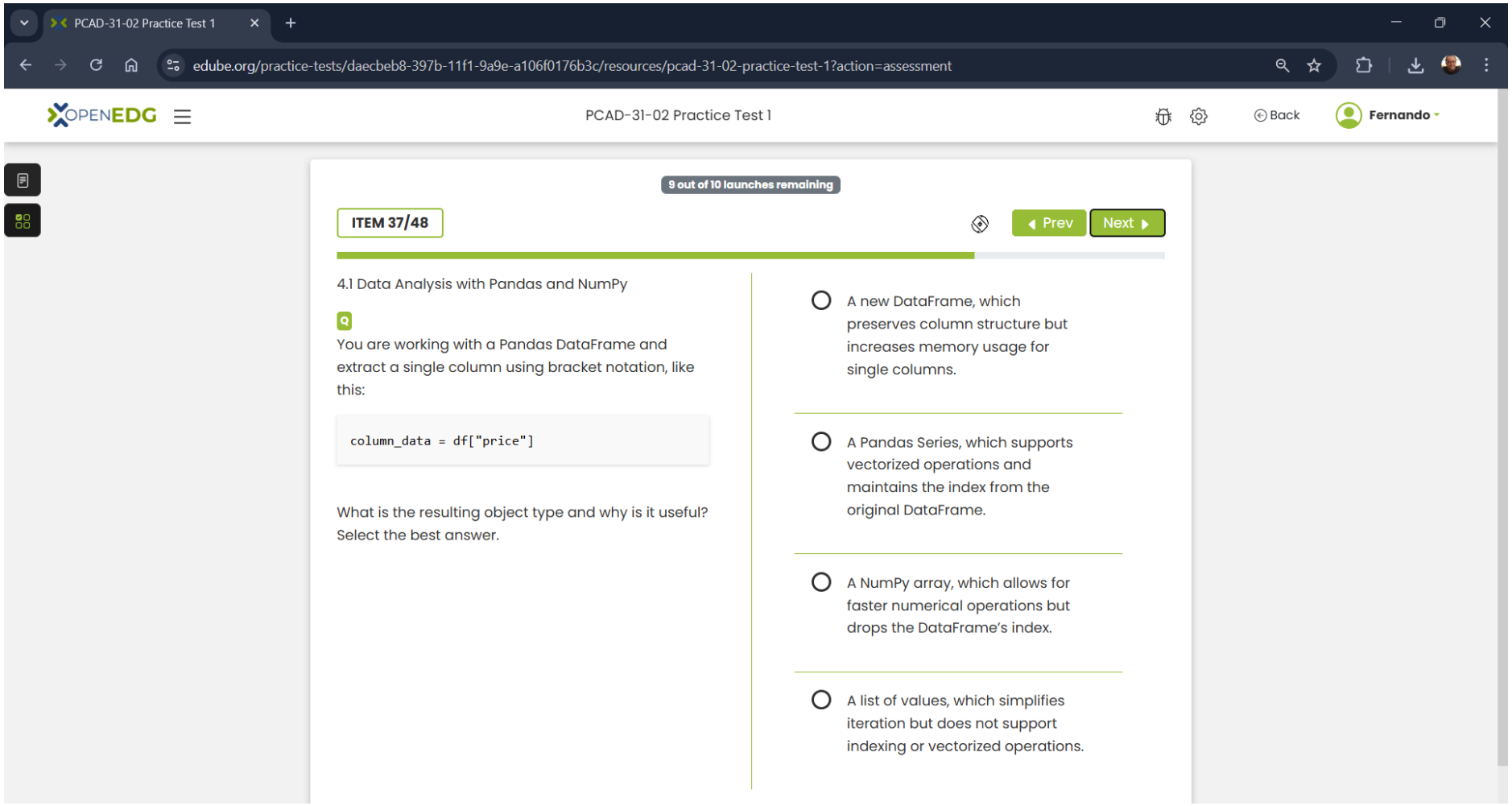
4.1 Data Analysis with Pandas and NumPy

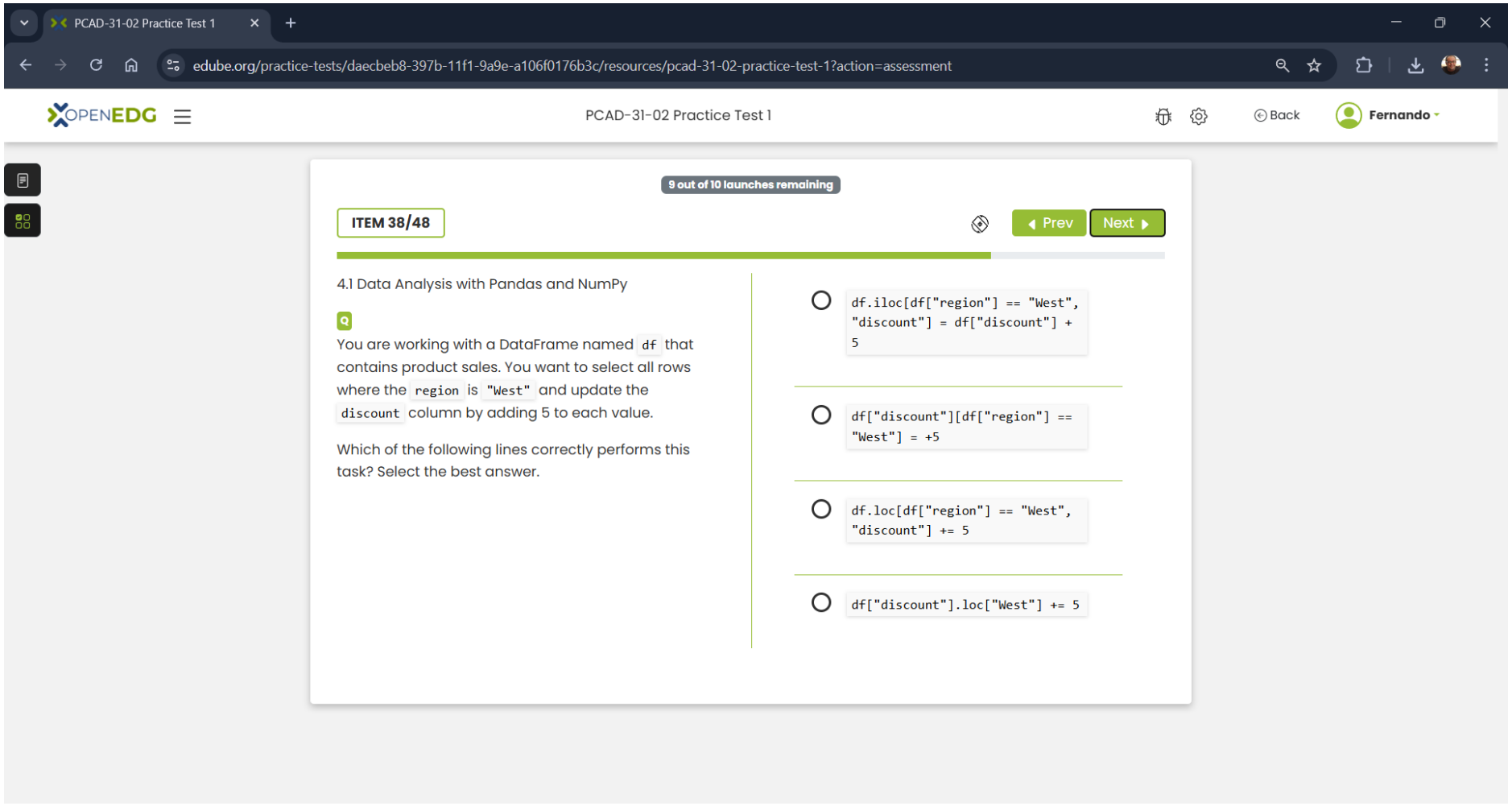
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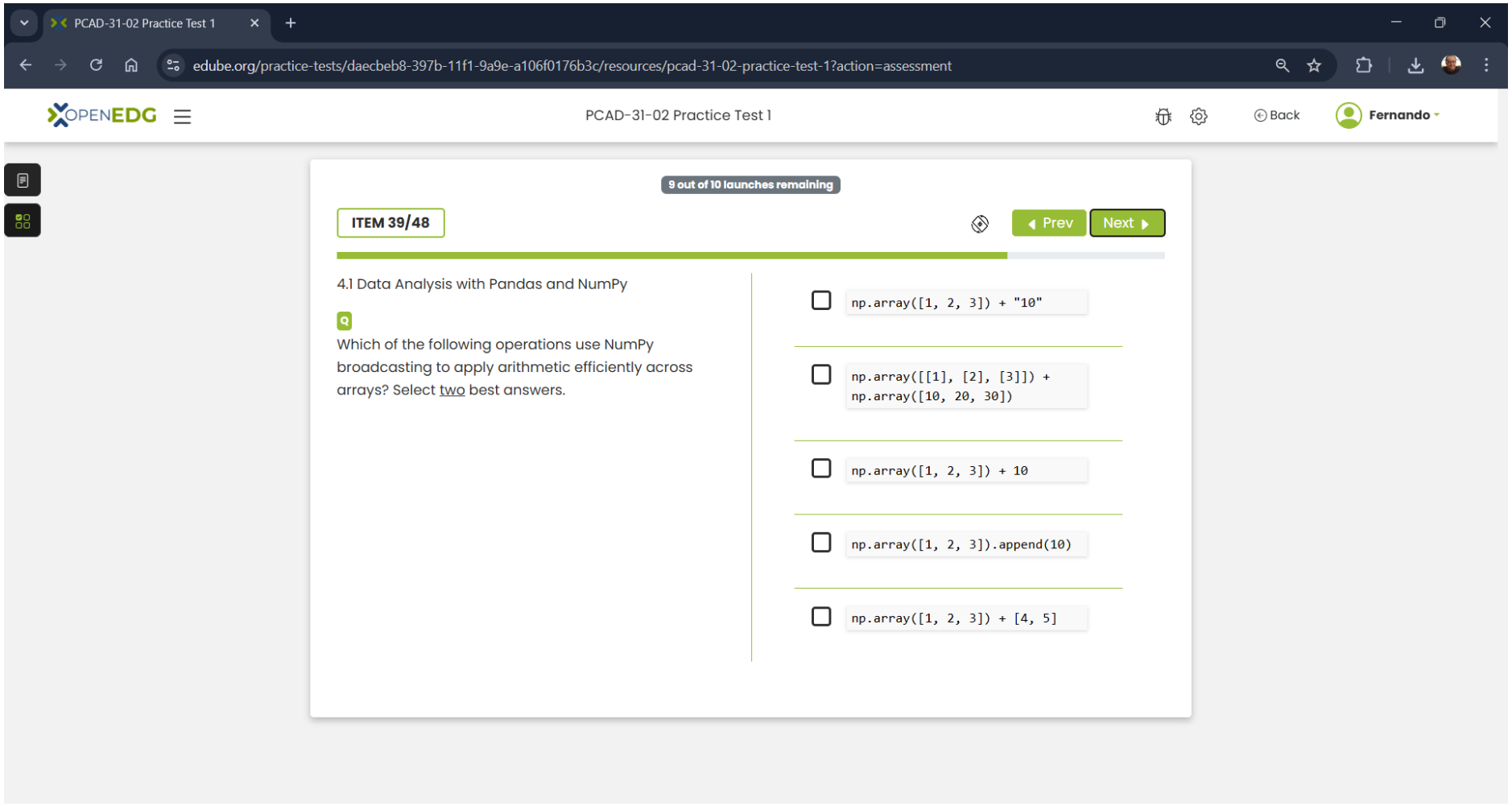
You are analyzing survey results stored in a DataFrame named `responses`. Each row includes a `respondent_id` and answers to multiple questions.

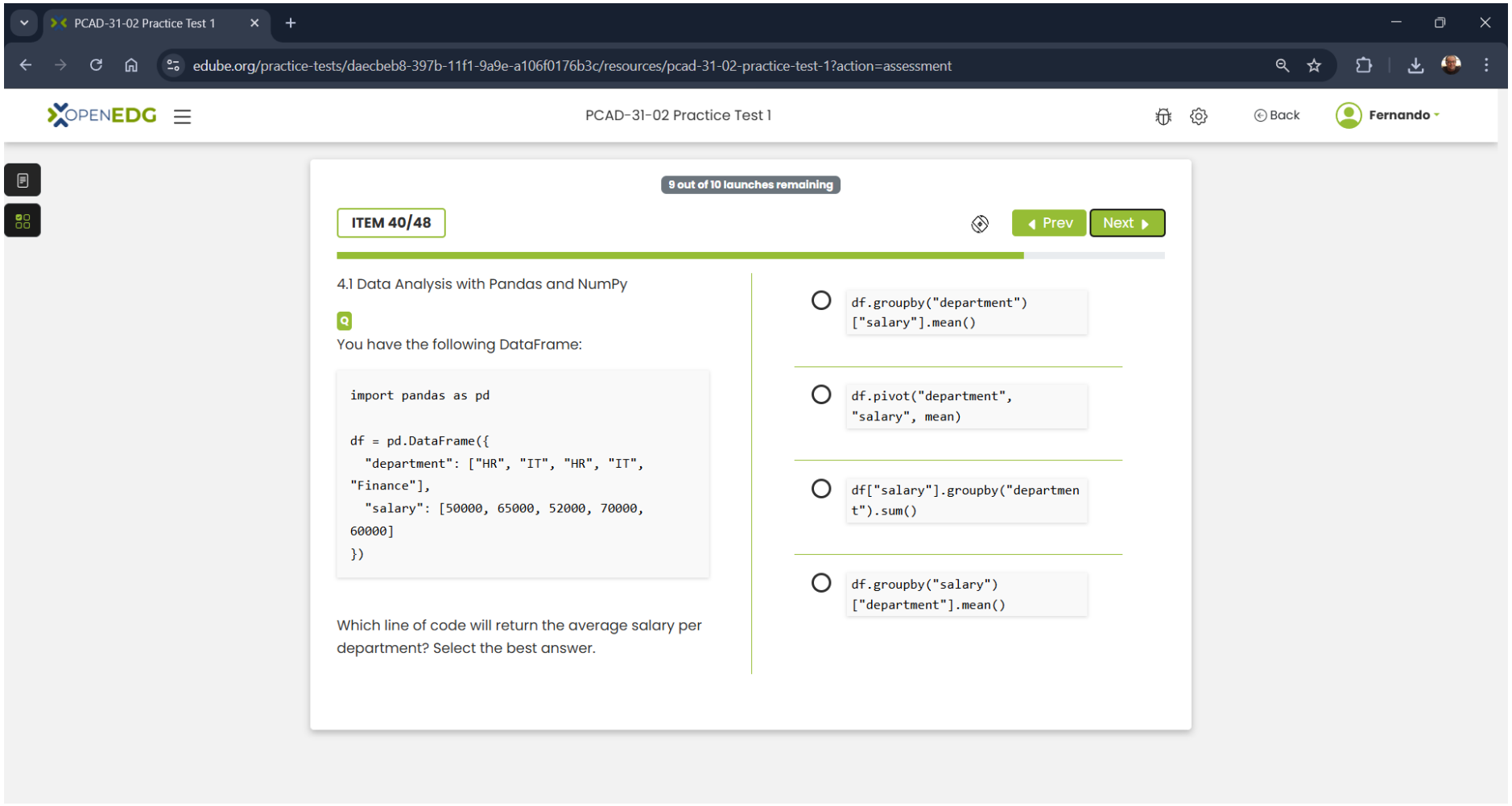
You want to reshape this wide-format data into a long format, where each question and its corresponding answer become a row. Which line of code correctly performs this transformation? Select the best answer.

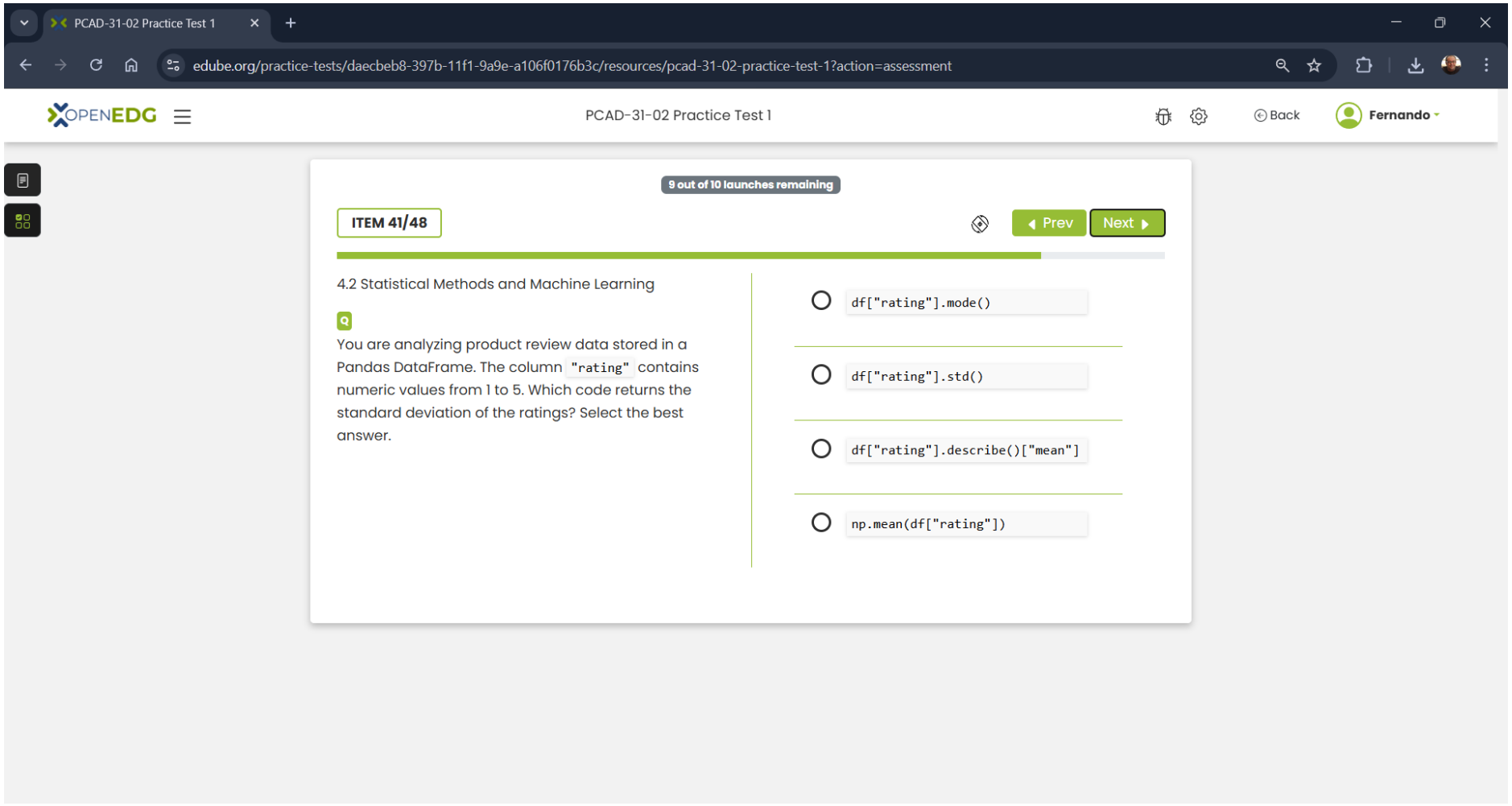
- `pd.melt(responses, id_vars="respondent_id", var_name="question", value_name="answer")`
- `responses.groupby("respondent_id").unstack()`
- `responses.pivot(index="respondent_id", columns="question", values="answer")`
- `pd.merge(responses, "respondent_id", on="question")`

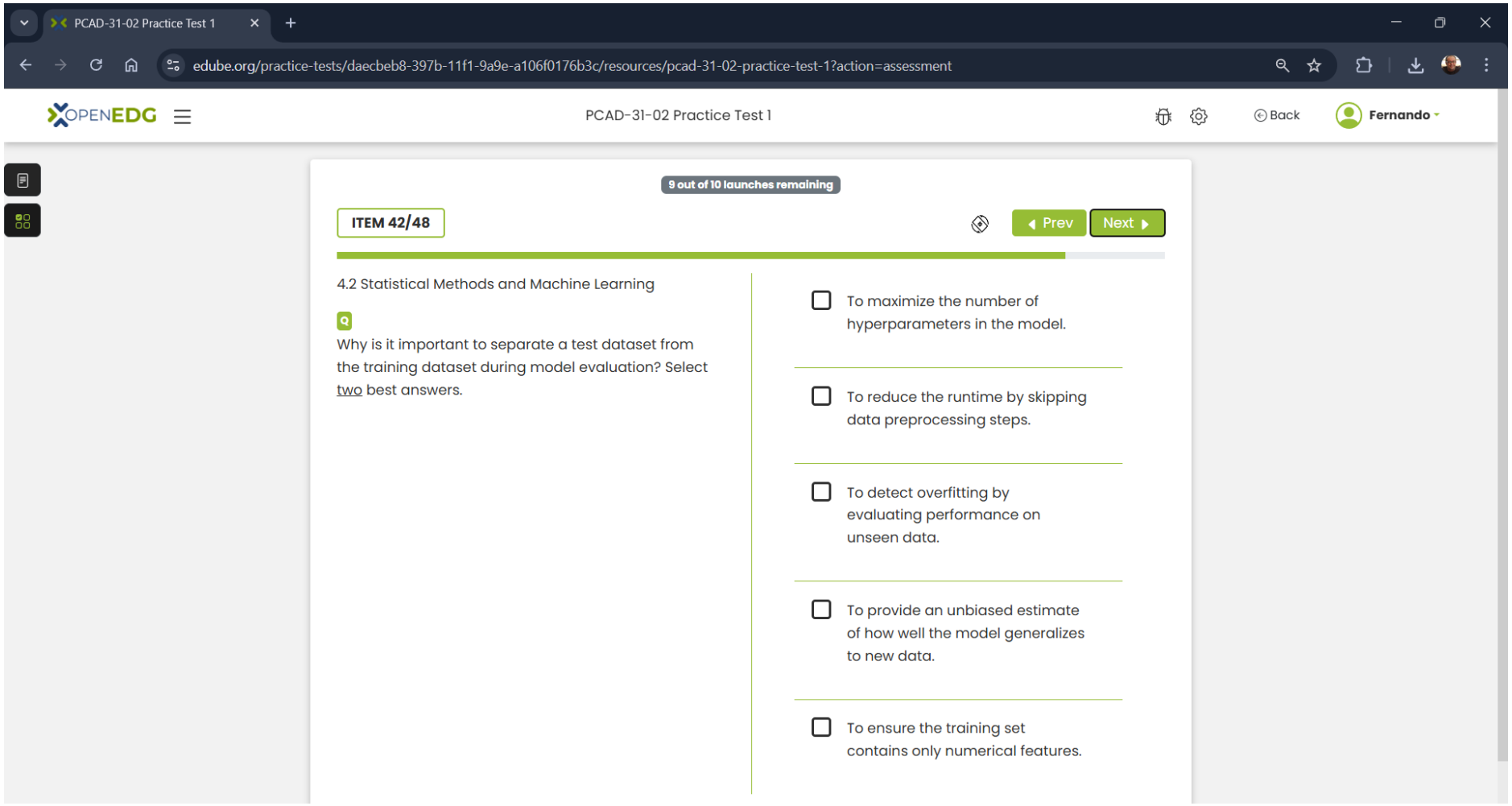

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4.1 Data Analysis with Pandas and NumPy
You are working with a Pandas DataFrame and extract a single column using bracket notation, like this:
column_data = df["price"]
What is the resulting object type and why is it useful? Select the best answer.
A new DataFrame, which preserves column structure but increases memory usage for single columns.
A Pandas Series, which supports vectorized operations and maintains the index from the original DataFrame.
A NumPy array, which allows for faster numerical operations but drops the DataFrame's index.
A list of values, which simplifies iteration but does not support indexing or vectorized operations.


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4.1 Data Analysis with Pandas and NumPy
Q
You are working with a DataFrame named df that contains product sales. You want to select all rows where the region is "West" and update the discount column by adding 5 to each value.
Which of the following lines correctly performs this task? Select the best answer.
df.iloc[df["region"] == "West", "discount"] = df["discount"] + 5
df["discount"][df["region"] == "West"] = +5
df.loc[df["region"] == "West", "discount"] += 5
df["discount"].loc["West"] += 5


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4.1 Data Analysis with Pandas and NumPy
Q
Which of the following operations use NumPy broadcasting to apply arithmetic efficiently across arrays? Select two best answers.
np.array([1, 2, 3]) + "10"
np.array([[1], [2], [3]]) + np.array([10, 20, 30])
np.array([1, 2, 3]) + 10
np.array([1, 2, 3]).append(10)
np.array([1, 2, 3]) + [4, 5]


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4.1 Data Analysis with Pandas and NumPy
You have the following DataFrame:
import pandas as pd
df = pd.DataFrame({
"department": ["HR", "IT", "HR", "IT", "Finance"],
"salary": [50000, 65000, 52000, 70000, 60000]
})
Which line of code will return the average salary per department? Select the best answer.
df.groupby("department")["salary"].mean()
df.pivot("department", "salary", mean)
df["salary"].groupby("department").sum()
df.groupby("salary")["department"].mean()


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4.2 Statistical Methods and Machine Learning
You are analyzing product review data stored in a Pandas DataFrame. The column "rating" contains numeric values from 1 to 5. Which code returns the standard deviation of the ratings? Select the best answer.
df["rating"].mode()
df["rating"].std()
df["rating"].describe()["mean"]
np.mean(df["rating"])


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4.2 Statistical Methods and Machine Learning
Q
Why is it important to separate a test dataset from the training dataset during model evaluation? Select two best answers.
To maximize the number of hyperparameters in the model.
To reduce the runtime by skipping data preprocessing steps.
To detect overfitting by evaluating performance on unseen data.
To provide an unbiased estimate of how well the model generalizes to new data.
To ensure the training set contains only numerical features.

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4.2 Statistical Methods and Machine Learning

Q

Which of the following statements best reflects the concept of the bias-variance tradeoff in supervised learning? Select the best answer.

- Bias and variance only apply to unsupervised models like clustering.
- As bias decreases, variance tends to increase, making it important to find a balance to optimize model accuracy.
- High bias and high variance are both desirable for robust models.
- Bias and variance are unrelated and can be optimized independently in all models.

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5.1 Data Visualization Techniques

You want to visualize the distribution of customer ratings stored in the `"rating"` column of a DataFrame using Seaborn.

Which of the following code snippets is the most appropriate to generate a histogram? Select the best answer.

- ```
  import seaborn as sns
  sns.boxplot(data=df,
  x="rating")
  ```
- ```
  import matplotlib.pyplot as plt
  plt.hist(df["rating"],
  kind="line")
  ```
- ```
  import seaborn as sns
  sns.histplot(data=df,
  x="rating")
  ```
- ```
  import seaborn as sns
  sns.histogram(data=df,
  x="rating")
  ```

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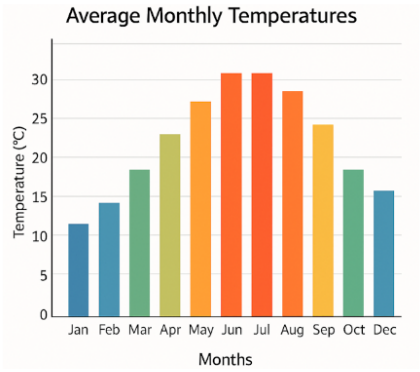
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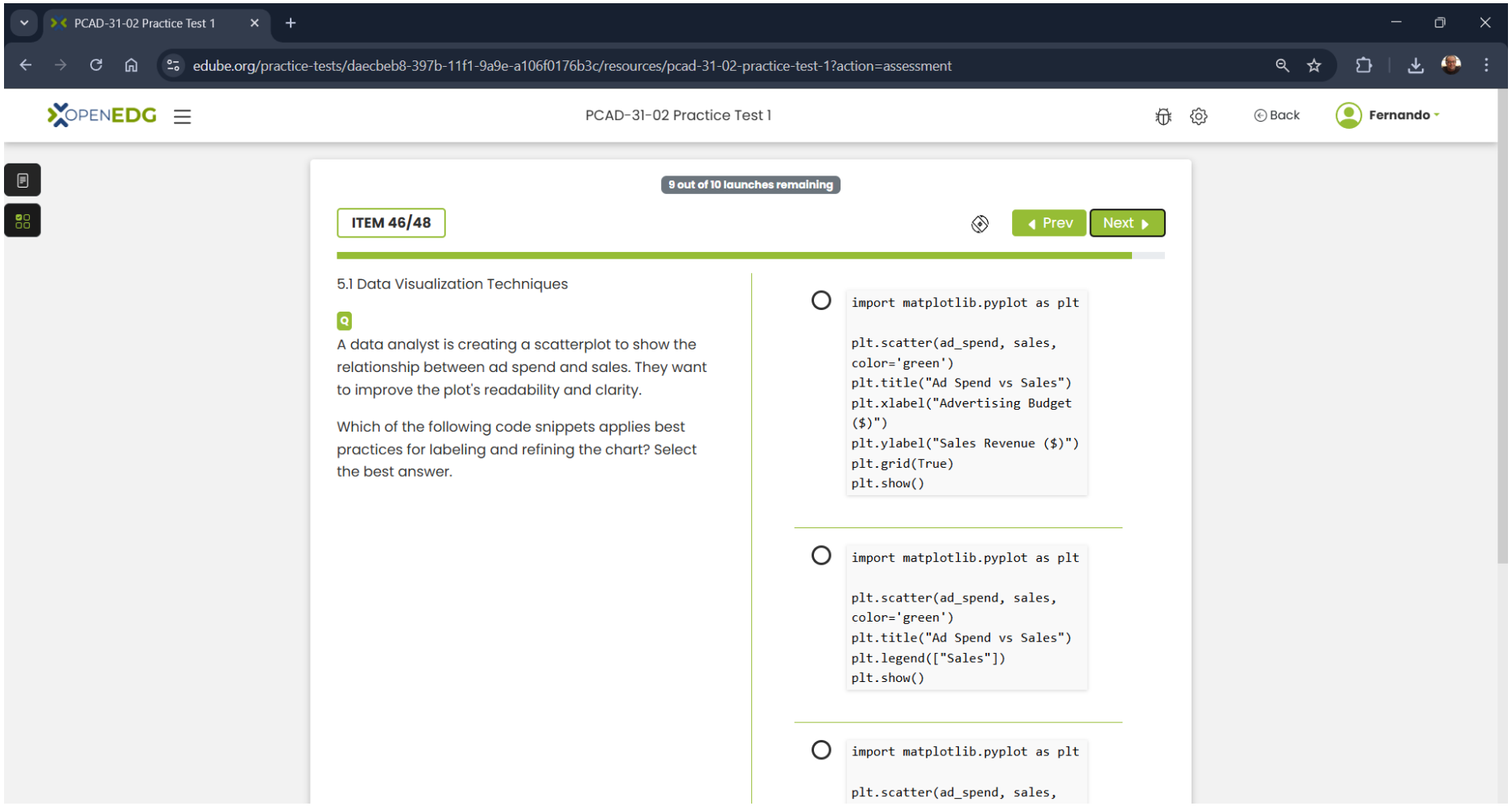
5.1 Data Visualization Techniques

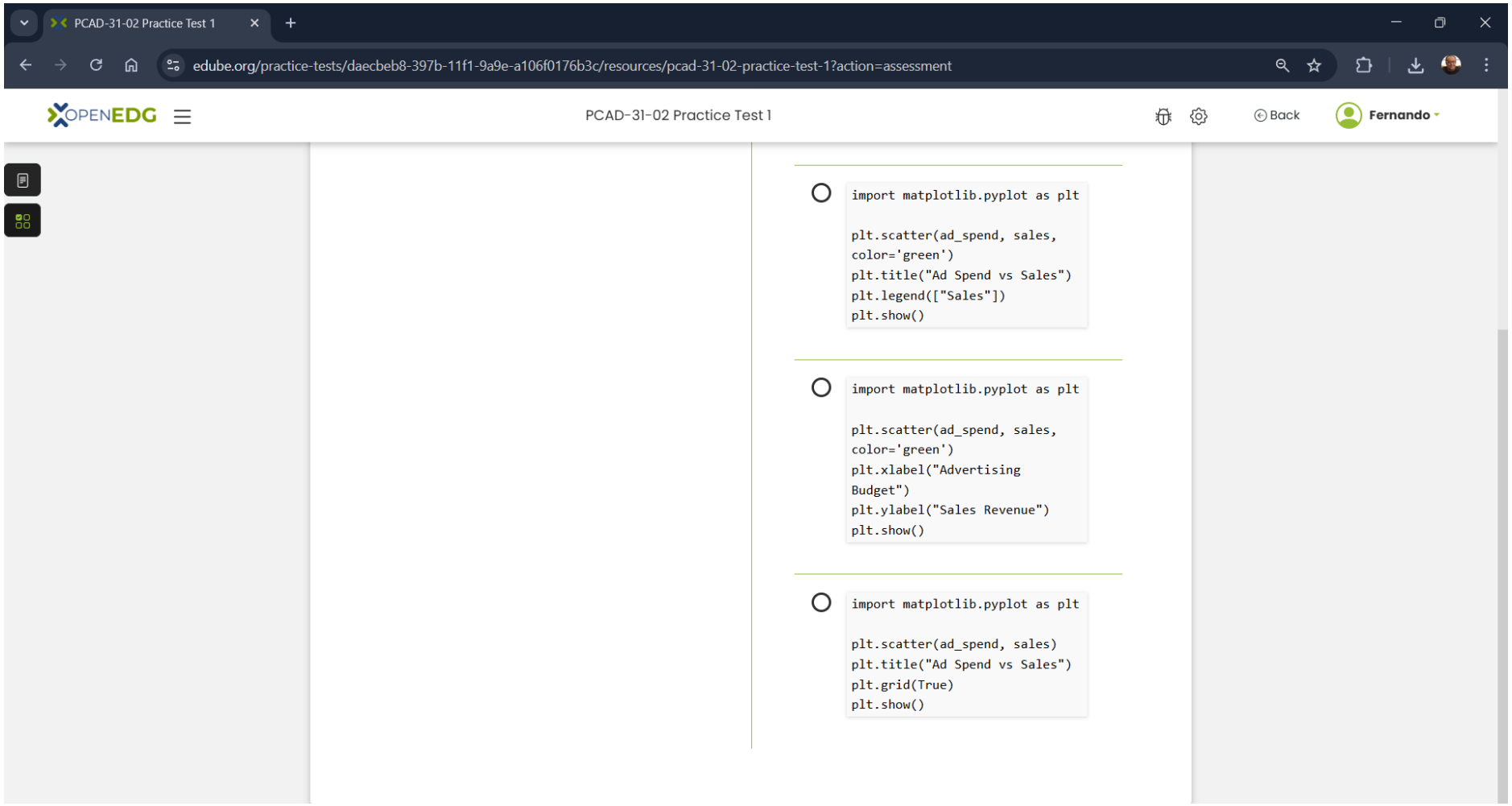
You are reviewing the chart below, which shows average monthly temperatures using vertical bars:

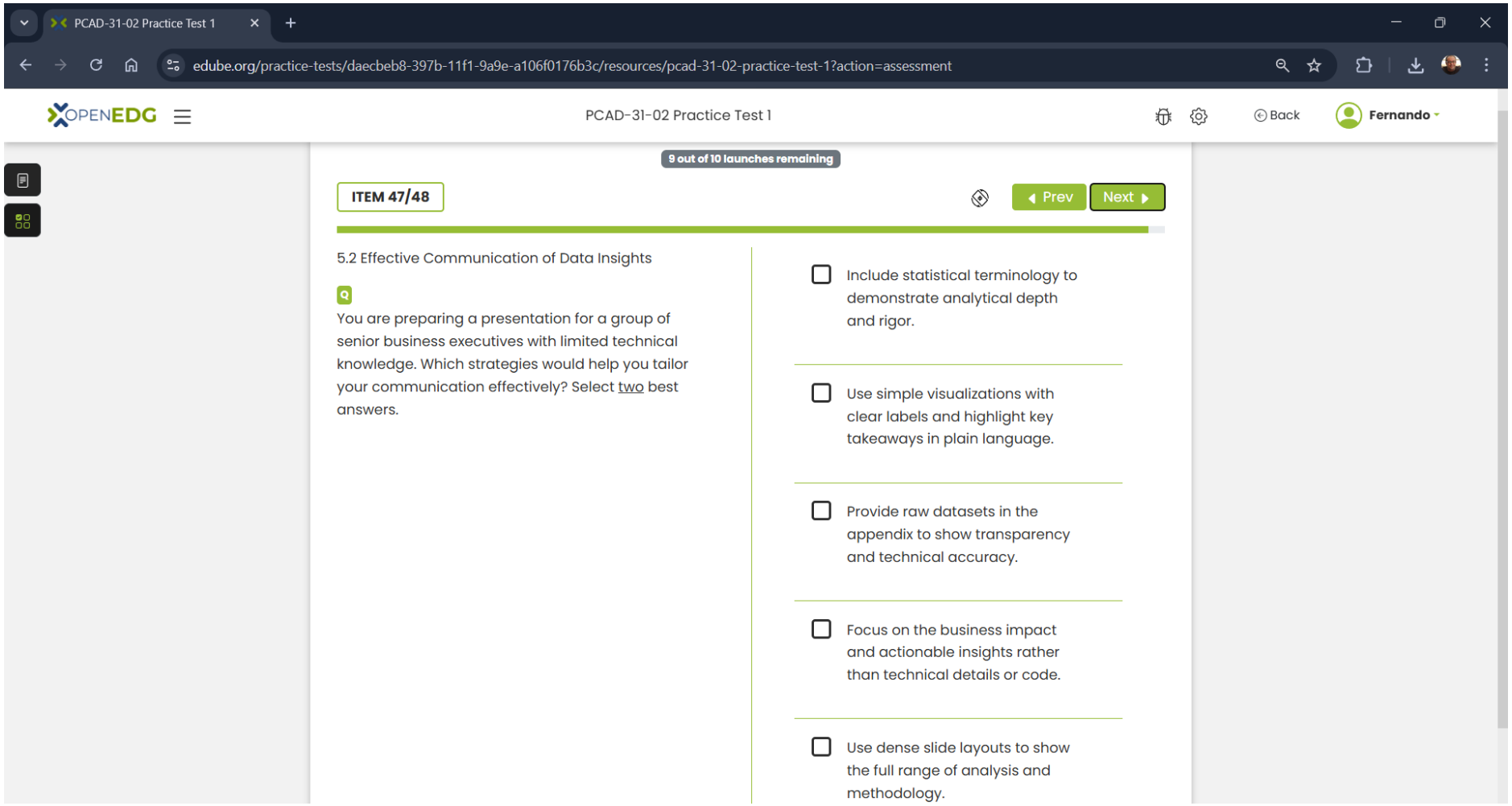


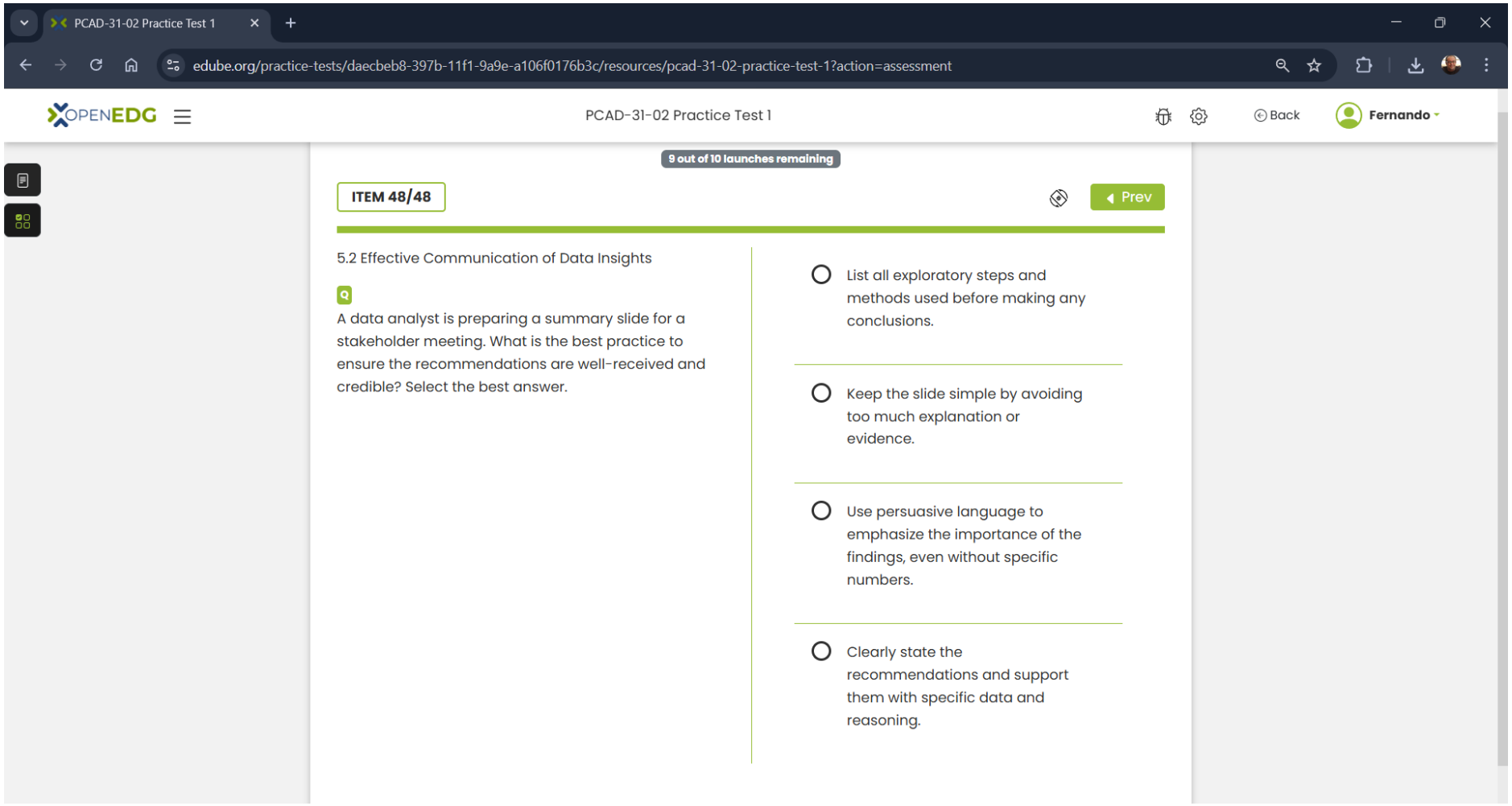
Based on the type of data and the chart style, which of the statements are correct? Select two best answers.

- [ ] Using color in the bars introduces a bias in interpreting average values.
- [ ] The chart is misleading because the bars are not sorted by value.
- [ ] The y-axis starts at 0, which exaggerates the differences between months.
- [ ] The chart does not clearly emphasize the continuity or seasonal trend in temperature changes.
- [ ] A line plot would better reflect the progression and smooth


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5.1 Data Visualization Techniques
A data analyst is creating a scatterplot to show the relationship between ad spend and sales. They want to improve the plot's readability and clarity.
Which of the following code snippets applies best practices for labeling and refining the chart? Select the best answer.
import matplotlib.pyplot as plt
plt.scatter(ad_spend, sales, color='green')
plt.title("Ad Spend vs Sales")
plt.xlabel("Advertising Budget ($)")
plt.ylabel("Sales Revenue ($)")
plt.grid(True)
plt.show()
import matplotlib.pyplot as plt
plt.scatter(ad_spend, sales, color='green')
plt.title("Ad Spend vs Sales")
plt.legend(["Sales"])
plt.show()
import matplotlib.pyplot as plt
plt.scatter(ad_spend, sales,


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import matplotlib.pyplot as plt
plt.scatter(ad_spend, sales, color='green')
plt.title("Ad Spend vs Sales")
plt.legend(["Sales"])
plt.show()
import matplotlib.pyplot as plt
plt.scatter(ad_spend, sales, color='green')
plt.xlabel("Advertising Budget")
plt.ylabel("Sales Revenue")
plt.show()
import matplotlib.pyplot as plt
plt.scatter(ad_spend, sales)
plt.title("Ad Spend vs Sales")
plt.grid(True)
plt.show()


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5.2 Effective Communication of Data Insights
Q
You are preparing a presentation for a group of senior business executives with limited technical knowledge. Which strategies would help you tailor your communication effectively? Select two best answers.
Include statistical terminology to demonstrate analytical depth and rigor.
Use simple visualizations with clear labels and highlight key takeaways in plain language.
Provide raw datasets in the appendix to show transparency and technical accuracy.
Focus on the business impact and actionable insights rather than technical details or code.
Use dense slide layouts to show the full range of analysis and methodology.


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5.2 Effective Communication of Data Insights
Q
A data analyst is preparing a summary slide for a stakeholder meeting. What is the best practice to ensure the recommendations are well-received and credible? Select the best answer.
List all exploratory steps and methods used before making any conclusions.
Keep the slide simple by avoiding too much explanation or evidence.
Use persuasive language to emphasize the importance of the findings, even without specific numbers.
Clearly state the recommendations and support them with specific data and reasoning.